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Chapter 1

A New Auxiliary Model Approach for Interacting Electrons: Mapping Impurity Models To Lattice Models

1.1 The Philosophy of Auxiliary Models

In the previous chapter, we extended the Anderson impurity model to host a correlation-driven delocalisation-localisation transition on the impurity site, from a local Fermi liquid state to a local moment state. We demonstrated that this transition captures very well the phenomenology of the Hubbard dimension in infinite dimensions (as seen from, say, dynamical mean-field theory (DMFT)). In this chapter, we extend this approach by developing a formalism to study the more realistic case of finite-dimensional interacting models by using appropriately chosen impurity models as auxiliary models. The key to an *auxiliary model method* is to construct an appropriate auxiliary model that while being tractable captures some essential property of the full lattice model [1]. This involves separating the complete degrees of freedom into a system part (H_S) which one treats exactly and the rest of the system (H_R) that one might simplify in order to be able to solve the problem. The non-trivial nature of the problem arises of course from the hybridisation H_{SR} between the system and the bath:

$$H = \begin{bmatrix} H_R & H_{RS} \\ H_{RS}^* & H_S \end{bmatrix}. \quad (1.1)$$

This separation can always be done exactly, if one does not care about tractability. For example, one can take the Hubbard model that describes electrons hopping on a lattice and interacting when two electrons reside on the same lattice site:

$$H_{\text{hub}} = -t \sum_{\langle i,j \rangle, \sigma} (c_{i,\sigma}^\dagger c_{j,\sigma} + \text{h.c.}) + U \sum_i n_{i\uparrow} n_{i\downarrow}, \quad (1.2)$$

and separate this into parts, the system being the local orbitals of a specific site l and the rest of the

system being all the other sites:

$$H_S = U n_{l\uparrow} n_{l\downarrow}, \quad H_R = -t \sum_{\langle i,j \rangle, \sigma} (c_{i,\sigma}^\dagger c_{j,\sigma} + \text{h.c.}) + U \sum_i n_{i\uparrow} n_{i\downarrow}, \quad H_{SR} + H_{RS} = -t \sum_{i,\sigma} (c_{l,\sigma}^\dagger c_{i,\sigma} + \text{h.c.}), \quad (1.3)$$

where the primed summation is carried out over all nearest-neighbour pairs that do not involve l and the summation in $H_{SR} + H_{RS}$ is over all sites adjacent to l .

The smaller system H_S (often called the *cluster*) is typically chosen such that its eigenstates are known exactly. Progress is then made by choosing a simpler form for H_R and its coupling H_{RS} with the smaller system. This combination of the cluster and the simpler bath is then called the *auxiliary system*. A tractable auxiliary system for the Hubbard model is the single-impurity Anderson model (SIAM); the correlated impurity site represents the set of local orbitals of any given site, while the conduction bath represents the rest of the lattice sites in an approximate fashion. Such a construction is shown in fig. 1.1.

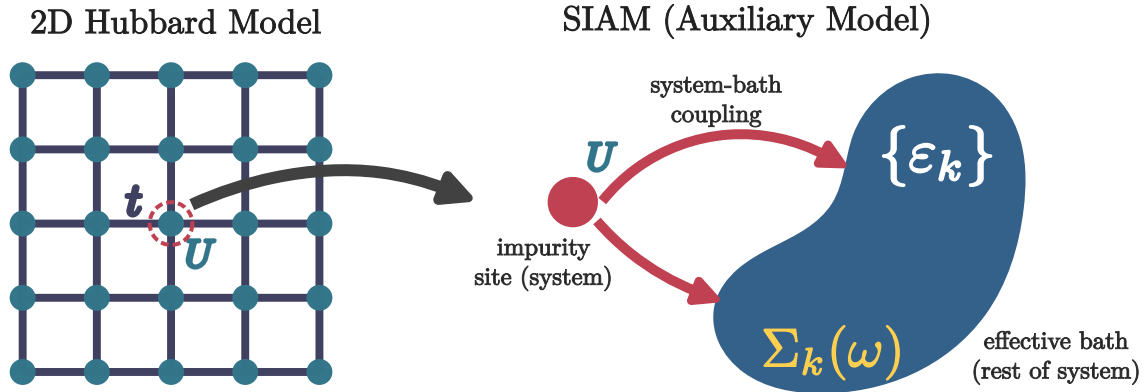


Figure 1.1: Left: Full Hubbard model lattice with onsite repulsion U^H on all sites and hopping between nearest neighbour sites with strength t^H . Right: Extraction of the auxiliary (cluster+bath) system from the full lattice. The central site on left becomes the impurity site (red) on the right (with an onsite repulsion ϵ_d), while the rest of the $N - 1$ sites on the left form a conduction bath (green circles) (with dispersion ϵ_k and correlation modelled by the self-energy $\Sigma_k(\omega)$) that hybridizes with the impurity through the coupling V .

It should be noted that any reasonable choice of the cluster and bath would break the translational symmetry of the full model: To allow computing quantities, one would need to make the bath (which is a much larger system) simpler than the cluster (which is a single site). This distinction breaks the translational symmetry of the Hubbard model. As a result, calculations that make use of the translation symmetry of the full model (such as k -space objects) will require a periodisation operation of an auxiliary model quantity.

The present work is aimed towards developing a new auxiliary model method to study correlated electronic systems. As described above, the first step is to identify the correct impurity model which can faithfully capture the physics of the lattice model we are trying to solve. The local behaviour of this impurity model should reflect the essential local physics of the lattice model. Typically,

we will consider impurity model geometries where the real-space bath site connected directly to the impurity hosts some local interaction. We will henceforth refer to this site as the *zeroth site*. This model must be solved using an impurity solver. In the present work, we use the recently-developed unitary renormalisation group method [2–7]. The leap to the bulk model is then made by applying **manybody lattice translation operators** on the auxiliary model. This process, referred to as *tiling* here, allows us to reconstruct the lattice model Hamiltonian from that of the auxiliary model and restores the translation invariance of the full lattice model. Once the Hamiltonians can be mapped onto one another, it is possible to relate the eigenstates and correlations functions between the auxiliary model and lattice model, using a manybody version of **Bloch’s theorem**.

1.2 What Is The Correct Auxiliary Model For Correlated Electrons on a 2D Square Lattice?

As discussed in more detail in the previous chapter, it was necessary to extend the single-impurity Anderson model (SIAM) in order to obtain a phase transition on the impurity site. Without any modification, the standard Anderson model (at particle-hole symmetry) leads to the emergence of an $\omega = 0$ Kondo resonance in the impurity ($d\sigma$) spectral function at any value of the impurity correlation U_d , at sufficiently low temperatures:

$$H_{\text{SIAM}} = \sum_{k,\sigma} \varepsilon_k n_{k,\sigma} - V \sum_{\sigma} (c_{0,\sigma}^\dagger c_{d,\sigma} + \text{h.c.}) + \varepsilon_d \sum_{\sigma} n_{d,\sigma} + U_d n_{d,\uparrow} n_{d,\downarrow}. \quad (2.1)$$

The operator $c_{0,\sigma}$ acts on the local conduction bath density that couples with the impurity site. The low-energy is one of local Fermi liquid for all parameter regimes. Increasing U_d only serves to reduce the width of the central peak at the cost of the appearance of side bands at energy scales of the order of U_d , but the resonance never disappears. The extended Anderson model that we developed augments the SIAM by allowing for local correlation in the local conduction bath charge density that couples with the impurity spin:

$$H_{\text{ESIAM}} = H_{\text{SIAM}} + J_K \mathbf{S}_d \cdot \mathbf{S}_0 - \frac{U_b}{2} (n_{0,\uparrow}^2 - n_{0,\downarrow}^2); \quad (2.2)$$

tuning this bath correlation to large attractive values (see [8] for some recent findings of non-local effective attractive interactions within the Hubbard model) *does* lead to a localisation transition on the impurity where the local Fermi liquid excitations are replaced by the physics of a gapped local moment on the impurity. We note that the generation of such correlations in the bath is something that occurs in DMFT as well – the conduction bath acquires a *non-trivial self-energy* during the convergence to self-consistency.

While this modified impurity model captures well the Mott metal-insulator transition of the Hubbard model in infinite dimensions, it *lacks any knowledge of the underlying lattice geometry*, and is therefore ill-suited to address the phenomenology of more realistic low-dimensional models. Examples of such phenomenology include the pseudogap phase of the cuprates (with nodal-antinodal dichotomy), strange metal (with T -linear resistivity) and high- T_c superconducting phases observed

in several families of quantum materials as well as graphene. This leads us to propose the framework of *lattice-embedded quantum impurity models*. Upon embedding the impurity site within the underlying lattice, the coupling between it and its environment must be consistent with the lowered symmetry of the lattice (see Fig. 1.2). Impurity-bath correlations calculated within such an impurity model are sensitive to the lattice geometry, and are much better suited to address the phenomenology mentioned above.

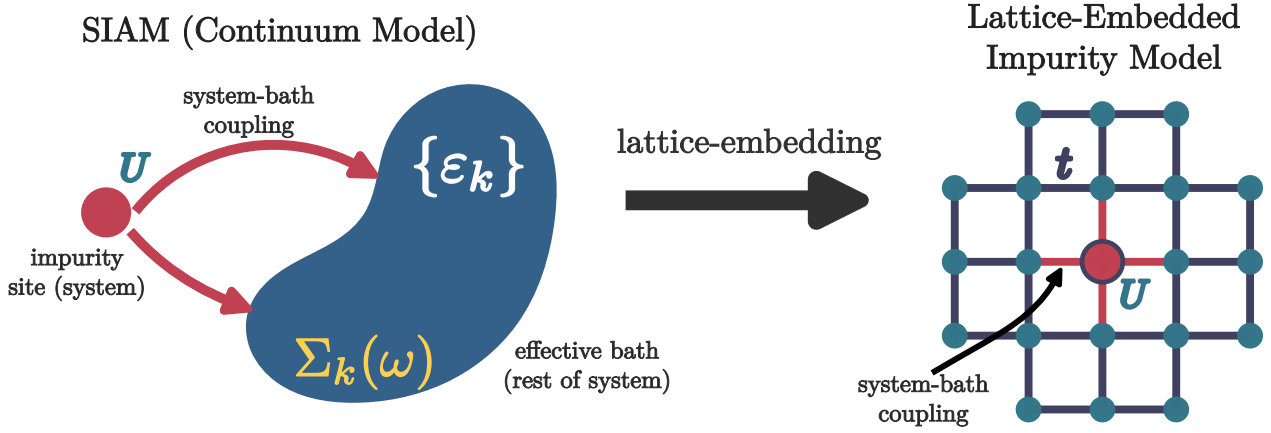


Figure 1.2

As a concrete example, we take the extended SIAM defined in eq. 2.2, and write down its lattice-embedded form on a 2D square lattice, with the impurity placed at the site r_d . This is done by allowing the impurity to hybridise with the four nearest-neighbours, through V and J_K . The same four sites also carry the local bath interaction. The complete auxiliary model is

$$\begin{aligned}
 H = & -t \sum_{\langle i,j \rangle, \sigma} (c_{i,\sigma}^\dagger c_{j,\sigma} + \text{h.c.}) + \sum_{\langle i, r_d \rangle} \left[-\frac{V}{Z} \sum_{\sigma} (c_{r_d, \sigma}^\dagger c_{i, \sigma} + \text{h.c.}) + \frac{J_K}{Z} \mathbf{S}_{r_d} \cdot \mathbf{S}_i - \frac{W}{2Z} (n_{i, \uparrow}^2 - n_{i, \downarrow}^2) \right] \\
 & + \varepsilon_d \sum_{\sigma} n_{r_d, \sigma} + U_d n_{r_d, \uparrow} n_{r_d, \downarrow},
 \end{aligned} \tag{2.3}$$

where Z is the coordination number of the lattice, and the second summation $\sum_{\langle i, r_d \rangle}$ is carried out over all sites i that are nearest-neighbour to the site r_d . Note that the one-particle hybridisation V and the spin-exchange coupling J_K have their *symmetry lowered* from the s -wave spherical symmetry of the continuum impurity models to a C_4 -symmetry consistent with the underlying lattice. This feature is crucial in explaining the phenomenology of the various low-dimensional materials.

Various enhancements can be applied to such a model. The lattice can be modified to study different materials. More complicated geometries such as a two-layer model can be studied using a two-impurity auxiliary model, where each impurity couples with its own conduction bath as well as to each other.

1.3 Mapping Auxiliary Model To Lattice Model

In this subsection, we provide an explicit example of constructing a lattice model from a lattice-embedded impurity model. This will involve the use of manybody lattice translation operators, so we begin with a brief discussion of them.

1.3.1 Manybody Translation Operators

In order to create the bulk model, we need to translate the local auxiliary model over the entire lattice. For this, we define *many-particle global* translation operators $T(\mathbf{a})$ that translate all positions by a vector $\mathbf{a}$. In terms of manybody states and operators, their action is defined as

$$\begin{aligned} T^\dagger(\mathbf{a}) |\mathbf{r}_1, \mathbf{r}_2, \dots\rangle &= |\mathbf{r}_1 + \mathbf{a}\rangle \otimes |\mathbf{r}_2 + \mathbf{a}\rangle \dots \otimes |\mathbf{r}_n + \mathbf{a}\rangle \\ T^\dagger(\mathbf{a}) \mathcal{O}(\mathbf{r}_1, \mathbf{r}_2, \dots) T(\mathbf{a}) &= \mathcal{O}(\mathbf{r}_1 + \mathbf{a}, \mathbf{r}_2 + \mathbf{a}, \dots), \end{aligned} \quad (3.1)$$

where $|\mathbf{r}_1, \mathbf{r}_2, \dots\rangle$ is a state in the manyparticle Fock-space basis with the particles localised at the specified positions. For example, for a local fermionic creation operator $c^\dagger(\mathbf{r})$, we have

$$T^\dagger(\mathbf{a}) c^\dagger(\mathbf{r}) T(\mathbf{a}) = c^\dagger(\mathbf{r} + \mathbf{a}). \quad (3.2)$$

It acts similarly on the auxiliary model Hamiltonian:

$$T^\dagger(\mathbf{a}) H_{\text{aux}}(\mathbf{r}_d) T(\mathbf{a}) = H_{\text{aux}}(\mathbf{r}_d + \mathbf{a}), \quad (3.3)$$

translating all sites by the vector $\mathbf{a}$. By introducing the Fourier transform to momentum space,

$$|\mathbf{r}_1, \mathbf{r}_2, \dots\rangle = \otimes_{j=1}^N \int d\mathbf{k}_j e^{i\mathbf{r}_j \cdot \mathbf{k}_j} |\mathbf{k}_j\rangle, \quad (3.4)$$

it is easy to see that the total momentum states are eigenstates of the global translation operators:

$$\begin{aligned} T^\dagger(\mathbf{a}) |\mathbf{k}_1, \mathbf{k}_2, \dots\rangle &= \otimes_{j=1}^N \int d\mathbf{r}_j e^{-i\mathbf{r}_j \cdot \mathbf{k}_j} T^\dagger(\mathbf{a}) |\mathbf{r}_j\rangle, \\ &= \otimes_{j=1}^N \int d\mathbf{r}_j e^{-i\mathbf{r}_j \cdot \mathbf{k}_j} |\mathbf{r}_j + \mathbf{a}\rangle \\ &= \otimes_{j=1}^N e^{i\mathbf{a} \cdot \mathbf{k}_j} \int d\mathbf{r}_j e^{-i\mathbf{r}_j \cdot \mathbf{k}_j} |\mathbf{r}_j\rangle \\ &= e^{i\mathbf{a} \cdot \mathbf{k}_{\text{tot}}} |\mathbf{k}_1, \mathbf{k}_2, \dots\rangle, \end{aligned} \quad (3.5)$$

where $\mathbf{k}_{\text{tot}} = \sum_j \mathbf{k}_j$ is the total momentum. This gives a compact form for the translation operator:

$$T^\dagger(\mathbf{a}) = e^{i\mathbf{a} \cdot \hat{\mathbf{k}}_{\text{tot}}}, \quad (3.6)$$

where $\hat{\mathbf{k}}_{\text{tot}}$ is the total momentum operator.

We end this section by calculating the effect of the translation operators on the k -space annihilation operators. We have

$$c(\mathbf{k}) = \frac{1}{\sqrt{N}} \sum_{\mathbf{r}} e^{-i\mathbf{k} \cdot \mathbf{r}} c(\mathbf{r}) . \quad (3.7)$$

Applying the translation operator to this decomposition, we get

$$\begin{aligned} T(\vec{a}) c(\mathbf{k}) T^\dagger(\vec{a}) &= \frac{1}{\sqrt{N}} \sum_{\mathbf{r}} e^{-i\mathbf{k} \cdot \mathbf{r}} T(\vec{a}) c(\mathbf{r}) T^\dagger(\vec{a}) \\ &= \frac{1}{\sqrt{N}} \sum_{\mathbf{r}} e^{-i\mathbf{k} \cdot \mathbf{r}} c(\mathbf{r} - \vec{a}) \\ &= e^{-i\mathbf{k} \cdot \vec{a}} c(\mathbf{k}) . \end{aligned} \quad (3.8)$$

1.3.2 Periodisation of Auxiliary Model Into Extended Hubbard Model

We recall the lattice-embedded impurity model defined in eq. 2.3. As discussed above, the route to restoring translation symmetry and reconstructing the lattice model is to translate the auxiliary model using manybody translation operators. Accordingly, we identify the “unit cell” of translation. Since we expect the impurity site to capture most of the physics arising from the correlated electrons, a good choice for the unit cell is the impurity site and its surrounding correlated environment. In order to retain information of the lattice geometry, we convert the hybridisation and bath interaction terms from real space to k -space. The Hamiltonian for the unit cell is therefore

$$\begin{aligned} H_{\text{clust}}(\mathbf{r}_d) &= - \sum_{\mathbf{k}, \sigma} e^{i\mathbf{k} \cdot \mathbf{r}_d} V(\mathbf{k}) (c_{\mathbf{r}_d, \sigma}^\dagger c_{\mathbf{k}, \sigma} + \text{h.c.}) + \sum_{\mathbf{k}_1, \mathbf{k}_2} e^{i(\mathbf{k}_1 - \mathbf{k}_2) \cdot \mathbf{r}_d} J_K(\mathbf{k}_1, \mathbf{k}_2) \mathbf{S}_{\mathbf{r}_d} \cdot \mathbf{S}_{\mathbf{k}_1, \mathbf{k}_2} + \varepsilon_d n_{\mathbf{r}_d} \\ &\quad + U_d n_{\mathbf{r}_d, \uparrow} n_{\mathbf{r}_d, \downarrow} - \frac{1}{2} \sum_{\{\mathbf{k}_i\}, \sigma, \sigma'} \sigma \sigma' e^{i(\mathbf{k}_1 - \mathbf{k}_2 + \mathbf{k}_3 - \mathbf{k}_4) \cdot \mathbf{r}_d} W(\{\mathbf{k}_i\}) c_{\mathbf{k}_1, \sigma}^\dagger c_{\mathbf{k}_2, \sigma} c_{\mathbf{k}_3, \sigma'}^\dagger c_{\mathbf{k}_4, \sigma'} , \end{aligned} \quad (3.9)$$

where $\mathbf{S}_{\mathbf{k}_1, \mathbf{k}_2} = \sum_{\alpha, \beta} \sigma_{\alpha, \beta} c_{\mathbf{k}_1, \alpha}^\dagger c_{\mathbf{k}_2, \beta}$, and the Fourier transforms are defined by the following relations:

$$\begin{aligned} \sum_{\mathbf{k}} e^{i\mathbf{k} \cdot \mathbf{r}_d} V(\mathbf{k}) c_{\mathbf{r}_d, \sigma}^\dagger c_{\mathbf{k}, \sigma} &= \frac{V}{Z} \sum_{\mathbf{r} \rightarrow \text{NN of } \mathbf{r}_d} c_{\mathbf{r}_d, \sigma}^\dagger c_{\mathbf{r}, \sigma} , \\ \sum_{\mathbf{k}_1, \mathbf{k}_2} e^{i(\mathbf{k}_1 - \mathbf{k}_2) \cdot \mathbf{r}_d} J_K(\mathbf{k}_1, \mathbf{k}_2) \vec{S}_{\mathbf{r}_d} \cdot \vec{S}_{\mathbf{k}_1, \mathbf{k}_2} &= \frac{J_K}{Z} \sum_{\mathbf{r} \rightarrow \text{NN of } \mathbf{r}_d} \mathbf{S}_{\mathbf{r}_d} \cdot \mathbf{S}_{\mathbf{r}} , \\ \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{q}_1, \mathbf{q}_2} e^{i(\mathbf{k}_1 - \mathbf{k}_2 + \mathbf{k}_3 - \mathbf{k}_4) \cdot \mathbf{r}_d} W(\{\mathbf{k}_i\}) c_{\mathbf{k}_1, \sigma}^\dagger c_{\mathbf{k}_2, \sigma} c_{\mathbf{k}_3, \sigma'}^\dagger c_{\mathbf{k}_4, \sigma'} &= \frac{W}{2Z} \sum_{\mathbf{r} \rightarrow \text{NN of } \mathbf{r}_d} n_{\mathbf{r}, \sigma} n_{\mathbf{r}, \sigma'} , \end{aligned} \quad (3.10)$$

We now translate this unit cell to all sites of the lattice, and reconstruct a translationally-invariant correlated lattice model:

$$H_{\text{latt}} = \sum_{\mathbf{r}} T^\dagger(\mathbf{r} - \mathbf{r}_d) H_{\text{clust}}(\mathbf{r}_d) T(\mathbf{r} - \mathbf{r}_d) . \quad (3.11)$$

We consider the effect of the translation operations on each part of the Hamiltonian. The purely local terms translate directly into a chemical potential term and a Hubbard interaction term:

$$\sum_{\mathbf{r}} T^\dagger(\mathbf{r} - \mathbf{r}_d) (\varepsilon_d \sum_{\sigma} n_{\mathbf{r}_d, \sigma} + U_d n_{\mathbf{r}_d, \uparrow} n_{\mathbf{r}_d, \downarrow}) T(\mathbf{r} - \mathbf{r}_d) = \varepsilon_d \sum_{\mathbf{r}, \sigma} n_{\mathbf{r}, \sigma} + U_d \sum_{\mathbf{r}} n_{\mathbf{r}, \uparrow} n_{\mathbf{r}, \downarrow}. \quad (3.12)$$

The one-particle hybridisation term maps onto a tight-binding hopping term:

$$\begin{aligned} \sum_{\mathbf{r}} T^\dagger(\mathbf{r} - \mathbf{r}_d) \sum_{\mathbf{k}} e^{i\mathbf{k} \cdot \mathbf{r}_d} V(\mathbf{k}) c_{\mathbf{r}_d, \sigma}^\dagger c_{\mathbf{k}, \sigma} T(\mathbf{r} - \mathbf{r}_d) &= \sum_{\mathbf{r}} \sum_{\mathbf{k}} e^{i\mathbf{k} \cdot \mathbf{r}} V(\mathbf{k}) c_{\mathbf{r}, \sigma}^\dagger c_{\mathbf{k}, \sigma} \\ &= \frac{V}{Z} \sum_{\mathbf{r}} \sum_{\mathbf{r}' \rightarrow \text{NN of } \mathbf{r}} c_{\mathbf{r}, \sigma}^\dagger c_{\mathbf{r}', \sigma} \\ &= \frac{2V}{Z} \sum_{\langle \mathbf{r}, \mathbf{r}' \rangle} c_{\mathbf{r}, \sigma}^\dagger c_{\mathbf{r}', \sigma}, \end{aligned} \quad (3.13)$$

where we used eq. 3.8 and the first equation from 3.10, and the fact that each nearest-neighbour pair $\mathbf{r}, \mathbf{r}'$ appears twice in the summation.

The Kondo terms map onto a nearest-neighbour spin-exchange term:

$$\begin{aligned} \sum_{\mathbf{r}, \mathbf{k}_1, \mathbf{k}_2} T^\dagger(\mathbf{r} - \mathbf{r}_d) e^{i(\mathbf{k}_1 - \mathbf{k}_2) \cdot \mathbf{r}_d} J_K(\mathbf{k}_1, \mathbf{k}_2) \vec{S}_{\mathbf{r}_d} \cdot \vec{S}_{\mathbf{k}_1, \mathbf{k}_2} T(\mathbf{r} - \mathbf{r}_d) &= \sum_{\mathbf{r}, \mathbf{k}_1, \mathbf{k}_2} e^{i(\mathbf{k}_1 - \mathbf{k}_2) \cdot \mathbf{r}} J_K(\mathbf{k}_1, \mathbf{k}_2) \vec{S}_{\mathbf{r}} \cdot \vec{S}_{\mathbf{k}_1, \mathbf{k}_2} \\ &= \frac{J_K}{Z} \sum_{\mathbf{r}} \sum_{\mathbf{r}' \rightarrow \text{NN of } \mathbf{r}} \vec{S}_{\mathbf{r}} \cdot \vec{S}_{\mathbf{r}'}, \\ &= \frac{2J_K}{Z} \sum_{\langle \mathbf{r}, \mathbf{r}' \rangle} \vec{S}_{\mathbf{r}} \cdot \vec{S}_{\mathbf{r}'}, \end{aligned} \quad (3.14)$$

where we used the second equation from 3.10. Finally, the bath interaction term W maps onto another Hubbard term

$$\begin{aligned} \sum_{\mathbf{r}, \mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} T^\dagger(\mathbf{r} - \mathbf{r}_d) e^{i(\mathbf{k}_1 - \mathbf{k}_2 + \mathbf{k}_3 - \mathbf{k}_4) \cdot \mathbf{r}_d} W(\{\mathbf{k}_i\}) c_{\mathbf{k}_1, \sigma}^\dagger c_{\mathbf{k}_2, \sigma} c_{\mathbf{k}_3, \sigma'}^\dagger c_{\mathbf{k}_4, \sigma'} T(\mathbf{r} - \mathbf{r}_d) \\ = \sum_{\mathbf{r}, \mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} e^{i(\mathbf{k}_1 - \mathbf{k}_2 + \mathbf{k}_3 - \mathbf{k}_4) \cdot \mathbf{r}} W(\{\mathbf{k}_i\}) c_{\mathbf{k}_1, \sigma}^\dagger c_{\mathbf{k}_2, \sigma} c_{\mathbf{k}_3, \sigma'}^\dagger c_{\mathbf{k}_4, \sigma'} \\ = \frac{W}{2Z} \sum_{\mathbf{r}} \sum_{\mathbf{r}' \rightarrow \text{NN of } \mathbf{r}} n_{\mathbf{r}', \sigma} n_{\mathbf{r}', \sigma'} \\ = \frac{W}{2} \sum_{\mathbf{r}} n_{\mathbf{r}, \sigma} n_{\mathbf{r}, \sigma'}, \end{aligned} \quad (3.15)$$

where we used the fact that each point $\mathbf{r}'$ (which we relabel as $\mathbf{r}$) appears Z times in the summation.

The complete lattice Hamiltonian, after applying the translation operators, is

$$H_{\text{latt}} = \varepsilon_d \sum_{\mathbf{r}, \sigma} n_{\mathbf{r}, \sigma} + U_d \sum_{\mathbf{r}} n_{\mathbf{r}, \uparrow} n_{\mathbf{r}, \downarrow} - \frac{2V}{Z} \sum_{\langle \mathbf{r}, \mathbf{r}' \rangle} c_{\mathbf{r}, \sigma}^\dagger c_{\mathbf{r}', \sigma} + \frac{2J_K}{Z} \sum_{\langle \mathbf{r}, \mathbf{r}' \rangle} \vec{S}_{\mathbf{r}} \cdot \vec{S}_{\mathbf{r}'} - \frac{W}{2} \sum_{\mathbf{r}} (n_{\mathbf{r}, \uparrow} - n_{\mathbf{r}, \downarrow})^2. \quad (3.16)$$

This is clearly an extended Hubbard model, with both 1-particle and spin-exchange nearest-neighbour hybridisation. The general form of the model is

$$H_{\text{ex-Hub}} = -t \sum_{\langle i,j \rangle, \sigma} (c_{i,\sigma}^\dagger c_{j,\sigma} + \text{h.c.}) + J \sum_{\langle i,j \rangle} \mathbf{S}_i \cdot \mathbf{S}_j + U \sum_i n_{i,\uparrow} n_{i,\downarrow} - \mu \sum_{i,\sigma} n_{i,\sigma} , \quad (3.17)$$

such that the effective couplings resulting from the tiling operation is

$$t = \frac{2V}{Z}, \quad J = \frac{2J_K}{Z}, \quad U = U_d - W, \quad \mu = -\varepsilon_d + \frac{W}{2} . \quad (3.18)$$

1.3.3 Form Of The Eigenstates: Bloch's Theorem

The tiled Hamiltonian defined in eq. 3.11 is manifestly symmetric under discrete translation, given that it's expressed in terms of the translation operators. The fact that the Hamiltonian H_{latt} commutes with the manybody translation operator implies that the total crystal momentum is a conserved quantity. We will now use this to write down a “tiling” relation between the eigenstates of the auxiliary model and those of the lattice model.

In the tight-binding approach to lattice problems, the full Hamiltonian is obtained by translating the dynamics of the localised orbitals at each site over the entire lattice. The full eigenstate $|\Psi\rangle$ is obtained by constructing linear combinations of the eigenstates $|\psi(i)\rangle$ of the local Hamiltonians. According to Bloch's theorem, $|\Psi\rangle$ is of the form $|\Psi_{n,K}\rangle = \sum_i e^{i\mathbf{K} \cdot \mathbf{r}_i} |\psi_n(i)\rangle$, where $\mathbf{K}$ is the single-particle crystal momentum, n is the band-index that labels various states within the local orbital, and $\mathbf{r}_i$ sums over the positions of the local Hamiltonians. Bloch's theorem ensures that eigenstates satisfy the following relation under a translation operation by an arbitrary number of lattice spacings $n\mathbf{a}$:

$$T^\dagger(l\mathbf{a}) |\Psi_{n,K}\rangle = \sum_i e^{i\mathbf{K} \cdot \mathbf{r}_i} |\psi_n(i+l)\rangle = e^{-il\mathbf{K} \cdot \mathbf{a}} |\Psi_{n,K}\rangle . \quad (3.19)$$

As discussed earlier, the eigenstates $|\Psi_{n,K}\rangle$ of the lattice Hamiltonian obtained using eq. 3.11 enjoy the discrete translation symmetry of the lattice, and therefore enjoy a *many-body* Bloch's theorem [9]. This means that the *local* eigenstates $|\psi_n(\mathbf{r}_d)\rangle$ (with the impurity located at an arbitrary position $\mathbf{r}_d$) of the unit cell auxiliary model Hamiltonian $H_{\text{clust}}(\mathbf{r}_d)$ defined in eq. 3.9 can be used to construct eigenstates of the lattice Hamiltonian. The index $n(= 0, 1, \dots)$ in the subscript indicates that it is the n^{th} eigenstate of the auxiliary model. The following combination of the auxiliary model eigenstates satisfies a many-particle equivalent of Bloch's theorem [9]:

$$|\Psi_s\rangle \equiv |\Psi_{K,n}\rangle = \frac{1}{\sqrt{N}} \sum_{\mathbf{r}_d} e^{i\mathbf{K} \cdot \mathbf{r}_d} |\psi_n(\mathbf{r}_d)\rangle , \quad (3.20)$$

where N is the total number of lattice sites and $\mathbf{r}_d$ is summed over all lattice spacings. The set of $n = 0$ states form the lowest band in the spectrum of the lattice, while higher values of n produce the more energetic bands.

1.4 Properties of the Bloch states

1.4.1 Translation invariance

It is easy to verify that $\Psi_{\vec{k}}(\{\mathbf{r}_k\})$ transforms like Bloch functions under translation by a displacement $\mathbf{r}$:

$$\begin{aligned}\Psi_{\vec{k}}(\{\mathbf{r}_k + \mathbf{r}\}) &= \frac{1}{\lambda_{|\vec{k}\rangle} \mathcal{Z} N} \sum_{\mathbf{r}_i, \mathbf{a}} e^{i\vec{k} \cdot \vec{R}_i} \psi_{\text{aux}}(\mathbf{r}_i - \mathbf{r}, \mathbf{a}) = \frac{1}{\lambda_{|\vec{k}\rangle} \mathcal{Z} N} \sum_{\mathbf{r}_j, \mathbf{a}} e^{i\vec{k} \cdot (\vec{R}_j + \mathbf{r})} \psi_{\text{aux}}(\mathbf{r}_j, \mathbf{a}) \\ &= e^{i\vec{k} \cdot \vec{R}} \Psi_{\vec{k}}(\{\mathbf{r}_k\})\end{aligned}\quad (4.1)$$

In the last equation, we transformed $\mathbf{r}_i \rightarrow \mathbf{r}_j = \mathbf{r}_i - \mathbf{r}$. Note that the argument $\mathbf{a}$ does not change under a translation of the system, because that vector always represents the difference between the impurity lattice position and its nearest neighbours, irrespective of the absolute position of the impurity.

The wavefunction can be even brought into the familiar Bloch function form:

$$\begin{aligned}\Psi_{\vec{k}}^n(\{\mathbf{r}_k\}) &= \sum_{\mathbf{r}_i, \mathbf{a}} \frac{e^{i\vec{k} \cdot \vec{R}_i}}{\lambda_{|\vec{k}\rangle} \mathcal{Z} N} \psi_{\text{aux}}(\{\mathbf{r}_d - \mathbf{r}_i, \mathbf{r}_0 - \mathbf{r}_i - \mathbf{a}\}) \\ &= \frac{e^{i\vec{k} \cdot \frac{1}{N} \sum_k \vec{r}_k}}{\lambda_{|\vec{k}\rangle} \mathcal{Z} N} \sum_{\mathbf{r}_i, \mathbf{a}} e^{-i\vec{k} \cdot (\frac{1}{N} \sum_k \mathbf{r}_k - \vec{R}_i)} \psi_{\text{aux}}(\{\mathbf{r}_d - \mathbf{r}_i, \mathbf{r}_0 - \mathbf{r}_i - \mathbf{a}\}) \\ &= e^{i\vec{k} \cdot \vec{r}_{\text{COM}}} \eta_{\vec{k}}(\{\mathbf{r}_k\})\end{aligned}\quad (4.2)$$

where $\mathbf{r}_{\text{COM}} = \frac{1}{N} \sum_k \mathbf{r}_k$ is the center-of-mass coordinate and

$$\eta_{\vec{k}}(\{\mathbf{r}_k\}) = \frac{1}{\lambda_{|\vec{k}\rangle} \mathcal{Z} N} \sum_{\mathbf{r}_i} e^{-i\vec{k} \cdot (\frac{1}{N} \sum_k \mathbf{r}_k - \vec{R}_i)} \psi_{\text{aux}}^n(\{\mathbf{r}_k - \mathbf{r}_i\}), \quad (4.3)$$

is the translation symmetric function. This form of the eigenstate allows the interpretation that tuning the Bloch momentum $\vec{k}$ corresponds to a translation of the center of mass of the system (or in this case, of the auxiliary models that comprise the system).

1.4.2 Orthonormality

It is straightforward to show that these states form an orthonormal basis. We start by writing down the inner product of two distinct such states:

$$\begin{aligned}\langle \Psi_{\vec{k}'} | \Psi_{\vec{k}} \rangle &= \frac{1}{\lambda_{|\vec{k}'\rangle}^* \lambda_{|\vec{k}\rangle} \mathcal{Z}^2 N^2} \sum_{\mathbf{r}_i, \mathbf{r}_j, \mathbf{a}, \mathbf{a}'} e^{i(\vec{k} \cdot \vec{R}_i - \vec{k}' \cdot \vec{R}_j)} \langle \psi_{\text{aux}}(\mathbf{r}_j, \mathbf{a}') | \psi_{\text{aux}}(\mathbf{r}_i, \mathbf{a}) \rangle \\ &= \frac{1}{\lambda_{|\vec{k}'\rangle}^* \lambda_{|\vec{k}\rangle} \mathcal{Z}^2 N^2} \sum_{\mathbf{r}_i, \mathbf{r}_j, \mathbf{a}, \mathbf{a}'} e^{i(\vec{k} \cdot \vec{R}_i - \vec{k}' \cdot \vec{R}_j)} \langle \psi_{\text{aux}}(\mathbf{r}_0, \mathbf{a}') T^\dagger(\mathbf{r}_0 - \mathbf{r}_j) T(\mathbf{r}_0 - \mathbf{r}_i) | \psi_{\text{aux}}(\mathbf{r}_0, \mathbf{a}) \rangle\end{aligned}\quad (4.4)$$

At this point, we insert a complete basis of momentum eigenkets $1 = \sum_{\vec{q}} |\vec{q}\rangle \langle \vec{q}|$ to resolve the translation operators:

$$\begin{aligned}
 \langle \Psi_{\vec{k}} | \Psi_{\vec{k}} \rangle &= \frac{1}{\lambda_{|\vec{k}\rangle}^* \lambda_{|\vec{k}\rangle} \mathcal{Z}^2 N^2} \sum_{\mathbf{r}_i, \mathbf{r}_j, \mathbf{a}, \mathbf{a}', \vec{q}} e^{i(\vec{k} \cdot \vec{R}_i - \vec{k}' \cdot \vec{R}_j)} \langle \psi_{\text{aux}}(\mathbf{r}_0, \mathbf{a}') T(\mathbf{r}_j - \mathbf{r}_i) |\vec{q}\rangle \langle \vec{q}| \psi_{\text{aux}}(\mathbf{r}_0, \mathbf{a}) \rangle \\
 &= \frac{1}{\lambda_{|\vec{k}\rangle}^* \lambda_{|\vec{k}\rangle} \mathcal{Z}^2 N^2} \sum_{\mathbf{r}_i, \mathbf{r}_j, \mathbf{a}, \mathbf{a}', \vec{q}} e^{i(\vec{k} \cdot \vec{R}_i - \vec{k}' \cdot \vec{R}_j)} \langle \psi_{\text{aux}}(\mathbf{r}_0, \mathbf{a}') e^{i\vec{q} \cdot (\mathbf{r}_j - \mathbf{r}_i)} |\vec{q}\rangle \langle \vec{q}| \psi_{\text{aux}}(\mathbf{r}_0, \mathbf{a}) \rangle \\
 &= \frac{1}{|\lambda_{|\vec{k}\rangle}|^2 \mathcal{Z}^2} \delta_{\vec{k}, \vec{k}'} \sum_{\mathbf{a}, \mathbf{a}'} \langle \psi_{\text{aux}}(\mathbf{r}_0, \mathbf{a}') |\vec{k}\rangle \langle \vec{k}| \psi_{\text{aux}}(\mathbf{r}_0, \mathbf{a}) \rangle
 \end{aligned} \tag{4.5}$$

At the last step, we used the summation form of the Kronecker delta function: $\sum_{\mathbf{r}_i} e^{i\mathbf{r}_i \cdot (\vec{k} - \vec{q})} \sum_{\mathbf{r}_j} e^{i\mathbf{r}_j \cdot (\vec{q} - \vec{k}')} = N \delta_{\vec{k}, \vec{q}} N \delta_{\vec{k}', \vec{q}}$. The final remaining step is to identify that the inner products inside the summation are actually independent of the direction $\mathbf{a}, \mathbf{a}'$ of the connecting vector, and both are equal to the normalisation factor $\lambda_{|\vec{k}\rangle}$:

$$\langle \Psi_{\vec{k}'} | \Psi_{\vec{k}} \rangle = \frac{1}{|\lambda_{|\vec{k}\rangle}|^2 \mathcal{Z}^2} \delta_{\vec{k}, \vec{k}'} \sum_{\mathbf{a}, \mathbf{a}'} |\lambda_{|\vec{k}\rangle}|^2 = \frac{1}{|\lambda_{|\vec{k}\rangle}|^2 \mathcal{Z}^2} \delta_{\vec{k}, \vec{k}'} w^2 |\lambda_{|\vec{k}\rangle}|^2 = \delta_{\vec{k}, \vec{k}'} \tag{4.6}$$

This concludes the proof of orthonormality.

1.4.3 Eigenstates

To demonstrate that these are indeed eigenstates of the bulk Hamiltonian, we will calculate the matrix elements of the full Hamiltonian between these states:

$$\begin{aligned}
 \langle \Psi_{\vec{k}'} | H_{\text{H-H}} | \Psi_{\vec{k}} \rangle &= \frac{1}{\lambda_{|\vec{k}\rangle}^* \lambda_{|\vec{k}\rangle} \mathcal{Z}^2 N^2} \sum_{\mathbf{r}_i, \mathbf{r}_j, \mathbf{r}_k, \mathbf{a}, \mathbf{a}'} e^{i(\vec{k} \cdot \vec{R}_i - \vec{k}' \cdot \vec{R}_j)} \langle \psi_{\text{aux}}(\mathbf{r}_j, \mathbf{a}') | H_{\text{aux}}(\mathbf{r}_k) | \psi_{\text{aux}}(\mathbf{r}_i, \mathbf{a}) \rangle \\
 &= \frac{1}{\lambda_{|\vec{k}\rangle}^* \lambda_{|\vec{k}\rangle} \mathcal{Z}^2 N^2} \sum_{\mathbf{r}_i, \mathbf{r}_j, \mathbf{r}_k, \mathbf{a}, \mathbf{a}'} e^{i(\vec{k} \cdot \vec{R}_i - \vec{k}' \cdot \vec{R}_j)} \langle \psi_{\text{aux}}(\mathbf{r}_0, \mathbf{a}') | T_{\mathbf{r}_0 - \mathbf{r}_j}^\dagger T_{\mathbf{r}_0 - \mathbf{r}_k} H_{\text{aux}}(\mathbf{r}_0) T_{\mathbf{r}_0 - \mathbf{r}_k}^\dagger T_{\mathbf{r}_0 - \mathbf{r}_i} | \psi_{\text{aux}}(\mathbf{r}_0, \mathbf{a}) \rangle \\
 &= \frac{1}{\lambda_{|\vec{k}\rangle}^* \lambda_{|\vec{k}\rangle} \mathcal{Z}^2 N^2} \sum_{\mathbf{r}_i, \mathbf{r}_j, \mathbf{r}_k, \mathbf{a}, \mathbf{a}'} e^{i(\vec{k} \cdot \vec{R}_i - \vec{k}' \cdot \vec{R}_j)} \langle \psi_{\text{aux}}(\mathbf{r}_0, \mathbf{a}') | T_{\mathbf{r}_k - \mathbf{r}_j}^\dagger H_{\text{aux}}(\mathbf{r}_0) T_{\mathbf{r}_k - \mathbf{r}_i} | \psi_{\text{aux}}(\mathbf{r}_0, \mathbf{a}) \rangle
 \end{aligned} \tag{4.7}$$

To resolve the translation operators, we will now insert the momentum eigenstates $|\vec{k}\rangle$ of the full model: $1 = \sum_{\vec{q}} |\vec{q}\rangle \langle \vec{q}|$ in between the operators.

$$\begin{aligned}
 \langle \Psi_{\vec{k}'} | H_{\text{H-H}} | \Psi_{\vec{k}} \rangle &= \sum_{\substack{r_i, r_j, r_k, \\ a, a', \vec{q}, \vec{q}'}} \frac{e^{i(\vec{k} \cdot \vec{R}_i - \vec{k}' \cdot \vec{R}_j)}}{\lambda_{|\vec{k}'\rangle}^* \lambda_{|\vec{k}\rangle} \mathcal{Z}^2 N^2} \langle \psi_{\text{aux}}(\mathbf{r}_0, \mathbf{a}') | \vec{q}' \rangle \langle \vec{q}' | T_{\mathbf{r}_k - \mathbf{r}_j}^\dagger H_{\text{aux}}(\mathbf{r}_0) T_{\mathbf{r}_k - \mathbf{r}_i} | \vec{q} \rangle \langle \vec{q} | \psi_{\text{aux}}(\mathbf{r}_0, \mathbf{a}) \rangle \\
 &= \sum_{\substack{r_i, r_j, r_k, \\ a, a', \vec{q}, \vec{q}'}} \frac{e^{i(\vec{k} \cdot \vec{R}_i - \vec{k}' \cdot \vec{R}_j)}}{\lambda_{|\vec{k}'\rangle}^* \lambda_{|\vec{k}\rangle} \mathcal{Z}^2 N^2} \langle \psi_{\text{aux}}(\mathbf{r}_0, \mathbf{a}') | \vec{q}' \rangle e^{-i\vec{q}' \cdot (\mathbf{r}_k - \mathbf{r}_j)} e^{i\vec{q} \cdot (\mathbf{r}_k - \mathbf{r}_i)} \langle \vec{q}' | H_{\text{aux}}(\mathbf{r}_0) | \vec{q} \rangle \langle \vec{q} | \psi_{\text{aux}}(\mathbf{r}_0, \mathbf{a}) \rangle \\
 &= \frac{N}{\lambda_{|\vec{k}'\rangle}^* \lambda_{|\vec{k}\rangle} \mathcal{Z}^2} \sum_{a, a', \vec{q}, \vec{q}'} \delta_{\vec{k}, \vec{q}} \delta_{\vec{q}, \vec{q}'} \delta_{\vec{q}', \vec{k}'} \langle \psi_{\text{aux}}(\mathbf{r}_0, \mathbf{a}') | \vec{q}' \rangle \langle \vec{q}' | H_{\text{aux}}(\mathbf{r}_0) | \vec{q} \rangle \langle \vec{q} | \psi_{\text{aux}}(\mathbf{r}_0, \mathbf{a}) \rangle \\
 &= \frac{N}{|\lambda_{|\vec{k}\rangle}|^2 \mathcal{Z}^2} \delta_{kk'} \sum_{a, a'} \langle \psi_{\text{aux}}(\mathbf{r}_0, \mathbf{a}') | \vec{k} \rangle \langle \vec{k} | H_{\text{aux}}(\mathbf{r}_0) | \vec{k} \rangle \langle \vec{k} | \psi_{\text{aux}}(\mathbf{r}_0, \mathbf{a}) \rangle \\
 &= \frac{N}{|\lambda_{|\vec{k}\rangle}|^2 \mathcal{Z}^2} \delta_{kk'} \langle \vec{k} | H_{\text{aux}}(\mathbf{r}_0) | \vec{k} \rangle \underbrace{\sum_{a, a'} \langle \psi_{\text{aux}}(\mathbf{r}_0, \mathbf{a}') | \vec{k} \rangle \langle \vec{k} | \psi_{\text{aux}}(\mathbf{r}_0, \mathbf{a}) \rangle}_{w^2 |\lambda_{|\vec{k}\rangle}|^2} \\
 &= N \delta_{kk'} \langle \vec{k} | H_{\text{aux}}(\mathbf{r}_0) | \vec{k} \rangle
 \end{aligned} \tag{4.8}$$

The momentum eigenket $|\vec{k}\rangle$ can be expressed as a sum over position eigenkets at varying distances from $\mathbf{r}_0$. This form allows a systematic method of improving the energy eigenvalue estimate, because the interacting in the impurity model is extremely localised, so the overlap between two auxiliary models decreases rapidly with distance. The majority of the contribution will come from the state at $\mathbf{r}_0$, and improvements are then made by considering auxiliary models at further distances.

1.5 Mapping Greens Functions from Auxiliary Model to Lattice Model

1.5.1 One-particle k -space Greens functions

In the previous part, we proposed a form for the eigenstates $|\Psi_s\rangle$ of the bulk Hamiltonian in terms of the ground-states $|\psi_n\rangle$ of the auxiliary models. In this section, we will relate one-particle Greens functions of the lattice model to those of the auxiliary model. We define the retarded time-domain lattice k -space Greens function at zero temperature as

$$G(\mathbf{k}\sigma; t) = -i\theta(t) \langle \Psi_{\text{gs}} | \{c_{\mathbf{k}\sigma}(t), c_{\mathbf{k}\sigma}^\dagger\} | \Psi_{\text{gs}} \rangle. \tag{5.1}$$

where $|\Psi_{\text{gs}}\rangle$ is the lattice groundstate. Time evolution of the operators is governed by the lattice Hamiltonian H_{latt} :

$$c_{\mathbf{k}\sigma}(t) = e^{itH_{\text{latt}}} c_{\mathbf{k}\sigma} e^{-itH_{\text{latt}}}. \tag{5.2}$$

By using eq. 3.8 and the fact that the translation operators commute with H_{latt} , we have the relation

$$T(\vec{a}) c(\mathbf{k})(t) T^\dagger(\vec{a}) = e^{-i\mathbf{k} \cdot \vec{a}} c(\mathbf{k})(t), \quad (5.3)$$

which combined with eq. 3.8 gives

$$T(\mathbf{a}) c(\mathbf{k})(t) c^\dagger(\mathbf{k}) T^\dagger(\mathbf{b}) = e^{i\mathbf{k} \cdot (\mathbf{b} - \mathbf{a})} c(\mathbf{k})(t) T^\dagger(\mathbf{b} - \mathbf{a}) c^\dagger(\mathbf{k}). \quad (5.4)$$

In order to make use of the mapping between auxiliary model and lattice model Hamiltonians, we would like to express this lattice Greens function in terms of auxiliary model Greens functions – objects that can be computed purely from within a single auxiliary model. This will be a major simplification in our computations, because it is easier to work with a quantum impurity model, as compared to an interacting lattice model.

To make progress towards this, we consider the equivalent object

$$G(\mathbf{k}\sigma; t) = -i\theta(t) \frac{1}{2} \left[\langle \Psi_{\text{gs}} | \{c_{\mathbf{k}\sigma}(t), c_{\mathbf{k}\sigma}^\dagger\} | \Psi_{\text{gs}} \rangle + \langle \Psi_{\text{gs}} | \{c_{\mathbf{k}\sigma}, c_{\mathbf{k}\sigma}^\dagger(-t)\} | \Psi_{\text{gs}} \rangle \right]. \quad (5.5)$$

We consider only the first term for now. We express the lattice groundstate in terms of the auxiliary model groundstates (using eq. 3.20):

$$\langle \Psi_{\text{gs}} | \{c_{\mathbf{k}\sigma}(t), c_{\mathbf{k}\sigma}^\dagger\} | \Psi_{\text{gs}} \rangle = \frac{1}{N} \sum_{\mathbf{r}_1, \mathbf{r}_2} \langle \psi_{\text{gs}}(\mathbf{r}_2) | \{c_{\mathbf{k}\sigma}(t), c_{\mathbf{k}\sigma}^\dagger\} | \psi_{\text{gs}}(\mathbf{r}_1) \rangle. \quad (5.6)$$

We take one of the terms in the anticommutator and proceed to simplify it:

$$\begin{aligned} \sum_{\mathbf{r}_1, \mathbf{r}_2} \langle \psi_{\text{gs}}(\mathbf{r}_2) | c_{\mathbf{k}\sigma}(t) c_{\mathbf{k}\sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_1) \rangle &= \sum_{\mathbf{r}_1, \mathbf{r}_2} \langle \psi_{\text{gs}}(\mathbf{r}_d) | T(\mathbf{r}_2 - \mathbf{r}_d) c_{\mathbf{k}\sigma}(t) c_{\mathbf{k}\sigma}^\dagger T^\dagger(\mathbf{r}_1 - \mathbf{r}_d) | \psi_{\text{gs}}(\mathbf{r}_d) \rangle \\ &= \sum_{\mathbf{r}_1, \mathbf{r}_2} e^{i\mathbf{k} \cdot (\mathbf{r}_1 - \mathbf{r}_2)} \langle \psi_{\text{gs}}(\mathbf{r}_d) | c_{\mathbf{k}\sigma}(t) T^\dagger(\mathbf{r}_1 - \mathbf{r}_2) c_{\mathbf{k}\sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle \\ &= N \sum_{\mathbf{r}} e^{i\mathbf{k} \cdot \mathbf{r}} \langle \psi_{\text{gs}}(\mathbf{r}_d) | c_{\mathbf{k}\sigma}(t) T^\dagger(\mathbf{r}) c_{\mathbf{k}\sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle. \end{aligned} \quad (5.7)$$

where $\mathbf{r}_d$ is a reference lattice site, and we used eq. 5.4 in the second step. The other term in the anticommutator gives

$$\begin{aligned} \sum_{\mathbf{r}_1, \mathbf{r}_2} \langle \psi_{\text{gs}}(\mathbf{r}_2) | c_{\mathbf{k}\sigma}^\dagger c_{\mathbf{k}\sigma}(t) | \psi_{\text{gs}}(\mathbf{r}_1) \rangle &= \sum_{\mathbf{r}_1, \mathbf{r}_2} \langle \psi_{\text{gs}}(\mathbf{r}_d) | T(\mathbf{r}_2 - \mathbf{r}_d) c_{\mathbf{k}\sigma}^\dagger c_{\mathbf{k}\sigma}(t) T^\dagger(\mathbf{r}_1 - \mathbf{r}_d) | \psi_{\text{gs}}(\mathbf{r}_d) \rangle \\ &= \sum_{\mathbf{r}_1, \mathbf{r}_2} e^{-i\mathbf{k} \cdot (\mathbf{r}_1 - \mathbf{r}_2)} \langle \psi_{\text{gs}}(\mathbf{r}_d) | c_{\mathbf{k}\sigma}^\dagger T^\dagger(\mathbf{r}_1 - \mathbf{r}_2) c_{\mathbf{k}\sigma}(t) | \psi_{\text{gs}}(\mathbf{r}_d) \rangle \\ &= N \sum_{\mathbf{r}} e^{-i\mathbf{k} \cdot \mathbf{r}} \langle \psi_{\text{gs}}(\mathbf{r}_d) | c_{\mathbf{k}\sigma}^\dagger T^\dagger(\mathbf{r}) c_{\mathbf{k}\sigma}(t) | \psi_{\text{gs}}(\mathbf{r}_d) \rangle. \end{aligned} \quad (5.8)$$

To resolve the time evolution of the operators, we insert $1 = \sum_m |\psi_m(\mathbf{r}_d)\rangle \langle \psi_m(\mathbf{r}_d)|$ into the equation. We again work with just the first term of the anticommutator for convenience:

$$\begin{aligned}
 & \sum_{\mathbf{r}_1, \mathbf{r}_2} \langle \psi_{\text{gs}}(\mathbf{r}_2) | c_{\mathbf{k}\sigma}(t) c_{\mathbf{k}\sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_1) \rangle \\
 &= N \sum_{\mathbf{r}, n_1, n_2} e^{i\mathbf{k} \cdot \mathbf{r}} \langle \psi_{\text{gs}}(\mathbf{r}_d) | c_{\mathbf{k}\sigma}(t) | \psi_{n_1}(\mathbf{r}_d) \rangle \langle \psi_{n_1}(\mathbf{r}_d) | T^\dagger(\mathbf{r}) | \psi_{n_2}(\mathbf{r}_d) \rangle \langle \psi_{n_2}(\mathbf{r}_d) | c_{\mathbf{k}\sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle \\
 &= N \sum_{\mathbf{r}, n_1, n_2} e^{i\mathbf{k} \cdot \mathbf{r}} \langle \psi_{\text{gs}}(\mathbf{r}_d) | c_{\mathbf{k}\sigma}(t) | \psi_{n_1}(\mathbf{r}_d) \rangle \langle \psi_{n_1}(\mathbf{r}_d) | \psi_{n_2}(\mathbf{r}_d + \mathbf{r}) \rangle \langle \psi_{n_2}(\mathbf{r}_d) | c_{\mathbf{k}\sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle .
 \end{aligned} \tag{5.9}$$

The overlap $\langle \psi_{n_1}(\mathbf{r}_d) | \psi_{n_2}(\mathbf{r}_d + \mathbf{r}) \rangle$ between the eigenstates of the auxiliary models at $\mathbf{r}_d$ and $\mathbf{r}_d + \mathbf{r}$ are appreciable only for $n_1 = n_2$. This is because, for small $\mathbf{r}$, the auxiliary models are very similar and the states are therefore almost orthogonal. For large $\mathbf{r}$, the overlap is again suppressed because the auxiliary model eigenstates are localised around the impurity. We therefore make the simplifying assumption: $\langle \psi_{n_1}(\mathbf{r}_d) | \psi_{n_2}(\mathbf{r}_d + \mathbf{r}) \rangle = \delta_{n_1, n_2} \langle \psi_{n_1}(\mathbf{r}_d) | \psi_{n_1}(\mathbf{r}_d + \mathbf{r}) \rangle = \delta_{n_1, n_2} F_{n_1}(\mathbf{r})$. Substituting this, we get

$$\sum_{\mathbf{r}_1, \mathbf{r}_2} \langle \psi_{\text{gs}}(\mathbf{r}_2) | c_{\mathbf{k}\sigma}(t) c_{\mathbf{k}\sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_1) \rangle = N \sum_n F_n(\mathbf{k}) \langle \psi_{\text{gs}}(\mathbf{r}_d) | c_{\mathbf{k}\sigma}(t) | \psi_n(\mathbf{r}_d) \rangle \langle \psi_n(\mathbf{r}_d) | c_{\mathbf{k}\sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle , \tag{5.10}$$

where we have defined the Fourier transform $F_n(\mathbf{k}) = \sum_{\mathbf{r}} e^{i\mathbf{k} \cdot \mathbf{r}} F_n(\mathbf{r})$ of the overlap function.

We now consider more carefully the transition operator $c_{\mathbf{k}\sigma}$ for the 1-particle excitation giving rise to the above Greens function. Within our auxiliary model approach, gapless excitations within the lattice model are represented by gapless excitations of the impurity site, specifically those that try to screen the impurity site. As a result, the uncoordinated scattering process $c_{\mathbf{k}\sigma}$ for the lattice model must be replaced by a coherent excitation $\mathcal{T}$ within the impurity model that captures those gapless excitations that occur in connection with the impurity, and projects out the uncorrelated excitations that take place even when the impurity site is decoupled from the bath.

In order to construct this auxiliary model $\mathcal{T}$ -matrix, we allow for all excitations of the impurity site in conjunction with those of the k -state:

$$\mathcal{T}_{\mathbf{k}, \sigma} = \left[c_{\mathbf{r}_d \sigma}^\dagger + S_{\mathbf{r}_d}^\sigma + C_{\mathbf{r}_d}^+ \right] c_{\mathbf{k}\sigma} , \tag{5.11}$$

where $S_{\mathbf{r}_d}^+$ and $C_{\mathbf{r}_d}^+$ are spin and charge flip operators on the impurity site that cause transition from $|\downarrow\rangle \rightarrow |\uparrow\rangle$ and $|0\rangle \rightarrow |\uparrow\downarrow\rangle$ respectively. The first term encodes the local component of the k -space Greens function by creating an excitation on the impurity site (which is then translated across the lattice), and the rest of the terms encode the non-local components by tracking impurity-bath excitations.

Apart from modifying the excitation operator, we will now simplify the time-evolution as well. The first overlap on the right-hand side of eq. 5.10 expands to

$$\langle \psi_{\text{gs}}(\mathbf{r}_d) | \mathcal{T}_{\mathbf{k}, \sigma}(t) | \psi_n(\mathbf{r}_d) \rangle = \langle \psi_{\text{gs}}(\mathbf{r}_d) | e^{i(H_{\text{aux}}(\mathbf{r}_d) + H_{\text{cav}}(\mathbf{r}_d))t} \mathcal{T}_{\mathbf{k}, \sigma} e^{-i(H_{\text{aux}}(\mathbf{r}_d) + H_{\text{cav}}(\mathbf{r}_d))t} | \psi_n(\mathbf{r}_d) \rangle , \tag{5.12}$$

where $H_{\text{aux}}(\mathbf{r}_d)$ (eq. 2.3) is the auxiliary model Hamiltonian consisting of the cluster Hamiltonian $H_{\text{clust}}(\mathbf{r}_d)$ along with all one-particle terms H_{1p} in the Hamiltonian (in a typical lattice model, this will be the tight-binding hopping terms), and $H_{\text{cav}}(\mathbf{r}_d)$ is the “cavity” Hamiltonian consisting of all interaction (2-particle and beyond) terms in the Hamiltonian. Together, they create the full lattice Hamiltonian: $H_{\text{latt}} = H_{\text{aux}}(\mathbf{r}_d) + H_{\text{cav}}(\mathbf{r}_d)$. Using the Zassenhaus form of the Baker-Campbell-Hausdorff formula, we have

$$e^{i(H_{\text{aux}}(\mathbf{r}_d) + H_{\text{cav}}(\mathbf{r}_d))t} \approx e^{iH_{\text{aux}}(\mathbf{r}_d)t} e^{iH_{\text{cav}}(\mathbf{r}_d)t} e^{[H_{\text{aux}}(\mathbf{r}_d), H_{\text{cav}}(\mathbf{r}_d)]t^2/2}, \quad (5.13)$$

where we have dropped the higher-order terms because the commutators involve only the boundary degrees of freedom and become subsequently smaller as the order increases. One can think of the operator $e^{iH_{\text{cav}}(\mathbf{r}_d)t} e^{[H_{\text{aux}}(\mathbf{r}_d), H_{\text{cav}}(\mathbf{r}_d)]t^2/2}$ as renormalising the auxiliary model $\mathcal{T}$ -matrix through hybridisation with the rest of the system and formally leading to a new $\mathcal{T}$ -matrix:

$$\mathcal{T}_{\mathbf{k},\sigma}^* = e^{iH_{\text{cav}}(\mathbf{r}_d)t} e^{[H_{\text{aux}}(\mathbf{r}_d), H_{\text{cav}}(\mathbf{r}_d)]t^2/2} \mathcal{T}_{\mathbf{k},\sigma} e^{-[H_{\text{aux}}(\mathbf{r}_d), H_{\text{cav}}(\mathbf{r}_d)]t^2/2} e^{-iH_{\text{cav}}(\mathbf{r}_d)t}. \quad (5.14)$$

One can also absorb the modulating phase factor $F_n(\mathbf{k})$ (defined above and appearing in eq. 5.10) into this renormalisation:

$$\mathcal{T}_{\mathbf{k},\sigma}^* = e^{iH_{\text{cav}}(\mathbf{r}_d)t} e^{[H_{\text{aux}}(\mathbf{r}_d), H_{\text{cav}}(\mathbf{r}_d)]t^2/2} \mathcal{T}_{\mathbf{k},\sigma} F_n(\mathbf{k}) \mathcal{P}_n(\mathbf{r}_d) e^{-[H_{\text{aux}}(\mathbf{r}_d), H_{\text{cav}}(\mathbf{r}_d)]t^2/2} e^{-iH_{\text{cav}}(\mathbf{r}_d)t}, \quad (5.15)$$

where $\mathcal{P}_n(\mathbf{r}_d)$ projects onto the n^{th} eigenstate $|\psi_n(\mathbf{r}_d)\rangle$.

Since the overlap is calculated in the eigenstates of the auxiliary model at $\mathbf{r}_d$, we adopt the zeroth order approximation of ignoring this renormalisation. Later, we will specify a prescription of systematically improving our computation. The overlap expression becomes

$$\langle \psi_{\text{gs}}(\mathbf{r}_d) | \mathcal{T}_{\mathbf{k},\sigma}(t) | \psi_n(\mathbf{r}_d) \rangle = \langle \psi_{\text{gs}}(\mathbf{r}_d) | e^{i(H_{\text{clust}}(\mathbf{r}_d) + H_{1p})t} \mathcal{T}_{\mathbf{k},\sigma} e^{-i(H_{\text{clust}}(\mathbf{r}_d) + H_{1p})t} | \psi_n(\mathbf{r}_d) \rangle. \quad (5.16)$$

To make the problem tractable, we treat the effect of H_{1p} at the level of an expectation value and note that the effect of the transition operator is to create a hole of momentum $\mathbf{k}$ (eq. 5.11). Let the Hamiltonian H_{1p} be diagonal in certain modes $c_{\nu,\sigma}$ with energies ϵ_ν , and the operator $c_{\mathbf{k},\sigma}$ can be expressed in terms of these modes: $c_{\mathbf{k},\sigma} = \sum_\nu C_{\nu,\mathbf{k}} c_{\nu,\sigma}$. Expanding the $\mathcal{T}$ -matrix in this fashion gives

$$\langle \psi_{\text{gs}}(\mathbf{r}_d) | \mathcal{T}_{\mathbf{k},\sigma}(t) | \psi_n(\mathbf{r}_d) \rangle = \sum_\nu C_{\nu,\mathbf{k}} \langle \psi_{\text{gs}}(\mathbf{r}_d) | e^{i(H_{\text{clust}}(\mathbf{r}_d) + H_{1p})t} \mathcal{T}_{\nu,\sigma} e^{-i(H_{\text{clust}}(\mathbf{r}_d) + H_{1p})t} | \psi_n(\mathbf{r}_d) \rangle. \quad (5.17)$$

Each excitation $\mathcal{T}_{\nu,\sigma}$ therefore leads to an energy cost $-\epsilon_\nu$. Moreover, $|\psi_n(\mathbf{r}_d)\rangle$ and $|\psi_{\text{gs}}(\mathbf{r}_d)\rangle$ are eigenstates of $H_{\text{clust}}(\mathbf{r}_d)$ with energies E_n and E_0 . This leads to the following simplified expression:

$$\langle \psi_{\text{gs}}(\mathbf{r}_d) | \mathcal{T}_{\mathbf{k},\sigma}(t) | \psi_n(\mathbf{r}_d) \rangle = \sum_\nu C_{\nu,\mathbf{k}} e^{-i(E_n - E_0 + \epsilon_\nu)t} \langle \psi_{\text{gs}}(\mathbf{r}_d) | \mathcal{T}_{\nu,\sigma} | \psi_n(\mathbf{r}_d) \rangle. \quad (5.18)$$

Substituting the expression back into 5.10 and then into 5.6, we get an expression for one half of the first part of the retarded Greens function (the full form is in eq. 5.5):

$$\begin{aligned} & -i\theta(t) \langle \Psi_{\text{gs}} | c_{\mathbf{k}\sigma}(t) c_{\mathbf{k}\sigma}^\dagger | \Psi_{\text{gs}} \rangle \\ & = -i\theta(t) \sum_{\nu,n} C_{\nu,\mathbf{k}} e^{-i(E_n - E_0 + \epsilon_\nu)t} \langle \psi_{\text{gs}}(\mathbf{r}_d) | \mathcal{T}_{\nu,\sigma} | \psi_n(\mathbf{r}_d) \rangle \langle \psi_n(\mathbf{r}_d) | \mathcal{T}_{\mathbf{k},\sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle. \end{aligned} \quad (5.19)$$

Working through the other term of eq. 5.6 with the same strategy, we get the full expression:

$$\begin{aligned}
 & -i\theta(t) \langle \Psi_{\text{gs}} | \{c_{\mathbf{k}\sigma}(t), c_{\mathbf{k}\sigma}^\dagger\} | \Psi_{\text{gs}} \rangle \\
 & = -i\theta(t) \sum_{v,n} C_{v,\mathbf{k}} e^{-i(E_n - E_0 + \epsilon_v)t} \langle \psi_{\text{gs}}(\mathbf{r}_d) | \mathcal{T}_{v,\sigma} | \psi_n(\mathbf{r}_d) \rangle \langle \psi_n(\mathbf{r}_d) | \mathcal{T}_{\mathbf{k},\sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle \\
 & \quad - i\theta(t) \sum_{v,n} C_{v,\mathbf{k}} e^{i(E_n - E_0 - \epsilon_v)t} \langle \psi_{\text{gs}}(\mathbf{r}_d) | \mathcal{T}_{\mathbf{k},\sigma}^\dagger | \psi_n(\mathbf{r}_d) \rangle \langle \psi_n(\mathbf{r}_d) | \mathcal{T}_{v,\sigma} | \psi_{\text{gs}}(\mathbf{r}_d) \rangle .
 \end{aligned} \tag{5.20}$$

Finally, we consider the other term in eq. 5.5, $-i\theta(t) \langle \Psi_{\text{gs}} | \{c_{\mathbf{k}\sigma}, c_{\mathbf{k}\sigma}^\dagger(-t)\} | \Psi_{\text{gs}} \rangle$. Proceeding in the same way, eq. 5.18 is replaced by

$$\begin{aligned}
 \langle \psi_n(\mathbf{r}_d) | \mathcal{T}_{\mathbf{k},\sigma}^\dagger(-t) | \psi_{\text{gs}}(\mathbf{r}_d) \rangle & = \langle \psi_{\text{gs}}(\mathbf{r}_d) | \mathcal{T}_{\mathbf{k},\sigma}(-t) | \psi_n(\mathbf{r}_d) \rangle^* \\
 & = \sum_v C_{v,\mathbf{k}}^* e^{-i(E_n - E_0 + \epsilon_v)t} \langle \psi_n(\mathbf{r}_d) | \mathcal{T}_{v,\sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle .
 \end{aligned} \tag{5.21}$$

Evaluating this term exactly as before and changing from time to frequency domain using $g(\omega) = \int dt e^{i\omega t} f(t)$ gives

$$\begin{aligned}
 G(\mathbf{k}, \sigma; \omega) & = \sum_{v,n} C_{v,\mathbf{k}} \left[\frac{\langle \psi_{\text{gs}}(\mathbf{r}_d) | \mathcal{T}_{v,\sigma} | \psi_n(\mathbf{r}_d) \rangle \langle \psi_n(\mathbf{r}_d) | \mathcal{T}_{\mathbf{k},\sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle}{\omega - (E_n - E_0 + \epsilon_v)} \right. \\
 & \quad \left. + \frac{\langle \psi_{\text{gs}}(\mathbf{r}_d) | \mathcal{T}_{\mathbf{k},\sigma}^\dagger | \psi_n(\mathbf{r}_d) \rangle \langle \psi_n(\mathbf{r}_d) | \mathcal{T}_{v,\sigma} | \psi_{\text{gs}}(\mathbf{r}_d) \rangle}{\omega + (E_n - E_0 - \epsilon_v)} \right] + \text{h.c.}
 \end{aligned} \tag{5.22}$$

We recognise the two parts of the lattice Greens function as a retarded off-diagonal Greens functions calculated within the auxiliary model at $\mathbf{r}_d$, but at shifted frequencies $\omega - \epsilon_v$:

$$G(\mathbf{k}, \sigma; \omega) = \sum_v \left[C_{v,\mathbf{k}} \mathcal{G}(\mathcal{T}_{v,\sigma}, \mathcal{T}_{\mathbf{k},\sigma}^\dagger; \omega - \epsilon_v) + C_{v,\mathbf{k}}^* \mathcal{G}(\mathcal{T}_{\mathbf{k},\sigma}, \mathcal{T}_{v,\sigma}^\dagger; \omega - \epsilon_v) \right] \tag{5.23}$$

We now discuss how to systematically improve this estimate. One way is to include the effect of other auxiliary models in the overlap function $F_n(\mathbf{r})$. This will of course require us to calculate an off-diagonal Greens function. As a concrete example, upon including the effects of the auxiliary models nearest to the reference point $\mathbf{r}_d$, the expression for the Greens function becomes

$$\begin{aligned}
 G(\mathbf{k}, \sigma; \omega) & = \sum_{v,n} C_{v,\mathbf{k}} \left[\frac{\langle \psi_{\text{gs}}(\mathbf{r}_d) | \mathcal{T}_{v,\sigma}^* | \psi_n(\mathbf{r}_d) \rangle \langle \psi_n(\mathbf{r}_d) | \mathcal{T}_{\mathbf{k},\sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle}{\omega - \epsilon_v - (E_n - E_0)} \right. \\
 & \quad \left. + \frac{\langle \psi_{\text{gs}}(\mathbf{r}_d) | \mathcal{T}_{\mathbf{k},\sigma}^\dagger | \psi_n(\mathbf{r}_d) \rangle \langle \psi_n(\mathbf{r}_d) | \mathcal{T}_{v,\sigma}^* | \psi_{\text{gs}}(\mathbf{r}_d) \rangle}{\omega - \epsilon_v + (E_n - E_0)} \right] ,
 \end{aligned} \tag{5.24}$$

with $\mathcal{T}_{v,\sigma}^* = \mathcal{T}_{v,\sigma}(F_n(0) + \sum_{\mathbf{a}} F_n(\mathbf{a}) e^{i\mathbf{k} \cdot \mathbf{a}})$ where $\mathbf{a}$ sums over the primitive lattice vectors. Another way to include more non-local contributions is by considering the effects of the cavity Hamiltonian in eq. 5.15:

$$\mathcal{T}_{\mathbf{k},\sigma}^*(t) = e^{iH_{\text{cav}}(\mathbf{r}_d)t} (\mathcal{T}_{\mathbf{k},\sigma} + t^2/2 [H_{\text{aux}}(\mathbf{r}_d), H_{\text{cav}}(\mathbf{r}_d)], \mathcal{T}_{\mathbf{k},\sigma}) e^{-iH_{\text{cav}}(\mathbf{r}_d)t} . \tag{5.25}$$

The time-domain Greens function associated with this operator will then need to be calculated, which can then be used to calculate frequency-domain functions.

1.5.2 One-particle real-space Greens functions

We now work with Greens functions for real-space excitations. The retarded Greens function in this case is

$$G(\mathbf{r}\sigma; t) = -i\theta(t) \langle \Psi_{\text{gs}} | \{c_{\mathbf{r}_d+\mathbf{r},\sigma}(t), c_{\mathbf{r}_d,\sigma}^\dagger\} | \Psi_{\text{gs}} \rangle . \quad (5.26)$$

Since our lattice model has discrete translation symmetry, we instead calculate the equivalent quantity

$$G(\mathbf{r}\sigma; t) = -i\theta(t) \frac{1}{2} \sum_{\mathbf{r}_d} \left[\langle \Psi_{\text{gs}} | \{c_{\mathbf{r}_d+\mathbf{r},\sigma}(t), c_{\mathbf{r}_d,\sigma}^\dagger\} | \Psi_{\text{gs}} \rangle + \langle \Psi_{\text{gs}} | \{c_{\mathbf{r}_d+\mathbf{r},\sigma}, c_{\mathbf{r}_d,\sigma}^\dagger(-t)\} | \Psi_{\text{gs}} \rangle \right] . \quad (5.27)$$

Similar to the k -space Greens functions, we consider only the first term for now, and express the groundstates as a superposition of auxiliary model Greens functions:

$$\begin{aligned} \sum_{\mathbf{r}_d} \langle \Psi_{\text{gs}} | c_{\mathbf{r}_d+\mathbf{r},\sigma}(t) c_{\mathbf{r}_d,\sigma}^\dagger | \Psi_{\text{gs}} \rangle &= \frac{1}{N} \sum_{\mathbf{R}, \Delta, \mathbf{r}_d} \langle \psi_{\text{gs}}(\mathbf{R} + \Delta) | c_{\mathbf{r}_d+\mathbf{r},\sigma}(t) c_{\mathbf{r}_d,\sigma}^\dagger | \psi_{\text{gs}}(\mathbf{R}) \rangle \\ &= \frac{1}{N} \sum_{\mathbf{R}, \Delta} \langle \psi_{\text{gs}}(-\mathbf{r}_d) | c_{\mathbf{r}-\mathbf{R}-\Delta,\sigma}(t) T(\Delta) c_{-\mathbf{R},\sigma}^\dagger | \psi_{\text{gs}}(-\mathbf{r}_d) \rangle . \end{aligned} \quad (5.28)$$

Local Greens function: $\mathbf{r} = 0$

We first consider the case of local Greens function, by setting $\mathbf{r} = 0$. We also change the dummy variable $\mathbf{r}_d$, $\mathbf{R}$ and Δ to $-\mathbf{r}_d$, $\mathbf{R}$ and $-\Delta$ to make the expressions tidier:

$$\sum_{\mathbf{r}_d} \langle \Psi_{\text{gs}} | c_{\mathbf{r}_d,\sigma}(t) c_{\mathbf{r}_d,\sigma}^\dagger | \Psi_{\text{gs}} \rangle = \frac{1}{N} \sum_{\mathbf{R}, \Delta, \mathbf{r}_d} \langle \psi_{\text{gs}}(\mathbf{r}_d) | c_{\mathbf{R}+\Delta,\sigma}(t) T^\dagger(\Delta) c_{\mathbf{R},\sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle . \quad (5.29)$$

We split the full summation into four parts, based on whether $\mathbf{R} = \mathbf{r}_d$ and whether $\Delta + \mathbf{R} = \mathbf{r}_d$:

$$\begin{aligned} \sum_{\mathbf{r}_d} \langle \Psi_{\text{gs}} | c_{\mathbf{r}_d,\sigma}(t) c_{\mathbf{r}_d,\sigma}^\dagger | \Psi_{\text{gs}} \rangle &= \langle \psi_{\text{gs}}(\mathbf{r}_d) | c_{\mathbf{r}_d,\sigma}(t) c_{\mathbf{r}_d,\sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle \\ &\quad + \sum_{\Delta \neq 0} \langle \psi_{\text{gs}}(\mathbf{r}_d) | c_{\mathbf{r}_d+\Delta,\sigma}(t) T^\dagger(\Delta) c_{\mathbf{r}_d,\sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle \\ &\quad + \sum_{\mathbf{R} \neq \mathbf{r}_d} \langle \psi_{\text{gs}}(\mathbf{r}_d) | c_{\mathbf{r}_d,\sigma}(t) T^\dagger(\mathbf{r}_d - \mathbf{R}) c_{\mathbf{R},\sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle \\ &\quad + \sum_{\mathbf{R} \neq \mathbf{r}_d, \Delta \neq \mathbf{r}_d} \langle \psi_{\text{gs}}(\mathbf{r}_d) | c_{\Delta,\sigma}(t) T^\dagger(\Delta - \mathbf{R}) c_{\mathbf{R},\sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle , \end{aligned} \quad (5.30)$$

where we have used the fact that the location of the auxiliary model does not affect the correlation calculated within the auxiliary model in order to cancel out the summation over $\mathbf{r}_d$ with the factor of $1/N$. The reason for splitting the expression into these parts is to explicitly write out which operators act on the impurity site $\mathbf{r}_d$ and which act on other real-space sites away from the impurity site. We

will now replace the real-space excitation operators away from the impurity site with the associated T -matrices, following the discussion above eq. 5.11:

$$\mathcal{T}_{r,\sigma} = \left[\left(\sum_{\sigma'} c_{r_d\sigma'}^\dagger + \text{h.c.} \right) + \left(S_{r_d}^+ + \text{h.c.} \right) + \left(C_{r_d}^+ + \text{h.c.} \right) \right] c_{r\sigma} . \quad (5.31)$$

We also insert a complete basis of states of the auxiliary model at r_d , $1 = \sum_n |\psi_n(r_d)\rangle \langle \psi_n(r_d)|$, generating an overlap factor F_n as before:

$$\begin{aligned} \sum_{r_d} \langle \Psi_{\text{gs}} | c_{r_d,\sigma}(t) c_{r_d,\sigma}^\dagger | \Psi_{\text{gs}} \rangle &= \langle \psi_{\text{gs}}(r_d) | c_{r_d,\sigma}(t) c_{r_d,\sigma}^\dagger | \psi_{\text{gs}}(r_d) \rangle \\ &+ \sum_{r_1, r_2 \neq r_d, n} F_n(\mathbf{r}_1 - \mathbf{r}_2)^* \langle \psi_{\text{gs}}(r_d) | \mathcal{T}_{r_1,\sigma}(t) | \psi_n(r_d) \rangle \langle \psi_n(r_d) | \mathcal{T}_{r_2,\sigma}^\dagger | \psi_{\text{gs}}(r_d) \rangle . \end{aligned} \quad (5.32)$$

Several terms have been dropped because those contributions vanish for systems where the eigenstates conserve total spin S_z .

We next treat the time-evolution of the operators exactly as in the k -space Greens function. We replace the one-particle Hamiltonian H_{1p} with its local component H_{loc} (such as any on-site term of inter-layer hopping, etc). As done previously, we will make use of the eigenstates $\nu(\mathbf{r})$ of H_{loc} , with energy $\epsilon_\nu^{\text{loc}}$. Without repeating the details, we state the final result here, for the frequency-domain real-space local Greens function:

$$\begin{aligned} G_{\text{loc}}(\omega) &= \frac{1}{2} \sum_{\nu} \left[C_{\nu,r_d} \mathcal{G}(\nu(r_d)\sigma, c_{r_d,\sigma}^\dagger; \tilde{\omega}) + C_{\nu,r_d}^* \mathcal{G}(c_{r_d,\sigma}, \nu^\dagger(r_d)\sigma; \tilde{\omega}) \right] \\ &+ \frac{1}{2} \sum_{r_1, r_2 \neq r_d, \nu} F_n(\mathbf{r}_1 - \mathbf{r}_2)^* \left[C_{r_1,\nu} \mathcal{G}(\mathcal{T}_{\nu(r_1)\sigma}, \mathcal{T}_{r_2\sigma}^\dagger; \tilde{\omega}) + C_{r_1,\nu}^* \mathcal{G}(\mathcal{T}_{r_1\sigma}, \mathcal{T}_{\nu(r_2)\sigma}^\dagger; \tilde{\omega}) \right] , \end{aligned} \quad (5.33)$$

where the hermitian conjugate in each part comes from considering the second term in eq. 5.27. The shifted frequency $\tilde{\omega} = \omega - \epsilon_\nu^{\text{loc}}$ captures the effect of local terms in H_{1p} ; the shift is defined as $e^{i\epsilon_\nu^{\text{loc}}t} c_\nu^\dagger = e^{iH_{\text{loc}}t} c_\nu^\dagger e^{-iH_{\text{loc}}t}$. The operator $\nu^\dagger(\mathbf{r})$ represents the eigenstates of H_{loc} at any site $\mathbf{r}$. The coefficients $C_{\nu,r}$ express the eigenstates in terms of local states: $c_r = \sum_{\nu} C_{\nu,r} c_{\nu(r)}$.

Of course, the states ν will differ from the local states c_r only if there are multiple local orbitals (say α) that can hybridise among each other. In such a case, one can also think of an inter-orbital Greens function, defined as

$$G_{\text{loc}}(\alpha, \alpha'; t) = -i\theta(t) \langle \Psi_{\text{gs}} | \{ c_{r_d,\sigma,\alpha}(t), c_{r_d,\sigma,\alpha'}^\dagger \} | \Psi_{\text{gs}} \rangle . \quad (5.34)$$

Such a Greens function can also be calculated from our approach:

$$\begin{aligned} G_{\text{loc}}(\alpha, \alpha'; \omega) &= \frac{1}{2} \sum_{\nu} \left[C_{\nu,\alpha} \mathcal{G}(\nu(r_d), \alpha\sigma, c_{r_d,\sigma,\alpha'}^\dagger; \tilde{\omega}) + C_{\nu,\alpha'}^* \mathcal{G}(c_{r_d,\sigma,\alpha}, \nu^\dagger(r_d), \alpha'\sigma; \tilde{\omega}) \right] \\ &+ \frac{1}{2} \sum_{r_1, r_2 \neq r_d, \nu} F_n(\mathbf{r}_1 - \mathbf{r}_2)^* \left[C_{\alpha,\nu} \mathcal{G}(\mathcal{T}_{\nu(r_1),\sigma,\alpha}, \mathcal{T}_{r_2,\sigma,\alpha'}^\dagger; \tilde{\omega}) + C_{\alpha',\nu}^* \mathcal{G}(\mathcal{T}_{r_1,\sigma,\alpha}, \mathcal{T}_{\nu(r_2),\sigma,\alpha'}^\dagger; \tilde{\omega}) \right] , \end{aligned} \quad (5.35)$$

Non-local Greens function: $\mathbf{r} \neq 0$

We next consider the case of non-local Greens function, where the excitations propagate a non-zero distance $\mathbf{r}$:

$$\sum_{\mathbf{r}_d} \langle \Psi_{\text{gs}} | c_{\mathbf{r}_d+\mathbf{r},\sigma}(t) c_{\mathbf{r}_d,\sigma}^\dagger | \Psi_{\text{gs}} \rangle = \frac{1}{N} \sum_{\mathbf{R}, \Delta, \mathbf{r}_d} \langle \psi_{\text{gs}}(\mathbf{r}_d) | c_{\mathbf{R}+\Delta+\mathbf{r},\sigma}(t) T^\dagger(\Delta) c_{\mathbf{R},\sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle . \quad (5.36)$$

To bring this expression closer to the one for the local Greens function, we redefine $\Delta + \mathbf{r} \rightarrow \Delta$:

$$\sum_{\mathbf{r}_d} \langle \Psi_{\text{gs}} | c_{\mathbf{r}_d+\mathbf{r},\sigma}(t) c_{\mathbf{r}_d,\sigma}^\dagger | \Psi_{\text{gs}} \rangle = \frac{1}{N} \sum_{\mathbf{R}, \Delta, \mathbf{r}_d} \langle \psi_{\text{gs}}(\mathbf{r}_d) | c_{\mathbf{R}+\Delta,\sigma}(t) T^\dagger(\Delta) T(\mathbf{r}) c_{\mathbf{R},\sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle . \quad (5.37)$$

This allows to again split the full summation based on whether $\mathbf{R} = \mathbf{r}_d$ and whether $\Delta + \mathbf{R} = \mathbf{r}_d$:

$$\begin{aligned} \sum_{\mathbf{r}_d} \langle \Psi_{\text{gs}} | c_{\mathbf{r}_d+\mathbf{r},\sigma}(t) c_{\mathbf{r}_d,\sigma}^\dagger | \Psi_{\text{gs}} \rangle &= \langle \psi_{\text{gs}}(\mathbf{r}_d) | c_{\mathbf{r}_d,\sigma}(t) T(\mathbf{r}) c_{\mathbf{r}_d,\sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle \\ &+ \sum_{\Delta \neq 0} \langle \psi_{\text{gs}}(\mathbf{r}_d) | c_{\mathbf{r}_d+\Delta,\sigma}(t) T^\dagger(\Delta - \mathbf{r}) c_{\mathbf{r}_d,\sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle \\ &+ \sum_{\mathbf{R} \neq \mathbf{r}_d} \langle \psi_{\text{gs}}(\mathbf{r}_d) | c_{\mathbf{r}_d,\sigma}(t) T^\dagger(\mathbf{r}_d - \mathbf{R} - \mathbf{r}) c_{\mathbf{R},\sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle \\ &+ \sum_{\mathbf{R} \neq \mathbf{r}_d, \Delta \neq \mathbf{r}_d} \langle \psi_{\text{gs}}(\mathbf{r}_d) | c_{\Delta,\sigma}(t) T^\dagger(\Delta - \mathbf{R} - \mathbf{r}) c_{\mathbf{R},\sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle . \end{aligned} \quad (5.38)$$

Introducing Fourier transforms of the operators and inserting eigenstates of the auxiliary models leads to

$$\begin{aligned} &\sum_{\mathbf{r}_d} \langle \Psi_{\text{gs}} | c_{\mathbf{r}_d+\mathbf{r},\sigma}(t) c_{\mathbf{r}_d,\sigma}^\dagger | \Psi_{\text{gs}} \rangle \\ &= \sum_n F_n^*(\mathbf{r}) \langle \psi_{\text{gs}}(\mathbf{r}_d) | c_{\mathbf{r}_d,\sigma}(t) | \psi_n(\mathbf{r}_d) \rangle \langle \psi_n(\mathbf{r}_d) | c_{\mathbf{r}_d,\sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle \\ &+ \sum_{\mathbf{k}, n} F'_n(\mathbf{k}) F_n^*(\mathbf{r}) \langle \psi_{\text{gs}}(\mathbf{r}_d) | \mathcal{T}_{\mathbf{k},\sigma}(t) | \psi_n(\mathbf{r}_d) \rangle \langle \psi_n(\mathbf{r}_d) | c_{\mathbf{r}_d,\sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle \\ &+ \sum_{\mathbf{k}, n} F'_n(\mathbf{k})^* F_n^*(\mathbf{r}) \langle \psi_{\text{gs}}(\mathbf{r}_d) | c_{\mathbf{r}_d,\sigma}(t) | \psi_n(\mathbf{r}_d) \rangle \langle \psi_n(\mathbf{r}_d) | \mathcal{T}_{\mathbf{k},\sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle \\ &+ \sum_{\mathbf{k}_1, \mathbf{k}_2} F'_n(\mathbf{k}_1) F'_n(\mathbf{k}_2)^* F_n^*(\mathbf{r}) \langle \psi_{\text{gs}}(\mathbf{r}_d) | \mathcal{T}_{\mathbf{k}_1,\sigma}(t) | \psi_n(\mathbf{r}_d) \rangle \langle \psi_n(\mathbf{r}_d) | \mathcal{T}_{\mathbf{k}_2,\sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle , \end{aligned} \quad (5.39)$$

The net difference from the diagonal Greens function is the presence of the overlap factor $F_n^*(\mathbf{r})$.

1.5.3 Equal-Time Ground State Correlators

Apart from dynamical correlations such as Greens functions, we would also like to use our approach to calculate static correlation functions in terms of auxiliary model quantities. We first consider a

real-space correlation that tracks the propagation of a general excitation described by the operator $O(\mathbf{r})$ over a distance Δ :

$$C_O(\mathbf{r}) = \langle \Psi_{\text{gs}} | O(\mathbf{r}_d + \mathbf{r}) O^\dagger(\mathbf{r}_d) | \Psi_{\text{gs}} \rangle . \quad (5.40)$$

We first consider a local correlation: $\mathbf{r} = 0$. To obtain a tractable expression for this, we note that the quantity being computed here is almost identical to the one computed in eq. 5.28, the only difference being the time-evolution of the operator. For the local correlation, we therefore assume the result of eq. 5.31, dropping the time evolution because the correlators in this section are calculated at equal time. We also transform from k -space back to real-space because there are no time-evolution operators which need extended states in order to be resolved. Putting all this together, we get

$$\begin{aligned} C_O^{\text{loc}} &= \langle \psi_{\text{gs}}(\mathbf{r}_d) | O(\mathbf{r}_d) O^\dagger(\mathbf{r}_d) | \psi_{\text{gs}}(\mathbf{r}_d) \rangle \\ &+ \sum_{\mathbf{r} \neq 0, n} F_n(\mathbf{r}) \langle \psi_{\text{gs}}(\mathbf{r}_d) | \mathcal{T}_{O(\mathbf{r}+\mathbf{r}_d)} | \psi_n(\mathbf{r}_d) \rangle \langle \psi_n(\mathbf{r}_d) | O^\dagger(\mathbf{r}_d) | \psi_{\text{gs}}(\mathbf{r}_d) \rangle \\ &+ \sum_{\mathbf{r} \neq 0, n} F_n(\mathbf{r}) \langle \psi_{\text{gs}}(\mathbf{r}_d) | O(\mathbf{r} + \mathbf{r}_d) | \psi_n(\mathbf{r}_d) \rangle \langle \psi_n(\mathbf{r}_d) | \mathcal{T}_{O(\mathbf{r}_d)}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle \\ &+ \sum_{\mathbf{r}_1 \neq 0, \mathbf{r}_2 \neq 0, n} F_n(\mathbf{r}) \langle \psi_{\text{gs}}(\mathbf{r}_d) | \mathcal{T}_{O(\mathbf{r}_1+\mathbf{r}_d)} | \psi_n(\mathbf{r}_d) \rangle \langle \psi_n(\mathbf{r}_d) | \mathcal{T}_{O(\mathbf{r}_2+\mathbf{r}_d)}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle , \end{aligned} \quad (5.41)$$

where the transition operator $\mathcal{T}_{O(\mathbf{r}+\mathbf{r}_d)}$ represents impurity-bath correlation:

$$\mathcal{T}_{O(\mathbf{r}+\mathbf{r}_d), \sigma} = \left[c_{\mathbf{r}_d \sigma}^\dagger + S_{\mathbf{r}_d}^\sigma + C_{\mathbf{r}_d}^+ \right] O(\mathbf{r} + \mathbf{r}_d) , \quad (5.42)$$

We next consider non-local correlations over a distance Δ . Adapting eq. 5.39 for the non-local real-space Greens function in the same manner, we get

$$\begin{aligned} C_O(\Delta) &= \sum_n F_n^*(\Delta) \langle \psi_{\text{gs}}(\mathbf{r}_d) | c_{\mathbf{r}_d, \sigma}(t) | \psi_n(\mathbf{r}_d) \rangle \langle \psi_n(\mathbf{r}_d) | c_{\mathbf{r}_d, \sigma}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle \\ &+ \sum_{\mathbf{r} \neq 0, n} F_n(\mathbf{r} + \Delta) \langle \psi_{\text{gs}}(\mathbf{r}_d) | \mathcal{T}_{O(\mathbf{r}+\Delta+\mathbf{r}_d)} | \psi_n(\mathbf{r}_d) \rangle \langle \psi_n(\mathbf{r}_d) | O^\dagger(\mathbf{r}_d) | \psi_{\text{gs}}(\mathbf{r}_d) \rangle \\ &+ \sum_{\mathbf{r} \neq 0, n} F_n(\mathbf{r} + \Delta) \langle \psi_{\text{gs}}(\mathbf{r}_d) | O(\mathbf{r} + \Delta + \mathbf{r}_d) | \psi_n(\mathbf{r}_d) \rangle \langle \psi_n(\mathbf{r}_d) | \mathcal{T}_{O(\mathbf{r}_d)}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle \\ &+ \sum_{\mathbf{r}_1 \neq 0, \mathbf{r}_2 \neq 0, n} F_n(\mathbf{r} + \Delta) \langle \psi_{\text{gs}}(\mathbf{r}_d) | \mathcal{T}_{O(\mathbf{r}_1+\Delta+\mathbf{r}_d)} | \psi_n(\mathbf{r}_d) \rangle \langle \psi_n(\mathbf{r}_d) | \mathcal{T}_{O(\mathbf{r}_2+\mathbf{r}_d)}^\dagger | \psi_{\text{gs}}(\mathbf{r}_d) \rangle , \end{aligned} \quad (5.43)$$

1.6 Entanglement Measures

We will now describe the prescription of calculating entanglement measures of the lattice model from within our auxiliary model treatment. In this section, we are interested mainly in two such measures, the entanglement entropy and the mutual information. Given a pure state $|\Psi\rangle$ describing

the complete system, the entanglement entropy $S_{\text{EE}}(\nu)$ of a subsystem ν quantifies the entanglement of ν with the rest of the subsystem, and is defined as

$$S_{\text{EE}}(\nu) = -\text{Tr} [\rho(\nu) \log \rho(\nu)], \quad \rho(\nu) = \text{Tr}_\nu [|\Psi\rangle \langle \Psi|] \quad (6.1)$$

where $\text{Tr} [\cdot]$ is the trace operator, and $\rho(\nu)$ is the reduced density matrix (RDM) for the subsystem ν obtained by taking the partial trace Tr_ν (over the states of ν) of the full density matrix $\rho = |\Psi\rangle \langle \Psi|$. If the subsystem ν describes local regions in real space (or states in k -space), we might be interested in the entanglement between two such subsystems ν_1 and ν_2 . The correct measure to quantify such entanglement is the mutual information:

$$I_2(\nu_1, \nu_2) = S_{\text{EE}}(\nu_1) + S_{\text{EE}}(\nu_2) - S_{\text{EE}}(\nu_1 \cup \nu_2), \quad (6.2)$$

where $\nu_1 \cup \nu_2$ is a larger subsystem formed by combining ν_1 and ν_2 .

We start by recalling the decomposition of entanglement entropy in terms of multipartite information as laid out in Subsection ???. Given a partitioning of the system into mutually exclusive subsystems $\{\nu_i\}$, the average entanglement entropy $S_{\text{EE}}^{(1)}$ over all subsystems can be expressed as

$$S_{\text{EE}}^{(1)} = \frac{1}{N} \sum_{n=2}^N (-1)^n \binom{N}{n} \bar{I}_n, \quad (6.3)$$

$$\bar{I}_n = \frac{(N-n)!n!}{N!} \sum_{S_n \in C_n(S_N)} I_n(S_n),$$

where $\bar{I}_n$ is the average multipartite information I_n over all distinct combinations of n subsystems.

We first take a look at real-space entanglement. Real-space entanglement measures can be used to probe delocalisation-localisation transitions. The simplest such measure is the entanglement of a lattice site in real-space. Since the local entanglement entropy will be uniform at each lattice site for a system with translation invariance, it suffices to calculate the real-space averaged entanglement entropy $S_{\text{EE}}^{(1)}$. The corresponding multipartite measures $\bar{I}_n$ involve collections of lattice sites. We write such a measure as

$$\bar{I}_n = \frac{(N-n)!n!}{N!} \frac{1}{N} \sum_{I(\mathbf{r}_d)} \sum_{B^{n-1}(\mathbf{r}_d)} I_n(I(\mathbf{r}_d); B^{n-1}(\mathbf{r}_d)), \quad (6.4)$$

where $I(\mathbf{r}_d)$ sums over all lattice sites in the form of an impurity site and $B^{n-1}(\mathbf{r}_d)$ considers all possible subsets of size $n-1$ of the remaining bath sites. Together, this choice corresponds to an auxiliary model with the impurity placed at $\mathbf{r}_d$. The additional factor of $1/N$ takes care of the overcounting arising from each choice appearing in N auxiliary models. We can therefore write the multipartite information as an average over auxiliary models, each auxiliary model contributing an average I_n over its bath sites:

$$\bar{I}_n = \frac{n}{N^2} \sum_{\mathbf{r}_d} I_n^{(\text{tot})}(\mathbf{r}_d). \quad (6.5)$$

The quantity $I_n^{(\text{tot})}(\mathbf{r}_d)$ is defined as $I_n^{(\text{tot})}(\mathbf{r}_d) = \frac{(N-n)!(n-1)!}{(N-1)!} \sum_{B^{n-1}(\mathbf{r}_d)} I_n(I(\mathbf{r}_d); B^{n-1}(\mathbf{r}_d))$. By translation invariance, such a measure will be uniform across auxiliary models and it suffices to

calculate the measure for a single auxiliary model located at a reference point $\mathbf{r}_d$; the whole expression therefore simplifies to

$$S_{\text{EE}}^{(1)} = \frac{1}{N} \sum_{n=2}^N (-1)^n \binom{N-1}{n-1} I_n^{(\text{tot})}(\mathbf{r}_d) = \frac{1}{N} \sum_{n=2}^N (-1)^n \sum_{B^{n-1}(\mathbf{r}_d)} I_n(I(\mathbf{r}_d); B^{n-1}(\mathbf{r}_d)) . \quad (6.6)$$

For example, truncating the summation to $n = 3$, we have the explicit expression

$$S_{\text{EE}}^{(1)} = \frac{N-1}{N} I_2^{(\text{tot})}(\mathbf{r}_d) - \frac{(N-1)(N-2)}{2N} I_3^{(\text{tot})}(\mathbf{r}_d) \quad (6.7)$$

Chapter 2

Mott Criticality as the Confinement Transition of of a Pseudogap-Mott Metal: Insights on Mott Transition and Pseudogap in Two Dimensions

2.1 Introduction

The Landau quasiparticle paradigm [10] underpins our understanding of metallic behaviour, where the screening of long-ranged repulsive interactions between delocalised electrons leads to well-defined excitations that carry the same quantum numbers as free electrons. This remarkable phenomenon - known as adiabatic continuity - is captured by excitations described by poles of the single-particle Greens function of the interacting system. But what defines metallicity when quasiparticles are no longer well-defined [11]? Can a surface of zeros of the Greens function [12], rather than poles forming a Fermi surface, define an entirely new kind of metal? Could such an exotic state serve as the precursor to Mott localisation driven purely by repulsion, and without invoking magnetic order [13]? These questions necessitate a radical departure in understanding conduction in correlated electron systems beyond the quasiparticle paradigm [14].

The anomalous pseudogap (PG) phase, observed in the cuprates, exhibits nodal-antinodal dichotomy with spectral weight concentrated on Fermi arcs around the nodal regions [15–17]. In the PG phase of such Mott systems [3, 18–46], the single-particle Green’s function develops surfaces of zeros, also known as Luttinger surfaces [12, 47], that signals a fundamental obstruction to quasiparticle formation. As interaction strength increases, these zero surfaces proliferate, fragmenting the Fermi surface into nodal arcs [27, 47], violating the Luttinger sum rule by reconfiguring Fermi volume [48], and breaking an emergent $\mathbb{Z}_2$ symmetry associated with the conservation of spin currents in Fermi liquids [49, 50]. Unlike conventional instabilities driven by poles, this phenomenon marks a topological breakdown of Fermi liquid theory [11, 51]. We show that this breakdown arises from a strongly entangled, scale-invariant state of quantum matter, one that serves as the parent metal to

the Mott insulator in a half-filled lattice model of strongly correlated electrons. This offers a concrete, controlled framework to understand both the non-quasiparticle character of the pseudogap phase and its continuous evolution into a repulsion-driven insulator, in full alignment with Mott's original vision [13].

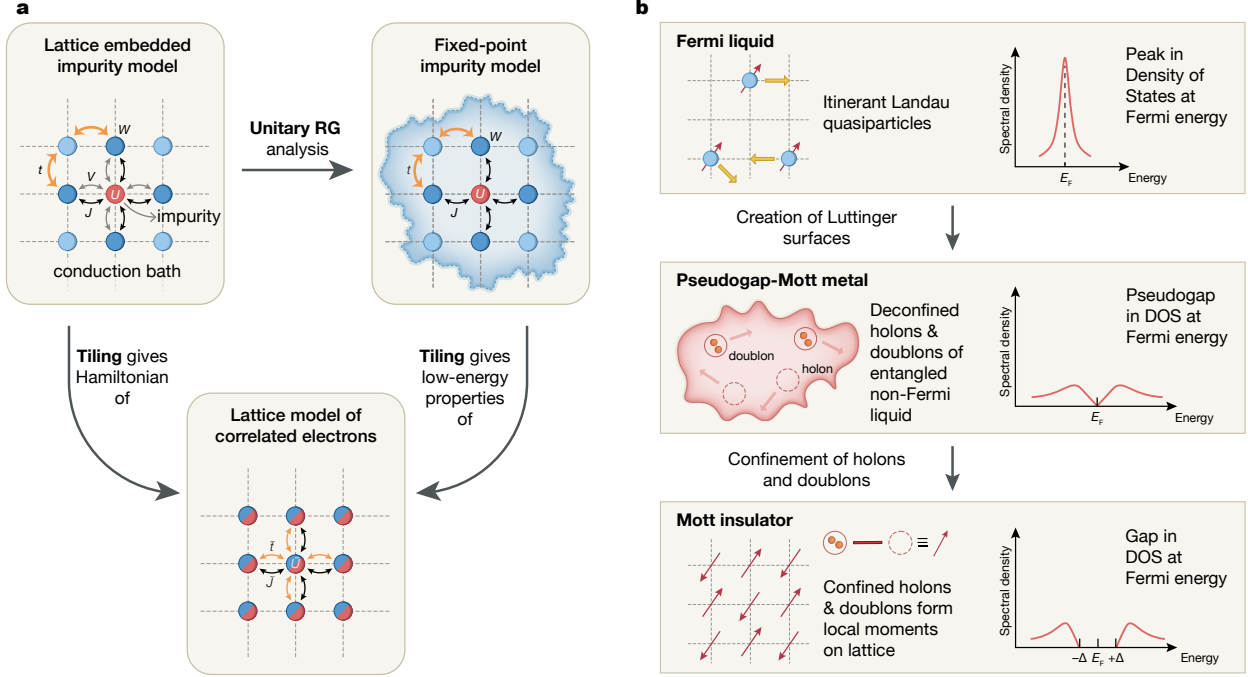


Figure 2.1: Schematic representation of (a) theoretical framework adopted by us (see main text for explanation of symbols), (b) phases obtained for the extended Hubbard model in two dimensions at $T = 0$, and passage between them.

Our approach involves developing a renormalization group (RG)-based impurity-to-lattice framework that reconstructs the low-energy physics of a strongly correlated system from the fixed-point structure of a quantum impurity model (Fig. 2.1(a)). Starting from a dynamical Anderson impurity embedded in a correlated bath, we solve the system using a unitary RG (URG) approach [4, 5]. We then construct a translation-invariant lattice Hamiltonian through a many-body tiling procedure that preserves non-local correlations and inherits momentum-resolved self-energies and renormalized couplings from the impurity solution. The resulting model is an *extended*-Hubbard Hamiltonian on a two-dimensional square lattice at half-filling, with both on-site repulsion and nearest-neighbour spin exchange interactions. These correlations, derived directly from the impurity dynamics, enable tracking the evolution from the FL to the Mott insulator (MI) via an intermediate PG phase.

A central insight of our work is that the PG phase hosts a NFL state governed by Kondo frustration and doublon-holon deconfinement [52] (Fig. 2.1(b)). A correlation scale (W) in the conduction bath suppresses Kondo screening (with coupling J) beyond a critical ratio of W/J , triggering the emergence of Luttinger surfaces in the lattice model. This breakdown of quasiparticles begins at the antinodes and progressively engulfs the nodal regions, driving a continuous transformation into

the MI. The resulting NFL exhibits a pseudogapped density of states that vanishes as ω^2 as $\omega \rightarrow 0$, multipartite long-range entanglement, and a universal scattering rate of the form $\sim (a + b\omega^2)^{-1}$ that grows as $\omega \rightarrow 0$. At the critical endpoint of this phase, the URG flow identifies a singular NFL described by the Hatsugai–Kohmoto (HK) model [50, 53, 54], with fully deconfined holon–doublon excitations.

The emergence of Luttinger surfaces alters the anomaly structure associated with the generalized symmetry of the FS [55–57]. The corresponding topological charge is defined via the Luttinger–Ward functional. This charge is modified by Green’s function zeros, signalling a shift in the underlying anomaly. Within our framework, this reorganization grants topological protection to the RG flows terminating in the PG regime, ensuring the stability of its NFL excitations. These are adiabatically connected to the Hatsugai–Kohmoto model [53, 54], reinforcing the interpretation of the PG as a distinct quantum phase. We thus identify the NFL state as a new gapless, long-range entangled phase of correlated matter - the *Mott metal* - whose deconfined holon-doublon excitations are eventually confined across the Mott transition [13].

While we demonstrate our approach for an extended Hubbard model, the underlying framework is broadly applicable to correlated quantum systems. The impurity-plus-tiling construction accommodates multi-orbital degrees of freedom, frustrated geometries, and nontrivial band topology. By deriving lattice Hamiltonians directly from renormalized local dynamics - without relying on mean-field approximations - this method preserves nonlocal correlations and enhances momentum-space resolution. Coupled with the unitary renormalisation group method as the impurity solver, this approach provides access to one-particle as well as two-particle correlations, entanglement measures and several dynamical quantities, making the method very versatile. It thus provides a versatile and controlled route for engineering strongly correlated models, particularly in regimes where PG formation, NFL scaling, and Mott physics intersect.

We now describe the layout of the work. In section 2.2, we formulate the lattice-embedded auxiliary quantum impurity model and sketch the unitary renormalisation group (URG) method employed to analyse the impurity model. In Chapter 1, we have already formulated the tiling procedure by which to map the impurity Hamiltonian, eigenstates and observables into a lattice model of strongly correlated electrons via the usage of many-body translation operators. Section 2.5 displays the destruction of the Fermi liquid from the breakdown of Kondo screening, leading to a pseudogapped non-Fermi liquid phase of matter. The novel properties of this phase are rigorously established in section 2.6. In section 2.7, we demonstrate that the continuous transition between the pseudogap and Mott insulating phases is described by the exactly solvable Hatsugai-Kohmoto model. We demonstrate in section 2.10 the universal organising principle that is responsible for the emergence of the pseudogap non-Fermi liquid phase. This also unveils the pseudogap as the strongly coupled long-range entangled parent metal of the Mott insulator. We conclude in section 2.11. Finally, we present appendices containing the derivation of the URG analysis of the impurity model, and the exactly solvable model at the critical point.

2.2 Auxiliary Model and URG analysis

To capture the interplay between Kondo physics and Mott localization, our approach relies on a two-dimensional auxiliary model consisting of an Anderson impurity embedded in a minimally-correlated bath, which we solve the system using a unitary RG (URG) approach [4,5]. The impurity model and URG analysis are laid out in this section. In order to make contact with a lattice model, we construct a translation-invariant Hamiltonian from the impurity model through a many-body tiling procedure: this preserves non-local correlations, as well as allows for momentum-resolved self-energies and renormalized couplings generated from the impurity solution. For the impurity model chosen in this work, tiling results in an *extended*-Hubbard Hamiltonian on a two-dimensional square lattice at particle-hole symmetry (half-filling). The interaction terms within the lattice model are a result of analogous correlations present in the auxiliary model. This is laid out next in Section ??.

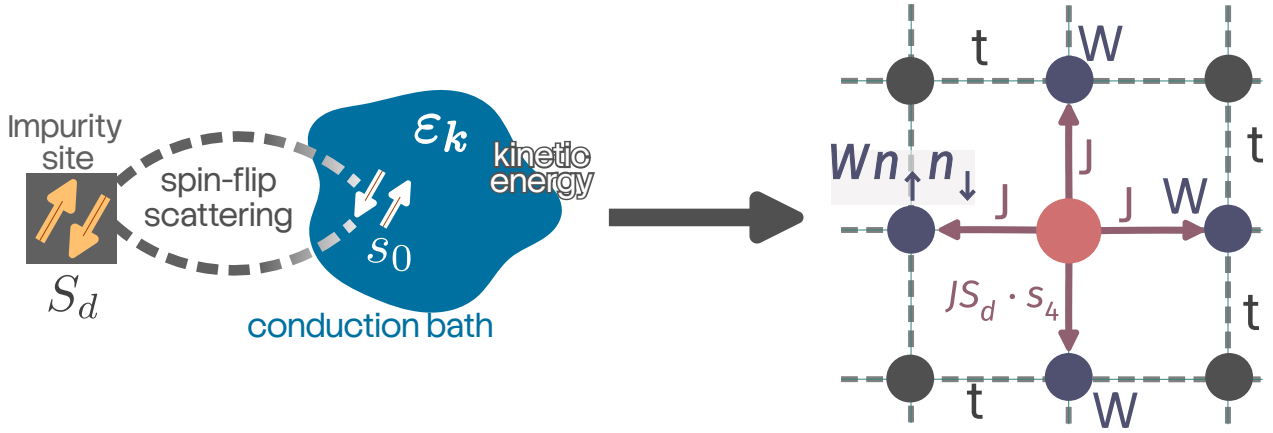


Figure 2.2: Particle and hole states.

2.2.1 Lattice-embedded Impurity Model

The auxiliary model consists of a correlated impurity site (with a double occupancy cost U) embedded within a correlated conduction bath defined on a square lattice (see Fig. 2.1(a)). The impurity site hybridises with the conduction bath sites adjacent to it through one-particle hybridisation V as well as a spin-exchange interaction J arising from fluctuations in local spin densities of the electrons. The conduction bath has minimal correlations in the form of a local correlation term W on the sites that are adjacent to the impurity.

The Hamiltonian of the auxiliary model studied here is

$$\mathcal{H}_{\text{aux}} = H_{\text{imp}} + H_{\text{coup}} + H_{\text{cbath}}, \quad (2.1)$$

where the particle-hole symmetric impurity term is

$$H_{\text{imp}} = -\frac{U}{2} (\hat{n}_{d\uparrow} - \hat{n}_{d\downarrow})^2, \quad (2.2)$$

describing on-site Coulomb interactions at the impurity. Here, $c_{d\sigma}^\dagger$ creates an electron with spin σ at the impurity site, and $\hat{n}_{d\sigma} = c_{d\sigma}^\dagger c_{d\sigma}$ denotes the corresponding number operator.

The impurity couples to four nearest-neighbour bath sites $c_{Z\sigma}$ through both hybridization and Kondo exchange:

$$H_{\text{coup}} = \frac{J}{2} \sum_{\sigma_1, \sigma_2, Z} \mathbf{S}_d \cdot c_{Z\sigma_1}^\dagger \boldsymbol{\tau}_{\sigma_1 \sigma_2} c_{Z\sigma_2} - V \sum_{\sigma, Z} \left(c_{Z\sigma}^\dagger c_{d\sigma} + \text{h.c.} \right), \quad (2.3)$$

where $\boldsymbol{\tau}$ are the Pauli matrices, and Z indexes the four neighbouring bath sites. The half-filled bath includes nearest neighbour hopping and on-site correlations on the site neighbouring the impurity:

$$H_{\text{cbath}} = \sum_{\mathbf{k}} \epsilon_{\mathbf{k}} c_{\mathbf{k},\sigma}^\dagger c_{\mathbf{k},\sigma} - \frac{W}{2} \sum_Z (\hat{n}_{Z\uparrow} - \hat{n}_{Z\downarrow})^2, \quad (2.4)$$

where $\epsilon_{\mathbf{k}} = -2t(\cos k_x + \cos k_y)$, and W parametrizes local repulsion in the bath that frustrates Kondo screening and drives the system towards local-moment formation. A crucial feature of this construction is that the Kondo coupling acquires a momentum structure upon Fourier transforming:

$$J_{\mathbf{k},\mathbf{k}'} = \frac{J}{2} [\cos(k_x - k'_x) + \cos(k_y - k'_y)], \quad (2.5)$$

which respects the C_4 symmetry of the square lattice.

2.2.2 The unitary renormalisation group method

In order to obtain the various low-energy phases of our impurity model, we perform a scaling analysis of the associated Hamiltonian using the recently developed unitary renormalisation group (URG) method [19,39,40]. The method has been applied successfully on a wide variety of problems of correlated fermions [?, 2, 6, 7, 58, 59]. The method proceeds by resolving quantum fluctuations in high-energy degrees of freedom, leading to a low-energy Hamiltonian with renormalised couplings and new emergent degrees of freedom. Typically, for a system with Fermi energy ϵ_F and bandwidth E_N , the sequence of isoenergetic shells $\{E_{(j)}\}$, $E_j \in [E_0, E_N]$ define the states whose quantum fluctuations we sequentially resolve. The momentum states lying on shells E_N that are far away from the Fermi surface comprise the UV states, while those on shells near the Fermi surface comprise the IR states.

As a result of the URG transformations, the Hamiltonian $H_{(j)}$ at a given RG step j involves scattering processes between the k -states that have energies lower than $D_{(j+1)}$. The unitary transformation $U_{(j)}$ is then defined so as to remove the number fluctuations of the currently most energetic set of states $D_{(j)}$ [4, 5]:

$$H_{(j-1)} = U_{(j)} H_{(j)} U_{(j)}^\dagger, \text{ such that } [H_{(j-1)}, \hat{n}_j] = 0. \quad (2.6)$$

The eigenvalue of $\hat{n}_j$ has, thus, been rendered an integral of motion (IOM) under the RG transformation.

The unitary transformations can be expressed in terms of a generator $\eta_{(j)}$ that has fermionic algebra [4, 5]:

$$U_{(j)} = \frac{1}{\sqrt{2}} \left(1 + \eta_{(j)} - \eta_{(j)}^\dagger \right), \quad \left\{ \eta_{(j)}, \eta_{(j)}^\dagger \right\} = 1, \quad (2.7)$$

where $\{\cdot\}$ is the anticommutator. The unitary operator $U_{(j)}$ that appears in Eq. (2.7) can be cast into the well-known general form $U = e^{\mathcal{S}}$, $\mathcal{S} = \frac{\pi}{4} \left(\eta_{(j)}^\dagger - \eta_{(j)} \right)$ that a unitary operator can take, defined by an anti-Hermitian operator $\mathcal{S}$. The generator $\eta_{(j)}$ is given by the expression [4, 5]

$$\eta_{(j)}^\dagger = \frac{1}{\hat{\omega}_{(j)} - \text{Tr}(H_{(j)} \hat{n}_j)} c_j^\dagger \text{Tr}(H_{(j)} c_j). \quad (2.8)$$

The operators $\eta_{(j)}, \eta_{(j)}^\dagger$ behave as the many-particle analogues of the single-particle field operators $c_j, c_j^\dagger$ - they change the occupation number of the single-particle Fock space $|n_j\rangle$. The important operator $\hat{\omega}_{(j)}$ originates from the quantum fluctuations that exist in the problem because of the non-commutation of the kinetic energy terms and the interaction terms in the Hamiltonian:

$$\hat{\omega}_{(j)} = H_{(j-1)} - H_{(j)}^i. \quad (2.9)$$

$H_{(j)}^i$ is the part of $H_{(j)}$ that commutes with $\hat{n}_j$ but does *not* commute with at least one $\hat{n}_l$ for $l < j$. The RG flow continues up to energy D^* , where a fixed point is reached from the vanishing of the RG function. Detailed comparisons of the URG with other methods (e.g., the functional RG, spectrum bifurcation RG etc.) can be found in Refs. [2, 4], and implementation on the present problem in Appendix A.

2.3 Derivation of unitary RG equations for the lattice-embedded impurity model

2.3.1 RG scheme

At any given step j of the RG procedure, we decouple the states $\{\mathbf{q}\}$ on the isoenergetic surface of energy ε_j . The diagonal Hamiltonian H_D for this step consists of all terms that do not change the occupancy of the states $\{\mathbf{q}\}$:

$$H_D^{(j)} = \varepsilon_j \sum_{q,\sigma} \tau_{q,\sigma} + \frac{1}{2} \sum_{\mathbf{q}} J_{\mathbf{q},\mathbf{q}} S_d^z (\hat{n}_{\mathbf{q},\uparrow} - \hat{n}_{\mathbf{q},\downarrow}) - \frac{1}{2} \sum_{\mathbf{q}} W_{\mathbf{q}} (\hat{n}_{\mathbf{q},\uparrow} - \hat{n}_{\mathbf{q},\downarrow})^2, \quad (3.1)$$

where $\tau = \hat{n} - 1/2$ and $W_{\mathbf{q}}$ is a shorthand for $W_{\mathbf{q},\mathbf{q},\mathbf{q}}$. The three terms, respectively, are the kinetic energy of the momentum states on the isoenergetic shell that we are decoupling, the spin-correlation energy between the impurity spin and the spins formed by these momentum states and, finally, the local correlation energy associated with these states arising from the W term. The off-diagonal part of the Hamiltonian on the other hand leads to scattering in the states $\{\mathbf{q}\}$. We now list these terms, classified by the coupling they originate from.

Arising from the Kondo spin-exchange term

$$\begin{aligned}
 T_{KZ1}^\dagger + T_{KZ1} &= \frac{1}{2} \sum_{\mathbf{k}, \mathbf{q}, \sigma} \sigma J_{\mathbf{k}, \mathbf{q}} S_d^z \left[c_{\mathbf{q}\sigma}^\dagger c_{\mathbf{k}, \sigma} + \text{h.c.} \right], \\
 T_{KZ2}^\dagger + T_{KZ2} &= \frac{1}{2} \sum_{\mathbf{q}, \sigma} \sigma J_{\mathbf{q}, \bar{\mathbf{q}}} S_d^z \left[c_{\mathbf{q}\sigma}^\dagger c_{\bar{\mathbf{q}}, \sigma} + \text{h.c.} \right], \\
 T_{KT1}^\dagger + T_{KT1} &= \frac{1}{2} \sum_{\mathbf{k}, \mathbf{q}} J_{\mathbf{k}, \mathbf{q}} \left[S_d^+ \left(c_{\mathbf{q}\downarrow}^\dagger c_{\mathbf{k}\uparrow} + c_{\mathbf{k}\downarrow}^\dagger c_{\mathbf{q}\uparrow} \right) + \text{h.c.} \right], \\
 T_{KT2}^\dagger + T_{KT2} &= \frac{1}{2} \sum_{\mathbf{q}} J_{\mathbf{q}, \bar{\mathbf{q}}} \left[S_d^+ \left(c_{\mathbf{q}\downarrow}^\dagger c_{\bar{\mathbf{q}}\uparrow} + c_{\bar{\mathbf{q}}\downarrow}^\dagger c_{\mathbf{q}\uparrow} \right) + \text{h.c.} \right],
 \end{aligned} \tag{3.2}$$

Arising from spin-preserving scattering within conduction bath

$$\begin{aligned}
 T_{P1}^\dagger + T_{P1} &= - \sum_{\mathbf{q} \in \mathcal{E}_j} \sum_{\mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4 < \mathcal{E}_j} \sum_{\sigma} \left[W_{\mathbf{q}, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} c_{\mathbf{q}, \sigma}^\dagger c_{\mathbf{k}_2, \sigma} c_{\mathbf{k}_3, \sigma}^\dagger c_{\mathbf{k}_4, \sigma} + \text{h.c.} \right] \\
 T_{P2}^\dagger + T_{P3} &= - \sum_{\mathbf{q} \in \mathcal{E}_j} \sum_{\mathbf{k}_2 < \mathcal{E}_j} \sum_{\sigma} W_{\mathbf{q}, \mathbf{k}_2, \bar{\mathbf{q}}, \bar{\mathbf{q}}} c_{\mathbf{q}, \sigma}^\dagger c_{\mathbf{k}_2, \sigma} n_{\bar{\mathbf{q}}, \sigma} - \sum_{\mathbf{q} \in \mathcal{E}_j} \sum_{\mathbf{k}_1 < \mathcal{E}_j} \sum_{\sigma} W_{\mathbf{k}_1, \mathbf{q}, \mathbf{q}, \bar{\mathbf{q}}} c_{\mathbf{k}_1, \sigma}^\dagger c_{\mathbf{q}, \sigma} n_{\bar{\mathbf{q}}, \sigma} \\
 T_{P4} &= - \sum_{\mathbf{q} \in \mathcal{E}_j} \sum_{\mathbf{k}_2, \mathbf{k}_3 < \mathcal{E}_j} \sum_{\sigma} W_{\mathbf{q}, \bar{\mathbf{q}}, \mathbf{k}_2, \mathbf{k}_3} c_{\mathbf{q}, \sigma}^\dagger c_{\bar{\mathbf{q}}, \sigma} c_{\mathbf{k}_2, \sigma}^\dagger c_{\mathbf{k}_3, \sigma} \\
 T_{P5} &= - \sum_{\mathbf{q} \in \mathcal{E}_j} \sum_{\mathbf{k}_2, \mathbf{k}_3 < \mathcal{E}_j} \sum_{\sigma} W_{\mathbf{q}, \mathbf{k}_2, \mathbf{k}_3, \bar{\mathbf{q}}} c_{\mathbf{q}, \sigma}^\dagger c_{\mathbf{k}_2, \sigma} c_{\mathbf{k}_3, \sigma}^\dagger c_{\bar{\mathbf{q}}, \sigma} \\
 &= + \sum_{\mathbf{q} \in \mathcal{E}_j} \sum_{\mathbf{k}_2, \mathbf{k}_3 < \mathcal{E}_j} \sum_{\sigma} W_{\mathbf{q}, \mathbf{k}_3, \mathbf{k}_2, \bar{\mathbf{q}}} c_{\mathbf{q}, \sigma}^\dagger c_{\bar{\mathbf{q}}, \sigma} c_{\mathbf{k}_2, \sigma}^\dagger c_{\mathbf{k}_3, \sigma} \\
 &= -T_{P4}
 \end{aligned} \tag{3.3}$$

Arising from spin-flip scattering within conduction bath

$$\begin{aligned}
 T_{F1}^\dagger + T_{F1} &= \sum_{\mathbf{q} \in \mathcal{E}_j} \sum_{\mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4 < \mathcal{E}_j} \sum_{\sigma} \left[W_{\mathbf{q}, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} c_{\mathbf{q}, \sigma}^\dagger c_{\mathbf{k}_2, \sigma} c_{\mathbf{k}_3, \bar{\sigma}}^\dagger c_{\mathbf{k}_4, \bar{\sigma}} + \text{h.c.} \right] \\
 T_{F2} &= \sum_{\mathbf{q}, \mathbf{q}' \in \mathcal{E}_j} \sum_{\mathbf{k}_2, \mathbf{k}_3 < \mathcal{E}_j} \sum_{\sigma} W_{\mathbf{q}, \mathbf{q}', \mathbf{k}_2, \mathbf{k}_3} c_{\mathbf{q}, \sigma}^\dagger c_{\mathbf{q}', \sigma} c_{\mathbf{k}_2, \bar{\sigma}}^\dagger c_{\mathbf{k}_3, \bar{\sigma}} \\
 T_{F3} &= \sum_{\mathbf{q}, \mathbf{q}' \in \mathcal{E}_j} \sum_{\mathbf{k}_2, \mathbf{k}_3 < \mathcal{E}_j} \sum_{\sigma} W_{\mathbf{q}, \mathbf{k}_2, \mathbf{k}_3, \mathbf{q}'} c_{\mathbf{q}, \sigma}^\dagger c_{\mathbf{k}_2, \sigma} c_{\mathbf{k}_3, \bar{\sigma}}^\dagger c_{\mathbf{q}', \bar{\sigma}} \\
 T_{F4}^\dagger + T_{F4} &= \sum_{\mathbf{q}, \mathbf{q}' \in \mathcal{E}_j} \sum_{\mathbf{k}_1 < \mathcal{E}_j} \sum_{\sigma} \left[W_{\mathbf{q}, \mathbf{q}, \mathbf{q}', \mathbf{k}_1} n_{\mathbf{q}, \sigma} c_{\mathbf{q}', \bar{\sigma}}^\dagger c_{\mathbf{k}_1, \bar{\sigma}} + \text{h.c.} \right]
 \end{aligned} \tag{3.4}$$

In all of the terms $T_{P[i]}$ and $T_{F[i]}$, the factor of 1/2 in front has been cancelled out by a factor of 2 coming from the multiple possibilities of arranging the momentum labels. We will henceforth ignore T_{P4} and T_{P5} because they cancel each other out.

The renormalisation of the Hamiltonian is constructed from the general expression

$$\Delta H^{(j)} = H_X \frac{1}{\omega - H_D} H_X . \quad (3.5)$$

The states on the isoenergetic shell $\pm|\varepsilon_j|$ come in particle-hole pairs $(\mathbf{q}, \bar{\mathbf{q}})$ with energies of opposite signs (relative to the Fermi energy). If $\mathbf{q}$ is defined as the hole state (unoccupied in the absence of quantum fluctuations), it will have positive energy, while the particle state $\bar{\mathbf{q}}$ will be of negative energy and hence below the Fermi surface. To be more specific, given a state $\mathbf{q}$ with energy $\pm|\varepsilon_j|$, we define its particle-hole transformed counterpart as the state $\bar{\mathbf{q}} = \pi + \mathbf{q}$, having energy $\mp|\varepsilon_j|$ and residing in the opposite quadrant of the Brillouin zone. Given this definition, we have the important property that

$$\begin{aligned} J_{\mathbf{k}, \bar{\mathbf{q}}} &= -J_{\mathbf{k}, \mathbf{q}}, \\ W_{\{\mathbf{k}\}, \bar{\mathbf{q}}} &= -W_{\{\mathbf{k}\}, \mathbf{q}} . \end{aligned} \quad (3.6)$$

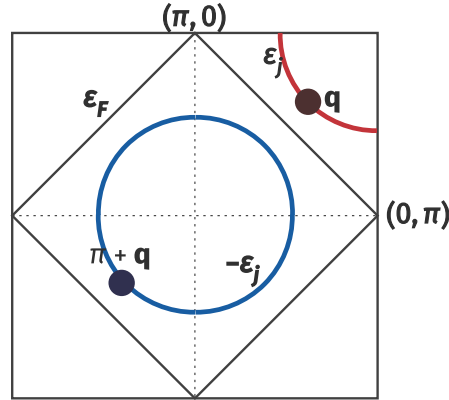


Figure 2.3: Particle and hole states.

2.3.2 Renormalisation of the bath correlation term W

The bath correlation term W can undergo renormalisation only via scattering processes arising from itself. Irrespective of whether the state $\mathbf{q}$ being decoupled is in a particle or hole configuration in the initial many-body state, the propagator $G = 1/(\omega - H_D)$ of the intermediate excited state is uniform, and equal to

$$G_W = 1/(\omega - |\varepsilon_j|/2 + W_{\mathbf{q}}/2) , \quad (3.7)$$

where $W_{\mathbf{q}}$ is the same whether $\mathbf{q}$ is above or below the Fermi surface. The $|\varepsilon_j|/2$ in H_D arises from the excited nature of the state after the initial scattering process.

Scattering arising purely from spin-preserving processes

In this subsection, we calculate the renormalisation to W arising from the terms T_{P1} , T_{P2} and T_{P3} . The first term is

$$\begin{aligned} T_{P1}^\dagger G_W T_{P3} &= \sum_{\sigma} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} \sum_{\mathbf{q}} W_{\mathbf{q}, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} c_{\mathbf{q}, \sigma}^\dagger c_{\mathbf{k}_2, \sigma} c_{\mathbf{k}_3, \sigma}^\dagger c_{\mathbf{k}_4, \sigma} G_W W_{\mathbf{k}_1, \mathbf{q}, \mathbf{q}, \mathbf{q}} c_{\mathbf{k}_1, \sigma}^\dagger c_{\mathbf{q}, \sigma} n_{\mathbf{q}, \sigma} \\ &= - \sum_{\sigma} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} c_{\mathbf{k}_1, \sigma}^\dagger c_{\mathbf{k}_2, \sigma} c_{\mathbf{k}_3, \sigma}^\dagger c_{\mathbf{k}_4, \sigma} \sum_{\mathbf{q} \in \text{PS}} W_{\mathbf{q}, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} G_W W_{\mathbf{k}_1, \mathbf{q}, \mathbf{q}, \mathbf{q}} . \end{aligned} \quad (3.8)$$

The operators acting on the states being decoupled contract to form a number operator $n_{\mathbf{q}, \sigma}$ which projects the sum over $\mathbf{q}$ into the states that are initial occupied (particle sector, PS).

The second such contribution is obtained by flipping the sequence of scattering processes:

$$\begin{aligned} T_{P3} G_W T_{P1}^\dagger &= \sum_{\sigma} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} \sum_{\mathbf{q}} W_{\mathbf{k}_1, \mathbf{q}, \mathbf{q}, \mathbf{q}} c_{\mathbf{k}_1, \sigma}^\dagger c_{\mathbf{q}, \sigma} n_{\mathbf{q}, \sigma} G_W W_{\mathbf{q}, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} c_{\mathbf{q}, \sigma}^\dagger c_{\mathbf{k}_2, \sigma} c_{\mathbf{k}_3, \sigma}^\dagger c_{\mathbf{k}_4, \sigma} \\ &= \sum_{\sigma} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} c_{\mathbf{k}_1, \sigma}^\dagger c_{\mathbf{k}_2, \sigma} c_{\mathbf{k}_3, \sigma}^\dagger c_{\mathbf{k}_4, \sigma} \sum_{\mathbf{q} \in \text{HS}} W_{\mathbf{q}, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} G_W W_{\mathbf{k}_1, \mathbf{q}, \mathbf{q}, \mathbf{q}} . \end{aligned} \quad (3.9)$$

By virtue of eq. 3.6, the product of couplings $W_{\mathbf{q}, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} G_W W_{\mathbf{k}_1, \mathbf{q}, \mathbf{q}, \mathbf{q}}$ is the same irrespective of whether $\mathbf{q}$ belongs to the particle or hole sector. The two contributions therefore cancel each other. Moreover, the remaining contributions $T_{P3}^\dagger G_W T_{P1}$ and $T_{P1} G_W T_{P2}^\dagger$ are effectively hermitian conjugates of the two contributions considered above, and therefore also cancel each other.

Scattering arising from spin-flip processes

We now come to the processes that involve spin-flips. Considering T_{F1} and T_{F4} first, we get

$$\begin{aligned} T_{F1}^\dagger G_W T_{F4} &= \sum_{\sigma} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} \sum_{\mathbf{q}} W_{\mathbf{q}, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} c_{\mathbf{q}, \sigma}^\dagger c_{\mathbf{k}_2, \sigma} c_{\mathbf{k}_3, \bar{\sigma}}^\dagger c_{\mathbf{k}_4, \bar{\sigma}} G_W W_{\mathbf{k}_1, \mathbf{q}, \mathbf{q}, \mathbf{q}} c_{\mathbf{k}_1, \sigma}^\dagger c_{\mathbf{q}, \sigma} n_{\mathbf{q}, \bar{\sigma}} \\ &= - \sum_{1,2,3,4} \sum_{\sigma} c_{\mathbf{k}_1, \sigma}^\dagger c_{\mathbf{k}_2, \sigma} c_{\mathbf{k}_3, \bar{\sigma}}^\dagger c_{\mathbf{k}_4, \bar{\sigma}} \sum_{\mathbf{q} \in \text{PS}} W_{\mathbf{q}, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} G_W W_{\mathbf{k}_1, \mathbf{q}, \mathbf{q}, \mathbf{q}} , \\ T_{F4} G_W T_{F1}^\dagger &= \sum_{\sigma} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} \sum_{\mathbf{q}} W_{\mathbf{k}_1, \mathbf{q}, \mathbf{q}, \mathbf{q}} c_{\mathbf{k}_1, \sigma}^\dagger c_{\mathbf{q}, \sigma} n_{\mathbf{q}, \bar{\sigma}} G_W W_{\mathbf{q}, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} c_{\mathbf{q}, \sigma}^\dagger c_{\mathbf{k}_2, \sigma} c_{\mathbf{k}_3, \bar{\sigma}}^\dagger c_{\mathbf{k}_4, \bar{\sigma}} \\ &= \sum_{1,2,3,4} \sum_{\sigma} c_{\mathbf{k}_1, \sigma}^\dagger c_{\mathbf{k}_2, \sigma} c_{\mathbf{k}_3, \bar{\sigma}}^\dagger c_{\mathbf{k}_4, \bar{\sigma}} \sum_{\mathbf{q} \in \text{HS}} W_{\mathbf{q}, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} G_W W_{\mathbf{k}_1, \mathbf{q}, \mathbf{q}, \mathbf{q}} . \end{aligned} \quad (3.10)$$

By the same arguments as in the previous subsection, these terms cancel each other out. Their hermitian conjugate contributions $T_{F1} G_W T_{F4}^\dagger$ and $T_{F4}^\dagger G_W T_{F1}$ also cancel out. The other two terms

are T_{F2} and T_{F3} , and their contributions also cancel out for the same reason:

$$\begin{aligned}
 T_{F2} G_W T_{F2} &= \sum_{\sigma} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} \sum_{\mathbf{q}} W_{\mathbf{q}, \bar{\mathbf{q}}, \mathbf{k}_3, \mathbf{k}_4} c_{\mathbf{q}, \sigma}^{\dagger} c_{\bar{\mathbf{q}}, \sigma} c_{\mathbf{k}_3, \bar{\sigma}}^{\dagger} c_{\mathbf{k}_4, \bar{\sigma}} G_W W_{\bar{\mathbf{q}}, \mathbf{q}, \mathbf{k}_1, \mathbf{k}_2} c_{\bar{\mathbf{q}}, \sigma}^{\dagger} c_{\mathbf{q}, \sigma} c_{\mathbf{k}_1, \bar{\sigma}}^{\dagger} c_{\mathbf{k}_2, \bar{\sigma}} \\
 &= \sum_{1,2,3,4} \sum_{\sigma} c_{\mathbf{k}_1 \sigma}^{\dagger} c_{\mathbf{k}_2 \sigma} c_{\mathbf{k}_3 \bar{\sigma}}^{\dagger} c_{\mathbf{k}_4 \bar{\sigma}} \sum_{\mathbf{q} \in \text{PS}} W_{\mathbf{q}, \bar{\mathbf{q}}, \mathbf{k}_3, \mathbf{k}_4} G_W W_{\bar{\mathbf{q}}, \mathbf{q}, \mathbf{k}_1, \mathbf{k}_2} , \\
 T_{F3} G_W T_{F3} &= \sum_{\sigma} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3, \mathbf{k}_4} \sum_{\mathbf{q}} W_{\mathbf{q}, \mathbf{k}_2, \mathbf{k}_3, \bar{\mathbf{q}}} c_{\mathbf{q}, \sigma}^{\dagger} c_{\mathbf{k}_2, \sigma} c_{\mathbf{k}_3, \bar{\sigma}}^{\dagger} c_{\bar{\mathbf{q}}, \bar{\sigma}} G_W W_{\bar{\mathbf{q}}, \mathbf{k}_4, \mathbf{k}_1, \mathbf{q}} c_{\bar{\mathbf{q}}, \bar{\sigma}}^{\dagger} c_{\mathbf{k}_4, \bar{\sigma}} c_{\mathbf{k}_1, \sigma}^{\dagger} c_{\mathbf{q}, \sigma} \\
 &= - \sum_{1,2,3,4} \sum_{\sigma} c_{\mathbf{k}_1 \sigma}^{\dagger} c_{\mathbf{k}_2 \sigma} c_{\mathbf{k}_3 \bar{\sigma}}^{\dagger} c_{\mathbf{k}_4 \bar{\sigma}} \sum_{\mathbf{q} \in \text{PS}} W_{\mathbf{q}, \mathbf{k}_2, \mathbf{k}_3, \bar{\mathbf{q}}} G_W W_{\bar{\mathbf{q}}, \mathbf{k}_4, \mathbf{k}_1, \mathbf{q}} ,
 \end{aligned} \tag{3.11}$$

Scattering involving both spin-flip and spin-preserving processes

These processes involve the combination of terms like T_{P1} with T_{F4} , and T_{P2} with T_{F1} . These again cancel each other out for the same reasons as outline above.

Net renormalisation for the bath correlation term

Since all the contributions cancel out in pairs, the bath correlation term W is *marginal*.

2.3.3 Renormalisation of the Kondo scattering term J

We focus on the renormalisation of the spin-flip part of the Kondo interaction. For these processes, the intermediate many-body state always involves the impurity spin being anti-correlated with the conduction electron spin, such that the propagator for that state is $G_J = 1/(\omega - |\varepsilon_j|/2 + J_{\mathbf{q}}/4 + W_{\mathbf{q}}/2)$.

Impurity-mediated spin-flip scattering purely through Kondo-like processes

The following processes arising from the Kondo term renormalise the spin-flip interaction:

$$\begin{aligned}
 T_{KT1}^{\dagger} G_J (T_{KZ1} + T_{KZ1}^{\dagger}) &= \frac{1}{4} \sum_{\mathbf{k}_1, \mathbf{k}_1, \mathbf{q}} J_{\mathbf{q}, \mathbf{k}_2} S_d^+ \left[-c_{\mathbf{q}\downarrow}^{\dagger} c_{\mathbf{k}_2\uparrow} G_J c_{\mathbf{k}_1\downarrow}^{\dagger} c_{\mathbf{q}\downarrow} + c_{\mathbf{k}_2\downarrow}^{\dagger} c_{\mathbf{q}\uparrow} G_J c_{\mathbf{q}\uparrow}^{\dagger} c_{\mathbf{k}_1\uparrow} \right] J_{\mathbf{k}_1, \mathbf{q}} S_d^z \\
 &= -\frac{1}{8} \sum_{\mathbf{k}_1, \mathbf{k}_1, \mathbf{q}} J_{\mathbf{q}, \mathbf{k}_2} S_d^+ \left[c_{\mathbf{k}_1\downarrow}^{\dagger} c_{\mathbf{k}_2\uparrow} G_J n_{\mathbf{q}\downarrow} + c_{\mathbf{k}_2\downarrow}^{\dagger} c_{\mathbf{k}_1\uparrow} (1 - n_{\mathbf{q}\uparrow}) G_J \right] J_{\mathbf{k}_1, \mathbf{q}} \\
 &= -\frac{1}{8} \sum_{\mathbf{k}_1, \mathbf{k}_1} S_d^+ c_{\mathbf{k}_1\downarrow}^{\dagger} c_{\mathbf{k}_2\uparrow} \sum_{\mathbf{q} \in \text{PS}} [J_{\mathbf{q}, \mathbf{k}_2} J_{\mathbf{k}_1, \mathbf{q}} + J_{\bar{\mathbf{q}}, \mathbf{k}_1} J_{\mathbf{k}_2, \bar{\mathbf{q}}}] G_J .
 \end{aligned} \tag{3.12}$$

In getting the final expression, we used the sigma matrix relation $S_d^+ S_d^z = -\frac{1}{2} S_d^+$, and absorbed the projector $1 - n_{\mathbf{q}\uparrow}$ into the sum over the particle sector by replacing q with its particle-hole transformed

counterpart $\bar{q}$. An identical contribution is obtained by switching the sequence of processes:

$$\begin{aligned} (T_{KZ1} + T_{KZ1}^\dagger) G_J T_{KT1}^\dagger &= \frac{1}{4} \sum_{\mathbf{k}_1, \mathbf{k}_1, \mathbf{q}} J_{\mathbf{k}_1, \mathbf{q}} S_d^z \left[-c_{\mathbf{k}_1 \downarrow}^\dagger c_{\mathbf{q} \downarrow} G_J c_{\mathbf{q} \downarrow}^\dagger c_{\mathbf{k}_2 \uparrow} + c_{\mathbf{q} \uparrow}^\dagger c_{\mathbf{k}_1 \uparrow} G_J c_{\mathbf{k}_2 \downarrow}^\dagger c_{\mathbf{q} \uparrow} \right] J_{\mathbf{q}, \mathbf{k}_2} S_d^+ \\ &= -\frac{1}{8} \sum_{\mathbf{k}_1, \mathbf{k}_1} S_d^+ c_{\mathbf{k}_1 \downarrow}^\dagger c_{\mathbf{k}_2 \uparrow} \sum_{\mathbf{q} \in \text{PS}} [J_{\bar{\mathbf{q}}, \mathbf{k}_2} J_{\mathbf{k}_1, \bar{\mathbf{q}}} + J_{\mathbf{q}, \mathbf{k}_1} J_{\mathbf{k}_2, \mathbf{q}}] G_J. \end{aligned} \quad (3.13)$$

Scattering processes involving interplay between the Kondo interaction and conduction bath interaction

Looking at $T_{KT1}^\dagger$ first, we have

$$T_{KT1}^\dagger G_J (T_{F4} + T_{F4}^\dagger) = \frac{1}{2} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{q}} J_{\mathbf{k}_2, \mathbf{q}} S_d^+ \left(c_{\mathbf{q} \downarrow}^\dagger c_{\mathbf{k}_2 \uparrow} G_J W_{\mathbf{q}, \mathbf{q}, \mathbf{k}_1, \mathbf{q}} n_{\mathbf{q} \uparrow} c_{\mathbf{k}_1 \downarrow}^\dagger c_{\mathbf{q} \downarrow} + c_{\mathbf{k}_2 \downarrow}^\dagger c_{\mathbf{q} \uparrow} G_J W_{\bar{\mathbf{q}}, \bar{\mathbf{q}}, \mathbf{q}, \mathbf{k}_1} n_{\bar{\mathbf{q}} \downarrow} c_{\mathbf{q} \uparrow}^\dagger c_{\mathbf{k}_1 \uparrow} \right). \quad (3.14)$$

For either of the two choices of the functional form of W , it is easy to show that $W_{\mathbf{q}, \mathbf{q}, \mathbf{k}_1, \mathbf{q}} = W_{\bar{\mathbf{q}}, \bar{\mathbf{q}}, \mathbf{q}, \mathbf{k}_1}$.

$$T_{KT1}^\dagger G_J (T_{F4} + T_{F4}^\dagger) = \frac{1}{2} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{q}} J_{\mathbf{k}_2, \mathbf{q}} W_{\mathbf{q}, \mathbf{q}, \mathbf{k}_1, \mathbf{q}} G_J S_d^+ \left[-c_{\mathbf{k}_1 \downarrow}^\dagger c_{\mathbf{k}_2 \uparrow} n_{\mathbf{q} \downarrow} n_{\mathbf{q} \uparrow} + c_{\mathbf{k}_2 \downarrow}^\dagger c_{\mathbf{k}_1 \uparrow} (1 - n_{\mathbf{q} \uparrow}) n_{\bar{\mathbf{q}} \downarrow} \right]. \quad (3.15)$$

Another contribution is obtained by switching the sequence of the scattering processes:

$$\begin{aligned} (T_{F4} + T_{F4}^\dagger) G_J T_{KT1}^\dagger &= \frac{1}{2} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{q}} \left(W_{\mathbf{q}, \mathbf{q}, \mathbf{k}_1, \mathbf{q}} n_{\bar{\mathbf{q}} \uparrow} c_{\mathbf{k}_1 \downarrow}^\dagger c_{\mathbf{q} \downarrow} G_J c_{\mathbf{q} \downarrow}^\dagger c_{\mathbf{k}_2 \uparrow} + W_{\bar{\mathbf{q}}, \bar{\mathbf{q}}, \mathbf{q}, \mathbf{k}_1} n_{\mathbf{q} \downarrow} c_{\mathbf{q} \uparrow}^\dagger c_{\mathbf{k}_1 \uparrow} G_J c_{\mathbf{k}_2 \downarrow}^\dagger c_{\mathbf{q} \uparrow} \right) J_{\mathbf{k}_2, \mathbf{q}} S_d^+ \\ &= \frac{1}{2} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{q}} \left(c_{\mathbf{k}_1 \downarrow}^\dagger c_{\mathbf{k}_2 \uparrow} n_{\bar{\mathbf{q}} \uparrow} (1 - n_{\mathbf{q} \downarrow}) - c_{\mathbf{k}_2 \downarrow}^\dagger c_{\mathbf{k}_1 \uparrow} n_{\mathbf{q} \downarrow} n_{\mathbf{q} \uparrow} \right) W_{\mathbf{q}, \mathbf{q}, \mathbf{k}_1, \mathbf{q}} G_J J_{\mathbf{k}_2, \mathbf{q}} S_d^+ \end{aligned} \quad (3.16)$$

The two contributions (eqs. 3.15 and 3.16) arising from T_{KT1} cancel each other.

We now consider the other spin-exchange process $T_{KT2}^\dagger$. One such contribution is

$$\begin{aligned} T_{KT2}^\dagger G_J T_{F3} &= \frac{1}{2} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{q}} J_{\mathbf{q}, \bar{\mathbf{q}}} S_d^+ \left(c_{\mathbf{q} \downarrow}^\dagger c_{\bar{\mathbf{q}} \uparrow} G_J c_{\bar{\mathbf{q}} \uparrow}^\dagger c_{\mathbf{k}_2 \uparrow} c_{\mathbf{k}_1 \downarrow}^\dagger c_{\mathbf{q} \downarrow} + c_{\bar{\mathbf{q}} \downarrow}^\dagger c_{\mathbf{q} \uparrow} G_J c_{\mathbf{q} \uparrow}^\dagger c_{\mathbf{k}_2 \uparrow} c_{\mathbf{k}_1 \downarrow}^\dagger c_{\bar{\mathbf{q}} \downarrow} \right) W_{\bar{\mathbf{q}}, \mathbf{k}_2, \mathbf{k}_1, \mathbf{q}} \\ &= -\frac{1}{2} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{q}} S_d^+ c_{\mathbf{k}_1 \downarrow}^\dagger c_{\mathbf{k}_2 \uparrow} \left[n_{\mathbf{q} \downarrow} (1 - n_{\bar{\mathbf{q}} \uparrow}) + n_{\bar{\mathbf{q}} \downarrow} (1 - n_{\mathbf{q} \uparrow}) \right] J_{\mathbf{q}, \bar{\mathbf{q}}} G_J W_{\bar{\mathbf{q}}, \mathbf{k}_2, \mathbf{k}_1, \mathbf{q}} \\ &= -\frac{1}{2} \sum_{\mathbf{k}_1, \mathbf{k}_2} S_d^+ c_{\mathbf{k}_1 \downarrow}^\dagger c_{\mathbf{k}_2 \uparrow} \sum_{\mathbf{q} \in \text{PS}} (J_{\mathbf{q}, \bar{\mathbf{q}}} W_{\bar{\mathbf{q}}, \mathbf{k}_2, \mathbf{k}_1, \mathbf{q}} + J_{\bar{\mathbf{q}}, \mathbf{q}} W_{\mathbf{q}, \mathbf{k}_2, \mathbf{k}_1, \bar{\mathbf{q}}}) G_J. \end{aligned} \quad (3.17)$$

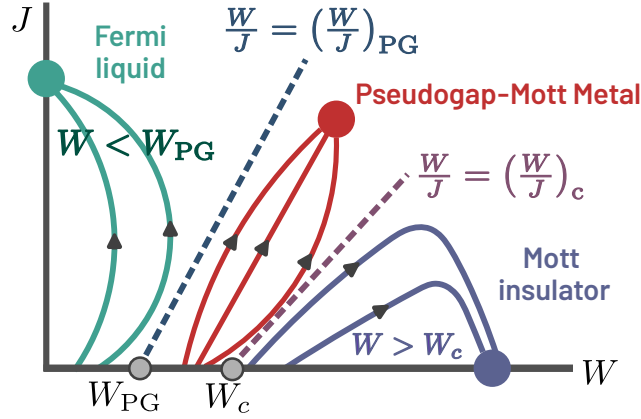


Figure 2.4: Renormalisation group (RG) flow diagram of the lattice model as obtained from a unitary RG analysis of the lattice-embedded impurity model. For $|W/J| < |W/J|_{PG}$, the flows lead to a Fermi liquid phase, while for $|W/J| < |W/J|_c$, they lead to a Mott insulator. For values in between, the system flows to a Mott metal phase that has a pseudogap in the density of states.

An identical contribution is obtained from the reversed term:

$$\begin{aligned}
 T_{F3} G_J T_{KT2}^\dagger &= \frac{1}{2} \sum_{\mathbf{k}_1, \mathbf{k}_2, \mathbf{q}} W_{\bar{\mathbf{q}}, \mathbf{k}_2, \mathbf{k}_1, \mathbf{q}} \left(c_{\bar{\mathbf{q}}\uparrow}^\dagger c_{\mathbf{k}_2\uparrow} c_{\mathbf{k}_1\downarrow}^\dagger c_{\mathbf{q}\downarrow} G_J c_{\mathbf{q}\downarrow}^\dagger c_{\bar{\mathbf{q}}\uparrow} + c_{\mathbf{q}\uparrow}^\dagger c_{\mathbf{k}_2\uparrow} c_{\mathbf{k}_1\downarrow}^\dagger c_{\bar{\mathbf{q}}\downarrow} G_J c_{\bar{\mathbf{q}}\downarrow}^\dagger c_{\mathbf{q}\uparrow} \right) J_{\mathbf{q}, \bar{\mathbf{q}}} S_d^+ \\
 &= -\frac{1}{2} \sum_{\mathbf{k}_1, \mathbf{k}_2} S_d^+ c_{\mathbf{k}_1\downarrow}^\dagger c_{\mathbf{k}_2\uparrow} \sum_{\mathbf{q} \in \text{PS}} (J_{\mathbf{q}, \bar{\mathbf{q}}} W_{\bar{\mathbf{q}}, \mathbf{k}_2, \mathbf{k}_1, \mathbf{q}} + J_{\bar{\mathbf{q}}, \mathbf{q}} W_{\mathbf{q}, \mathbf{k}_2, \mathbf{k}_1, \bar{\mathbf{q}}}) G_J.
 \end{aligned} \tag{3.18}$$

Net renormalisation to the Kondo interaction

Combining the results from eqs. 3.12, 3.13, 3.17 and 3.18, as well as using the properties $J_{\bar{\mathbf{q}}, \mathbf{k}_1} J_{\mathbf{k}_2, \bar{\mathbf{q}}} = J_{\mathbf{q}, \mathbf{k}_2} J_{\mathbf{k}_1, \mathbf{q}} = J_{\mathbf{k}_2, \mathbf{q}} J_{\mathbf{q}, \mathbf{k}_1}$ and $J_{\mathbf{q}, \bar{\mathbf{q}}} W_{\bar{\mathbf{q}}, \mathbf{k}_2, \mathbf{k}_1, \mathbf{q}} = J_{\bar{\mathbf{q}}, \mathbf{q}} W_{\mathbf{q}, \mathbf{k}_2, \mathbf{k}_1, \bar{\mathbf{q}}}$, the total renormalisation in the momentum-resolved Kondo coupling $J^{(j)}$ at the j^{th} step amounts to

$$\Delta J_{\mathbf{k}_1, \mathbf{k}_2}^{(j)} = - \sum_{\mathbf{q} \in \text{PS}} \frac{J_{\mathbf{k}_2, \mathbf{q}}^{(j)} J_{\mathbf{q}, \mathbf{k}_1}^{(j)} + 4 J_{\mathbf{q}, \bar{\mathbf{q}}}^{(j)} W_{\bar{\mathbf{q}}, \mathbf{k}_2, \mathbf{k}_1, \mathbf{q}}}{\omega - \frac{1}{2} |\varepsilon_j| + J_{\mathbf{q}}^{(j)} / 4 + W_{\mathbf{q}} / 2} \tag{3.19}$$

A representation of the RG phase diagram in the $-W/t$ vs. J/t plane of couplings has been provided in Fig.2.5(a) from detailed numerical computations of eq.(??). We also provide a schematic representation of RG flows to various fixed point theories in the J vs. W plane of couplings in Fig.2.4. This schematic figure shows the existence of three distinct phases (Fermi liquid metal, Mott Insulator, and Pseudogap-Mott metal) described by distinct fixed point theories of the RG flows. The Fermi liquid metal and Mott Insulator phases are distinguished from the Pseudogap-Mott metal by separatrices of the RG flows.

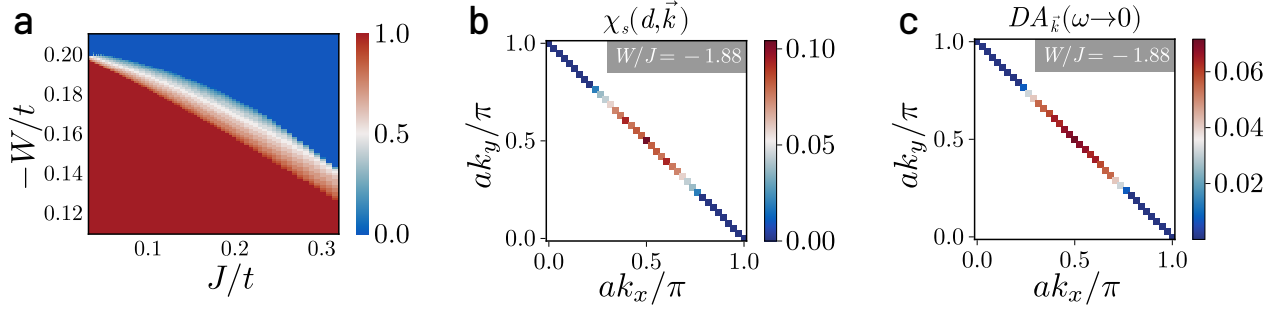


Figure 2.5: (a) Phase diagram of impurity model at strong coupling in U in terms of competing dimensionless Kondo (J/t) and bath correlation (W/t) couplings. Colourbar represents the fraction of states on the Fermi surface replaced by zeros of the Greens function (Luttinger surface). A pseudogap phase (shaded region) is observed between the local Fermi liquid (red region) and local moment (blue region) phases (b) k -space-resolved spin-spin correlation $\chi_s(d, \mathbf{k}) = \langle \mathbf{S}_d \cdot \mathbf{S}_{\mathbf{k}} \rangle$ in the pseudogap phase of the impurity model. Antinodal regions are observed to decouple from Kondo screening of the impurity. (c) Upon tiling, this leads to a k -space-resolved antinodal gap in the electronic density of states of the lattice model, corresponding to Luttinger surfaces of zeros.

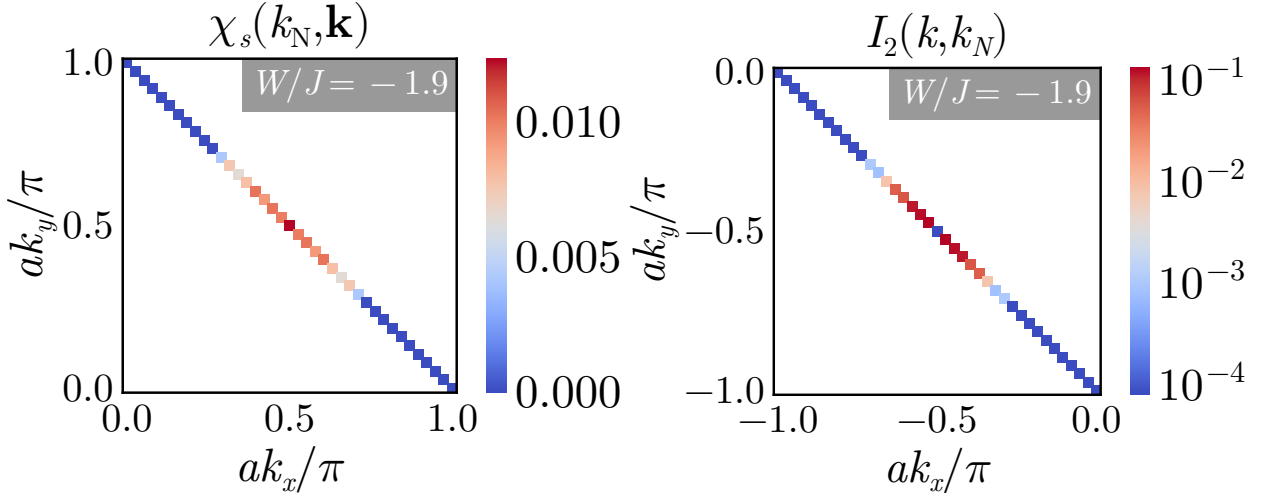


Figure 2.6: Left: Spin correlations χ_s for the tiled model, between the nodal point $k_N = (-\pi/2, -\pi/2)$ and an arbitrary k -point on the Fermi surface. In the PG, the spin-correlations near the antinode vanish, indicating that they have been removed from the metallic excitations, while the correlations near the nodal point become enhanced because of the increasingly correlated nature of the metal. Right: Mutual information I_2 between an arbitrary k -state and the nodal point. The nodal points remain strongly entangled with each other all the way through the PG.

2.4 Reconstructing of lattice model from auxiliary model

To reconstruct a translationally invariant lattice Hamiltonian from the impurity model (with Hamiltonian $\mathcal{H}_{\text{aux}}$), we define a many-body tiling prescription that embeds the auxiliary impurity system uniformly across the two-dimensional square lattice. Thus, the tiling process systematically derives

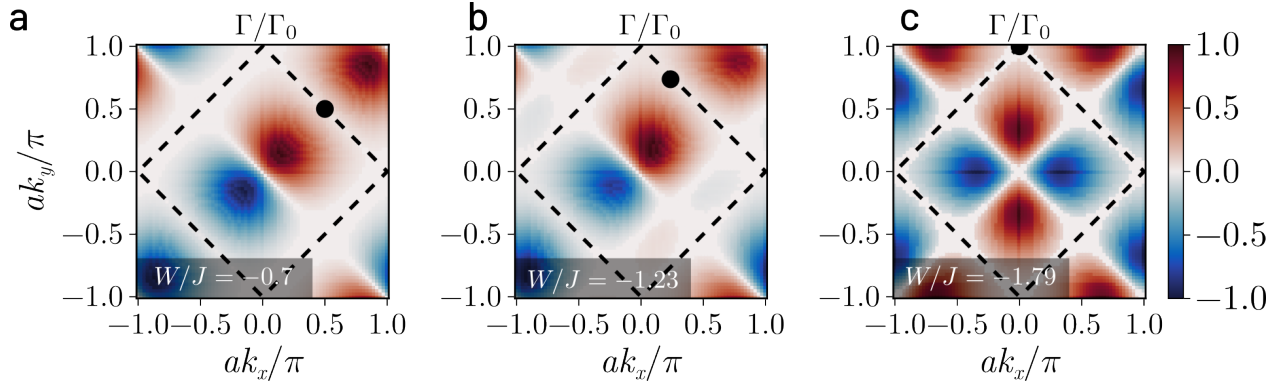


Figure 2.7: Initiation of the decoupling of $J_{\mathbf{k},\mathbf{k}'}$ (positive, negative and zeros shown in red, blue and white respectively), with $\mathbf{k}$ (black circle) at low-energies for $\mathbf{k}$ corresponding to the node (a), antinode (b) and a point mid-way between them (c) on the top right arm of the FS respectively with tuning W/J . As dictated by the symmetry of $J_{\mathbf{k},\mathbf{k}'}$, the decoupling for a given $\mathbf{k}$ proceeds via the appearance of zeros (white patches) of $J_{\mathbf{k},\mathbf{k}'}$ for $\mathbf{k}'$ initially on the nodal regions of adjacent arms, and progresses gradually towards the antinodes. The decoupling ends with the onset of the pseudogap.

the full lattice dynamics from the renormalized fixed-point structure of the impurity. As described in Section 2.2, the auxiliary model consists of a single impurity embedded in a correlated bath at the location $\mathbf{r}_d$:

$$H(\mathbf{r}_d) = H_{\text{imp}} + H_{\text{cbath}} + H_{\text{imp-cbath}}. \quad (4.1)$$

The form of various terms in eq.(4.1) have been defined in eqs.(2.2), (2.3) and (2.4) in Section 2.2.

The tiled lattice Hamiltonian is generated by symmetrically translating this impurity model:

$$\mathcal{H}_{\text{tiled}} = \mathcal{T}[H(\mathbf{r}_d)] = \sum_{\mathbf{r}} T^\dagger(\mathbf{r} - \mathbf{r}_d) H(\mathbf{r}_d) T(\mathbf{r} - \mathbf{r}_d), \quad (4.2)$$

where $T(\mathbf{a})$ are many-body translation operators [9]:

$$T^\dagger(\mathbf{a}) O(\{\mathbf{r}_i\}) T(\mathbf{a}) = O(\{\mathbf{r}_i + \mathbf{a}\}). \quad (4.3)$$

Applying this to each term yields:

$$\begin{aligned} \mathcal{T}[H_{\text{cbath}}] &= -\frac{(N-2)t}{\sqrt{\mathcal{Z}}} \sum_{\langle \mathbf{r}_i, \mathbf{r}_j \rangle; \sigma} \left(c_{\mathbf{r}_i \sigma}^\dagger c_{\mathbf{r}_j \sigma} + \text{h.c.} \right) \\ &\quad - \frac{1}{2} W \sum_{\mathbf{r}} (\hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow})^2, \\ \mathcal{T}[H_{\text{imp}}] &= -\frac{U}{2} \sum_{\mathbf{r}} (\hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow})^2, \\ \mathcal{T}[H_{\text{imp-cbath}}] &= \sum_{\langle \mathbf{r}_i, \mathbf{r}_j \rangle} \left[\frac{2J}{\mathcal{Z}} \mathbf{S}_{\mathbf{r}_i} \cdot \mathbf{S}_{\mathbf{r}_j} - \frac{2V}{\sqrt{\mathcal{Z}}} \sum_{\sigma} \left(c_{\mathbf{r}_i \sigma}^\dagger c_{\mathbf{r}_j \sigma} + \text{h.c.} \right) \right], \end{aligned} \quad (4.4)$$

with local spin operators defined as

$$\mathbf{S}_{\mathbf{r}} = \sum_{\sigma, \sigma'} c_{\mathbf{r}\sigma}^\dagger \boldsymbol{\tau}_{\sigma\sigma'} c_{\mathbf{r}\sigma'}. \quad (4.5)$$

To avoid overcounting of the non-interacting bath terms, we subtract extra copies of the kinetic energy term. The reconstructed extended-Hubbard model then reads:

$$\begin{aligned} \mathcal{H}_{\text{tiled}} = & -\frac{1}{\sqrt{\tilde{Z}}} \tilde{t} \sum_{\langle \mathbf{r}_i, \mathbf{r}_j \rangle; \sigma} \left(c_{\mathbf{r}_i \sigma}^\dagger c_{\mathbf{r}_j \sigma} + \text{h.c.} \right) \\ & + \frac{\tilde{J}}{\tilde{Z}} \sum_{\langle \mathbf{r}_i, \mathbf{r}_j \rangle} \mathbf{S}_{\mathbf{r}_i} \cdot \mathbf{S}_{\mathbf{r}_j} - \frac{1}{2} \tilde{U} \sum_{\mathbf{r}} (\hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow})^2, \end{aligned} \quad (4.6)$$

where the effective lattice couplings are given by:

$$\tilde{t} = t + 2V, \quad \tilde{U} = U + W, \quad \tilde{J} = 2J. \quad (4.7)$$

A non-zero chemical potential in the bath and/or the impurity can be included to study the effects of doping. Crucially, the eigenstates of $\mathcal{H}_{\text{tiled}}$ obey a many-body generalization of Bloch's theorem [9], which allows for the exact computation of momentum-resolved observables (presented here for a 77×77 $\mathbf{k}$ -space Brillouin zone grid). We shall demonstrate below that the lattice embedding and tiling procedures provide a controlled route to capture low-energy NFL behavior and Mott criticality directly from a quantum impurity model.

2.5 Low-energy theory for the lattice-embedded impurity model

2.5.1 RG Phase Diagram

A detailed picture of the PG in the impurity model is obtained from a momentum-resolved breakdown of Kondo screening in the strong-coupling regime $U \gg t$ phase of H_{aux} . A unitary renormalisation group (URG) scaling analysis [4] obtains a flow equation of the Kondo coupling $J_{\mathbf{k}_1, \mathbf{k}_2}^{(j)}$ (see Appendix A for details)

$$\Delta J_{\mathbf{k}_1, \mathbf{k}_2}^{(j)} = - \sum_{\mathbf{q} \in \text{PS}} \frac{J_{\mathbf{k}_2, \mathbf{q}}^{(j)} J_{\mathbf{q}, \mathbf{k}_1}^{(j)} + 4J_{\mathbf{q}, \bar{\mathbf{q}}}^{(j)} W_{\bar{\mathbf{q}}, \mathbf{k}_2, \mathbf{k}_1, \mathbf{q}}}{\omega - \frac{1}{2}|\varepsilon_j| + J_{\mathbf{q}}^{(j)}/4 + W_{\mathbf{q}}/2}, \quad (5.1)$$

where ε_j is the energy of the shell being decoupled at the j^{th} step, the sum is over all occupied momentum states $\mathbf{q}$ of the energy shell ε_j , and $\bar{\mathbf{q}} = \mathbf{q} + \boldsymbol{\pi}$ is the particle-hole transformed state associated with $\mathbf{q}$. The bath interaction coupling $W_{\bar{\mathbf{q}}, \mathbf{k}_2, \mathbf{k}_1, \mathbf{q}}$ is found to be marginal under these transformations. While we present a detailed numerical evaluation of the RG equation for $J_{\mathbf{k}_1, \mathbf{k}_2}$ (eq.(5.1)) below, it is clear that the frustration of Kondo screening due to charge fluctuations (for attractive bath interactions $W < 0$) leads to the Mott transition [60].

A detailed numerical computation of the URG equation in eq. (5.1) leads to the phase diagram shown in Fig. 2.5(a). Upon tuning the ratio of the bath and Kondo interactions (W/J) from zero to negative values, the following phases emerge in the impurity model from the competition between J and W in eq. (5.1): (i) for $W/J < (W/J)_{\text{PG}}$, an LFL phase (red region), where the entire FS participates in Kondo screening, (ii) for $\frac{W}{J} \in [(\frac{W}{J})_{\text{PG}}, (\frac{W}{J})_c]$ (shaded region), a local PG phase where disconnected parts of the FS around the node participate in Kondo screening, and (iii) a local moment phase for $\frac{W}{J} > (\frac{W}{J})_c$ (blue region), where the impurity remains unscreened at low-energies. These can be visualised from spin correlations, $\chi_s(d, \mathbf{k}) = \langle \mathbf{S}_d \cdot \mathbf{S}_{\mathbf{k}} \rangle$, as shown in Fig. 2.5(b) for the PG. The values $(W/J)_{\text{PG}}$ and $(W/J)_c$ are therefore the entry into and exit from the PG phase.

Mapping onto the lattice model via tiling, we observe that RG-induced nodal–antinodal dichotomy in $J_{\mathbf{k}, \mathbf{k}'}$ in the impurity model is the microscopic origin of the PG in the lattice model: the extinction of Kondo coherence translates into spectral zeros in the lattice Green’s function (i.e., Luttinger surfaces) at antinodal momenta (Fig. 2.5(c)). This establishes that the $T = 0$ Mott transition of the 2D *extended*-Hubbard model proceeds from FL to MI through an intervening PG phase. In Fig. 2.6, similar insights obtain from the spin-correlation $\chi_s(\mathbf{k}, k_N) = \langle \mathbf{S}_{\mathbf{k}} \cdot \mathbf{S}_{k_N} \rangle$ in k -space (where k_N is the nodal momentum) and mutual information $I_2(\mathbf{k} : k_N)$ between two momentum states: the PG displays Fermi arcs in the nodal regions along with vanishing correlation and entanglement in the antinodal regions.

2.5.2 Unravelling of Kondo Screening

The $\mathbf{k}$ -space anisotropy of Kondo breakdown can be visualized in terms of zeros of $J_{\mathbf{k}_N, \mathbf{k}}$, involving spin-flip scattering between the node $\mathbf{k}_N = (\pi/2, \pi/2)$ and a general wavevector $\mathbf{k}$. For any W/J , the C_4 lattice symmetry dictates that $J_{\mathbf{k}_N, \mathbf{k}}$ vanishes if $\mathbf{k}$ belongs to any of the antinodes or adjacent nodes. Tuning W/J towards $(W/J)_{\text{PG}}$ leads to an unravelling of the Kondo screening: $J_{\mathbf{k}_N, \mathbf{k}}$ for $\mathbf{k}$ close to the adjacent nodes turns RG-irrelevant first, and a patch of zeros subsequently appears in $J_{\mathbf{k}_N, \mathbf{k}}$ around this point (Fig. 2.7(a)). Tuning W/J further extends the patch of zeros towards the antinodes (Fig. 2.7(b) and 2.7(c)). Kondo screening thus unravels by a systematic decoupling of all $J_{\mathbf{k}_1, \mathbf{k}_2}$ that connect adjacent quadrants of the Brillouin zone.

Precisely at $W/J = (W/J)_{\text{PG}}$, the antinode joins this connected region of zeros in $J_{\mathbf{k}_1, \mathbf{k}_2}$, marking the decoupling of the antinodes from all other points in the neighbourhood of the FS. This is an interaction-driven Lifshitz transition of the FS, and marks the entry into a PG phase possessing Fermi arcs [61]. Importantly, it coincides with an emergent two-channel Kondo (2CK) impurity model, where each channel corresponds to a pair of Fermi arcs on opposite faces of the conduction bath FS (see Subection 2.5.3). The 2CK nature of the PG is guaranteed by the symmetry of $J_{\mathbf{k}, \mathbf{k}'}$:

$$J_{\mathbf{k}, \mathbf{k}'} = -J_{\mathbf{k}+\mathbf{Q}, \mathbf{k}'} = -J_{\mathbf{k}, \mathbf{k}'+\mathbf{Q}}, \quad \mathbf{Q} = (\pi, \pi). \quad (5.2)$$

The PG expands by shrinking these disconnected Fermi arcs towards the respective nodes, leading to nodal metals whose disappearance heralds the Mott transition.

2.5.3 Emergence of a two-channel Kondo Model

As the bath interaction W is tuned through the L-PG phase, the four nodal points in the Brillouin zone are the last to decouple from the impurity. This allows us to write down a simpler Kondo model near the transition, focusing on scattering processes involving the nodal region. Specifically, for the Kondo term $J_{\mathbf{k}_1, \mathbf{k}_2}$, we find that only the following classes of scattering processes survive close to the critical point:

- Both momenta in a neighbourhood around $\mathbf{k}_N$, where $\mathbf{k}_N$ can be any of the four nodal points,
- One momentum from the neighbourhood around $\mathbf{k}_N$, while the other from the neighbourhood around $\mathbf{k}_N + (\pi, \pi)$.

The second class is tied to the first through a symmetry of the Hamiltonian (eq. ??). The crucial feature of these processes is that each node only interacts with the other node directly opposite to it across the origin. That is, $(\pi/2, \pi/2)$ and $(-\pi/2, -\pi/2)$ only interact between themselves, while $(\pi/2, -\pi/2)$ and $(-\pi/2, \pi/2)$ form their own pair of connected regions.

Let $S^\pm$ be the set of momentum states in a small window around nodes along $k_x = \pm k_y$. As discussed immediately above, the Kondo spin-exchange term close to the transition does not lead to scattering of any k -state from S^+ to S^- . This is verified by calculating the ratio $\max \{J_{k_1^+, k_2^-}\} / \max \{J_{k_1^+, k_2^+}\}$, where the maximum in the numerator is calculated from all the low-energy (fixed point) Kondo scattering processes that connect the two sets $S^\pm$, while the denominator maximum is from within the set S^+ . This ratio vanishes (left panel of Fig. 2.10) within the local pseudogap regime, indicating that no direct Kondo scattering process persists between the two sectors $S^\pm$ at low energies, deep within the pseudogap regime. We call the value of W at which this ratio vanishes W^* .

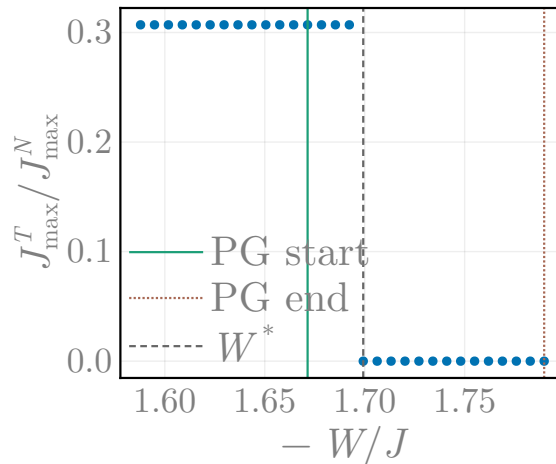


Figure 2.8: Ratio between the maximum values of the fixed point values of the Kondo couplings for scattering processes between and within the regions S^+ and S^- . The vanishing of the fixed point ratio indicates that the pseudogap regime features a singular low-energy effective theory, where the Fermi surface neighbourhood splits into two disconnected regions with no explicit term in the Hamiltonian connecting them directly.

The decoupling of the k -space regions $S^\pm$ is captured by the following simplified low-energy Hamiltonian:

$$H_{\text{eff}} = H(S^+) + H(S^-); H(S) = \sum_{\mathbf{q} \in S, \sigma} \varepsilon_{\mathbf{q}} c_{\mathbf{q}\sigma}^\dagger c_{\mathbf{q}\sigma} + \sum_{\mathbf{q}_1 \in S, \mathbf{q}_2 \in S} \sum_{\alpha, \beta} J^*(\mathbf{q}_1, \mathbf{q}_2) \mathbf{S}_d \cdot \sigma_{\alpha\beta} c_{\mathbf{q}_1\alpha}^\dagger c_{\mathbf{q}_2\beta}, \quad (5.3)$$

where $H(S^\pm)$ represents the dynamics of momentum states residing in regions $S^\pm$. The only source of information transfer between the two regions is the fact that they interact with the same impurity local moment. Eq. 5.3 tells us that the low-energy dynamics of the embedded eSIAM close to the Kondo-breakdown transition is governed by an emergent **two-channel Kondo model**, with the conduction bath states in the sets $S^\pm$ making up the two channels.

2.5.4 Momentum-resolved Dynamical Spectral Weight Transfer

Passage through the PG phase is accompanied by a highly structured transfer of spectral weight across the FS. Strong charge fluctuations develop between the nodal and antinodal regions of the FS in the PG regime of the impurity model (Fig. 2.9(a)), as captured by the correlator:

$$\chi_c(\mathbf{k}_1, \mathbf{k}_2) = \left\langle c_{\mathbf{k}_1\uparrow}^\dagger c_{\mathbf{k}_1\downarrow}^\dagger c_{\mathbf{k}_2\downarrow} c_{\mathbf{k}_2\uparrow} + \text{h.c.} \right\rangle. \quad (5.4)$$

These fluctuations dynamically redistribute low-energy spectral weight from the antinodes to higher energies, leading to selective gap formation. Accordingly, the Luttinger surfaces of the PG [12] coincides with the appearance of poles of the lattice model self-energy $\Sigma(\mathbf{k}, \omega = 0)$ near the antinodes; these poles approach the nodes on tuning towards the Mott transition (Fig. 2.9(b)). This mirrors the coalescing of finite-frequency poles of the self-energy poles towards zero frequency in the underlying impurity model (Fig. 2.9(c)).

2.6 Non-Fermi liquid excitations within the Pseudogap

2.6.1 Pseudogapped spectral function and singular self-energy

In the PG regime, the nature of gapless Fermi arcs changes dramatically. We have already argued that the low-energy dynamics of these gapless Fermi arcs are governed by an underlying two-channel Kondo (2CK) impurity model [62, 63]. This is consistent with the rapid fall of the impurity quasi-particle residue Z_{imp} (Fig. 2.10(a)) from finite values in the FL phase to vanishingly small values just before the onset of the PG. Inset shows the simultaneous emergence of increasingly uncompensated local magnetic moments upon traversing the PG phase. The accompanying impurity spectral function of the gapless arcs show a pseudogapped behaviour for $\omega \simeq 0$, with a rapid fall in the spectral weight at $\omega = 0$ upon traversing the PG (Fig. 2.10(c)). The collapse of the Kondo resonance into a pseudogapped spectral function is accompanied by the redistribution of spectral weight [12] in the impurity spectral function (Fig. 2.10(b)) from $\omega \sim 0$ to the Hubbard sidebands at finite frequencies $\omega \simeq \pm 3$ (in units of the bandwidth).

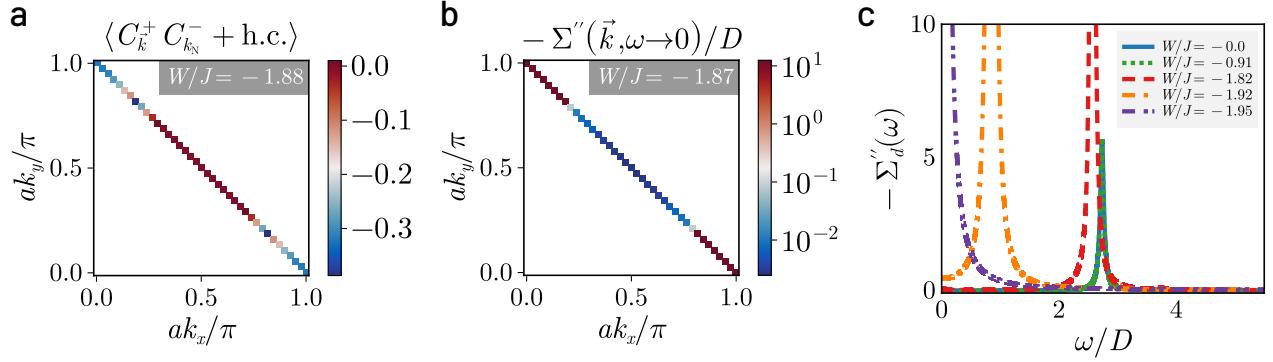


Figure 2.9: (a) Enhanced charge correlations $\chi_c(\mathbf{k}_1, \mathbf{k}_2) = \langle c_{\mathbf{k}_1\uparrow}^\dagger c_{\mathbf{k}_1\downarrow}^\dagger c_{\mathbf{k}_2\downarrow} c_{\mathbf{k}_2\uparrow} + \text{h.c.} \rangle$ between the nodal and antinodal regions, signalling Kondo breakdown in the pseudogap phase of the impurity model. (b) In turn, the breakdown leads to the gapping of the antinodal regions in the lattice model, seen from the appearance of poles in the imaginary part of the self-energy. (c) The imaginary part of the impurity self-energy $\Sigma''(\omega > 0)$ possesses a pole at non-zero ω in the PG that moves towards $\omega = 0$ as the Mott transition is approached.

Concomitant with this is the emergence of a zero-frequency peak in the imaginary part of the self-energy of the NFL in the PG phase(Fig. 2.11(a)),

$$-1/\Sigma''(\omega) \sim 1/\Sigma''(0) + \omega^\beta. \quad (6.1)$$

Remarkably, we find that the exponent $\beta = 2$ characterises the NFL for the entire PG phase (see Fig. 2.11(b)), including the critical end-point. This is in stark contrast with the $\Sigma''(\omega) \sim \omega^2$ for the FL (Fig. 2.11(b)). A similar analysis reveals the power-law scaling for the low-frequency behaviour of the impurity spectral function in Fig.2.12. We observe that $A_d(\omega) \sim A_d(0) + \omega^2$ for the pseudogap phase, while $1/A_d(\omega) \sim 1/A_d(0) + \omega^2$ for the Fermi liquid (inset of Fig. 2.12).

2.6.2 Finite-Frequency Scattering Rate and Optical Response Through the Pseudogap

All $\Sigma''(\omega)$ for the NFL are observed to lie above the Mott-Ioffe-Regel (MIR) bound [64,65] (dashed blue line in Fig. 2.11(a)), while those for the FL lie below. The MIR bound is the maximum expected scattering rate in metals when the mean free path approaches the lattice spacing:

$$-2\Sigma''_{\text{MIR}} = 1/\tau_{\text{MIR}}, \quad \tau_{\text{MIR}} = l_{\text{min}}/v_F, \quad (6.2)$$

where l_{min} is the minimum mean free path in metals (equal to one lattice spacing) and τ_{MIR} is the associated lifetime of quasiparticles close to the FS (with Fermi velocity v_F). The height of the zero-frequency peak rises by almost 4 orders of magnitude from the start of the PG till its end at the Mott transition point (Fig. 2.11(c)); the dramatic growth of the peak height very near the Mott critical point coincides with the coalescing of the finite-frequency poles of the self-energy into a single pole at zero-frequency, signalling the singular nodal NFL present at the Mott quantum critical point.

To further probe the breakdown of coherent transport in the pseudogap (PG) regime, we examine the product of the impurity spectral function $A(\omega)$ and the scattering lifetime $\tau(\omega)$, which approximates the optical conductivity $\sigma(\omega)$ via a Drude-type relation. As shown in the main panel of Fig. ??, this quantity exhibits a sharp Drude peak at $\omega = 0$ in the Fermi liquid (FL) regime (blue), consistent with long-lived quasiparticles. In stark contrast, the PG regime (green, red, orange) shows a pronounced suppression of low-frequency conductivity and the emergence of a finite-frequency peak centered near $\omega/D \sim 0.3\text{--}0.4$ (where D is the kinetic energy bandwidth).

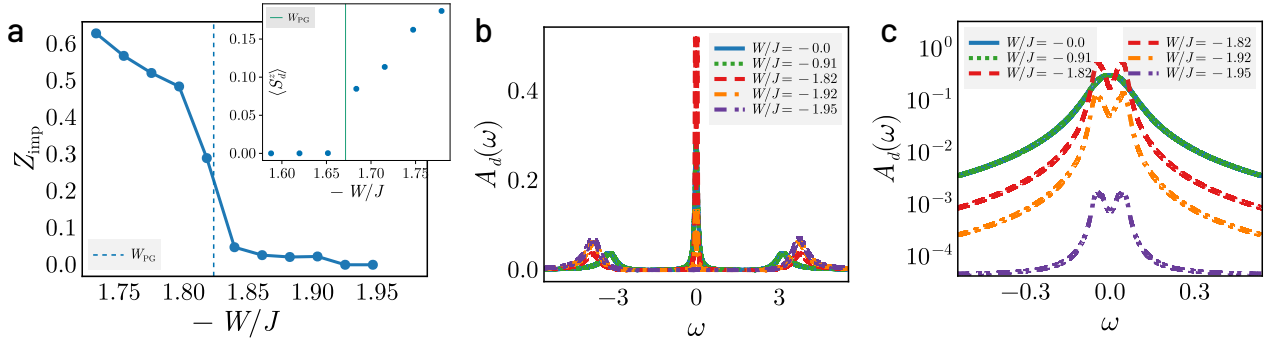


Figure 2.10: (a) Suppression of quasiparticle residue (Z_{imp}) as the impurity model is tuned towards the Mott transition. An initial drastic fall in Z_{imp} is observed for $W/J \lesssim (W/J)_{\text{PG}}$ from 0.3 to around 0.05, signalling the destruction of the FL with unravelling of Kondo screening. A steady decrease in Z_{imp} is observed in passage through the PG, and is vanishingly small close to the Mott transition due to a divergent self-energy. Inset shows the growth of unscreened impurity magnetic moment in the pseudogap phase, signalling the breakdown of Kondo screening. (b) Impurity Spectral function in the FL and PG phases. The evolution of the central peak from Kondo resonance to pseudogap is accompanied by dynamical spectral weight transfer to the Hubbard side bands at finite frequencies $\omega \approx \pm 3$ (in units of the bandwidth). (c) Zoom-in of the central (Kondo) resonance of the impurity spectral function. While undisturbed in the Fermi liquid phase, it splits into a pseudogap in the PG phase, with its height diminishing rapidly as the Mott transition is approached.

Strikingly, this peak shifts very little across the PG regime (see inset), despite a deepening spectral gap and diverging self-energy. Its position lies well below the Mott gap ($\sim 2.5D$), suggesting that it represents an emergent mid-infrared (MIR) excitation intrinsic to the Mott metal characterising the PG. This feature closely mirrors the experimentally observed MIR peak in high- T_c cuprates [66–68], which has been variously attributed to excitations of electrons bound to vacancies [69], or the dressing of holes by spin excitations [70]. Here, its robust energy scale and insensitivity to increasing incoherence imply a universality persisting through the PG, likely rooted in short-range entanglement between deconfined doublons and holons excitations of the Mott metal. This is further evidence that the PG metal is not simply a degraded Fermi liquid, but a distinct phase whose optical response is driven by emergent degrees of freedom.

2.6.3 Non-local correlations in the pseudogap

In Fig. 2.13(a) and 2.13(b), the spin-flip correlations and mutual information between the impurity spin and conduction bath sites respectively are observed to undergo a crossover within the PG, from a short-ranged behaviour at its onset, to a long-ranged behaviour as the Mott transition approaches. The entanglement is also observed to be multipartite in nature: in Fig. 2.13(c), the quantum Fisher information (QFI) [71] computed for the ground state wavefunction using an operator corresponding to the sum of local spin-flip exchange processes shows a jump at the onset of the PG. Further, the FL is observed to possess bi-partite entanglement while the NFL of the PG phase displays pentapartite entanglement [?, ?].

These striking results imply that the Mott transition observed by us lies beyond the local quantum criticality scenario [72]. Instead, we observe the PG phase to be a novel state of strongly interacting quantum matter emergent from the breakdown of local Kondo screening. This state is described by a quantum critical Fermi surface [73] with NFL Fermi arcs that display increasingly critical behaviour, i.e., dynamics described by non-local quantum fluctuations, and excitations that become truly long-ranged close to the transition. It is tempting to conjecture that the long-range entangled critical state captured in Fig. 2.13(b) reflects the existence of a highly efficient scrambler [74–76], i.e., a state in which entanglement spreads among its constituents at the fastest rate permissible by quantum mechanics.

2.7 Exactly solvable nodal Non-Fermi liquid at Mott Criticality

Very close to the transition, the excitations of the nodal NFL correspond to those of a Hatsugai-Kohmoto model [53, 54]. This insight is obtained from a perturbation-theoretic treatment of the RG fixed point Hamiltonian of the impurity model for $W/J \lesssim (W/J)_{PG}$, by considering the effects of a small fixed point Kondo scattering probability J^* in the backdrop of a larger bath interaction parameter $|W|$. This yields the HK model [53, 54] as the singular part of the effective Hamiltonian arising from forward scattering processes.

As the bath interaction W is tuned through the L-PG phase, the nodal region is the last to decouple from the impurity. This allows us to write down a simpler Kondo model near the transition, where only the nodal region is hybridising with the impurity spin through Kondo interactions. This is done by retaining only those scattering processes $\mathbf{k}_1 \rightarrow \mathbf{k}_2$ that originate from and end at k -points within a small neighborhood of width $|\mathbf{q}|$ around the four nodal points: $\mathbf{k}_1, \mathbf{k}_2 \in \mathbf{N} + \mathbf{q}$, $|\mathbf{q}| \ll \pi$, where $\mathbf{N}$ can be any one of the four nodal points $\mathbf{N}_1 = (\pi/2, \pi/2)$, $\mathbf{N}_2 = (-\pi/2, \pi/2)$ and $\mathbf{N}_1 + \mathbf{Q}_1$ and $\mathbf{N}_2 + \mathbf{Q}_2$, where $\mathbf{Q}_1 = (-\pi, -\pi)$ and $\mathbf{Q}_2 = (\pi, -\pi)$ are the two nesting vectors. We have already seen, in Subsection 2.5.3, that the pseudogap regime displays a decoupling of the Brillouin zone into sets of points that have no explicit connecting term in the Hamiltonian. Near the transition, the neighbourhoods of $\mathbf{N}_1$ and $\mathbf{N}_1 + \mathbf{Q}_1$ form one such connected set, while the other two nodes $\mathbf{N}_2$ and $\mathbf{N}_2 + \mathbf{Q}_2$ form another connected set, but these two sets are disconnected from each other, in the sense of a two-channel model. We also assume that the window of $\mathbf{q}$ is small enough so that the fixed point Kondo coupling values $J^*(\mathbf{q}_1, \mathbf{q}_2)$ for the scattering processes involving $\mathbf{q}_1$ and $\mathbf{q}_2$ can be replaced by an average value J^* .

With these considerations, the simplified low-energy model near the transition describing the Kondo scattering processes can be written as

$$\begin{aligned} \tilde{H}_{\text{imp-cbath}} = J^* \frac{1}{2} \sum_{l=1,2} \sum_{\mathbf{q}_1, \mathbf{q}_2} \sum_{\alpha, \beta} \mathbf{S}_d \cdot \boldsymbol{\sigma}_{\alpha\beta} & \left(c_{\mathbf{N}_l+\mathbf{q}_1, \alpha}^\dagger c_{\mathbf{N}_l+\mathbf{q}_2, \beta} + c_{\mathbf{N}_l+\mathbf{Q}_l+\mathbf{q}_1, \alpha}^\dagger c_{\mathbf{N}_l+\mathbf{Q}_l+\mathbf{q}_2, \beta} \right. \\ & \left. - c_{\mathbf{N}_l+\mathbf{Q}_l+\mathbf{q}_1, \alpha}^\dagger c_{\mathbf{N}_l+\mathbf{q}_2, \beta} - c_{\mathbf{N}_l+\mathbf{q}_1, \alpha}^\dagger c_{\mathbf{N}_l+\mathbf{Q}_l+\mathbf{q}_2, \beta} \right) . \end{aligned} \quad (7.1)$$

For scattering processes that involve two opposite nodal points (that is, connected by one of the nesting vectors $\mathbf{Q}_l$, the Kondo coupling is $-J^*$, which explains the negative signs in the third and fourth terms. An explanation of the various dummy indices is in order. The label l can take values 1 or 2, allowing us to consider both the decoupled sets (associated with $\mathbf{N}_1$ and $\mathbf{N}_2$). It also labels the nesting vectors $\mathbf{Q}_l$ associated with the two sets. α and β indicate spin indices, and $\mathbf{q}_1$ and $\mathbf{q}_2$ represent incoming and outgoing momenta in the scattering processes.

Because of the decoupling of the channel $l = 1$ and $l = 2$, we consider only the $l = 1$ channel for the rest of the calculations in this section, and we comment at the end on what the presence of two channels means for the results. In order to simplify the Hamiltonian, we define new fermionic operators

$$\psi_{\mathbf{q}, \sigma} = \frac{1}{\sqrt{2}} (c_{\mathbf{N}_1+\mathbf{q}, \sigma} + c_{\mathbf{N}_1+\mathbf{Q}_1-\mathbf{q}, \sigma}) , \quad \phi_{\mathbf{q}, \sigma} = \frac{1}{\sqrt{2}} (c_{\mathbf{N}_1+\mathbf{q}, \sigma} - c_{\mathbf{N}_1+\mathbf{Q}_1-\mathbf{q}, \sigma}) , \quad (7.2)$$

The operator satisfy fermionic anticommutation relations:

$$\begin{aligned} \left\{ \psi_{\mathbf{q}, \sigma}, \psi_{\mathbf{q}', \sigma'}^\dagger \right\} &= \left[\delta_{\mathbf{q}, \mathbf{q}'} + \frac{1}{2} (\delta_{\mathbf{Q}_1-\mathbf{q}, \mathbf{q}'} + \delta_{\mathbf{q}, \mathbf{Q}_1-\mathbf{q}'}) \right] \delta_{\sigma, \sigma'} \\ &= \delta_{\mathbf{q}, \mathbf{q}'} \delta_{\sigma, \sigma'} , \\ \left\{ \phi_{\mathbf{q}, \sigma}, \phi_{\mathbf{q}', \sigma'}^\dagger \right\} &= \left[\delta_{\mathbf{q}, \mathbf{q}'} - \frac{1}{2} (\delta_{\mathbf{Q}_1-\mathbf{q}, \mathbf{q}'} + \delta_{\mathbf{q}, \mathbf{Q}_1-\mathbf{q}'}) \right] \delta_{\sigma, \sigma'} \\ &= \delta_{\mathbf{q}, \mathbf{q}'} \delta_{\sigma, \sigma'} , \\ \left\{ \psi_{\mathbf{q}, \sigma}, \phi_{\mathbf{q}', \sigma'}^\dagger \right\} &= \left\{ \phi_{\mathbf{q}, \sigma}, \psi_{\mathbf{q}', \sigma'}^\dagger \right\} = 0 , \end{aligned} \quad (7.3)$$

where both $\mathbf{q}$ and $\mathbf{q}'$ are chosen from a small window around the nodes, so their difference is also small and it is therefore guaranteed that $|\mathbf{q} + \mathbf{q}'| \neq |\mathbf{Q}_1|$, rendering both $\delta_{\mathbf{q}, \mathbf{Q}_1-\mathbf{q}'}$ and $\delta_{\mathbf{Q}_1-\mathbf{q}, \mathbf{q}'}$ zero. For convenience, we define new number operators for the sum and relative degrees of freedom:

$$s_{\mathbf{q}, \sigma} = \psi_{\mathbf{q}, \sigma}^\dagger \psi_{\mathbf{q}, \sigma} , \quad r_{\mathbf{q}, \sigma} = \phi_{\mathbf{q}, \sigma}^\dagger \phi_{\mathbf{q}, \sigma} . \quad (7.4)$$

We then have the following useful relation between the corresponding number operators $n = c^\dagger c$:

$$n_{\mathbf{N}_1+\mathbf{q}, \sigma} + n_{\mathbf{N}_1+\mathbf{Q}_1-\mathbf{q}, \sigma} = s_{\mathbf{q}, \sigma} + r_{\mathbf{q}, \sigma} . \quad (7.5)$$

We find that the sum degrees of freedom $s_{\mathbf{q}, \sigma}$ do not enter the Kondo scattering Hamiltonian $\tilde{H}_{\text{imp-cbath}}$ (for the single channel $l = 1$), and it is expressed purely in terms of the relative degrees of freedom:

$$\tilde{H}_{\text{imp-cbath}} = J^* \sum_{\mathbf{q}_1, \mathbf{q}_2, \alpha, \beta} \mathbf{S}_d \cdot \boldsymbol{\sigma}_{\alpha\beta} \phi_{\mathbf{q}_1 \alpha}^\dagger \phi_{\mathbf{q}_2 \beta} . \quad (7.6)$$

We now proceed to rewrite the other parts of the Hamiltonian in terms of these new degrees of freedom. The kinetic energy can be written as

$$\begin{aligned}\tilde{H}_{\text{cbath}} &= \sum_{\mathbf{q},\sigma} \varepsilon_{\mathbf{N}_1+\mathbf{q}} (n_{\mathbf{N}_1+\mathbf{q},\sigma} + n_{\mathbf{N}_1+\mathbf{Q}_1-\mathbf{q},\sigma}) \\ &= \sum_{\mathbf{q},\sigma} \varepsilon_{\mathbf{N}_1+\mathbf{q}} \left(\psi_{\mathbf{q},\sigma}^\dagger \phi_{\mathbf{q},\sigma} + \phi_{\mathbf{q},\sigma}^\dagger \psi_{\mathbf{q},\sigma} \right),\end{aligned}\quad (7.7)$$

We also consider a simplified form of the bath interaction (for the channel $l = 1$), taking into account the density-density correlations in k -space:

$$\begin{aligned}\tilde{H}_{\text{cbath-int}} &= -W \sum_{\mathbf{q},\sigma} (n_{\mathbf{N}_1+\mathbf{q},\sigma} + n_{\mathbf{N}_1+\mathbf{Q}_1-\mathbf{q},\sigma}) + W \sum_{\mathbf{q}_1,\mathbf{q}_2} (n_{\mathbf{N}_1+\mathbf{q}_1,\uparrow} + n_{\mathbf{N}_1+\mathbf{Q}_1-\mathbf{q}_1,\uparrow}) (n_{\mathbf{N}_1+\mathbf{q}_2,\downarrow} + n_{\mathbf{N}_1+\mathbf{Q}_1-\mathbf{q}_2,\downarrow}) \\ &= -W \sum_{\mathbf{q},\sigma} (s_{\mathbf{q},\sigma} + r_{\mathbf{q},\sigma}) + W \sum_{\mathbf{q}_1,\mathbf{q}_2} (s_{\mathbf{q}_1,\uparrow} + r_{\mathbf{q}_1,\uparrow}) (s_{\mathbf{q}_2,\downarrow} + r_{\mathbf{q}_2,\downarrow}).\end{aligned}\quad (7.8)$$

We used eq. 7.5 in order to obtain the final form.

Since both $\tilde{H}_{\text{imp-cbath}}$ and $\tilde{H}_{\text{cbath-int}}$ commute with $s_{\mathbf{q},\sigma}$, we replace the local objects $\sum_{\mathbf{q}} s_{\mathbf{q},\sigma}$ in $\tilde{H}_{\text{cbath-int}}$ with $\frac{1}{2}$ since the bath is always at half-filling. This results in a simplified Hamiltonian for just the $\phi_{\mathbf{q},\sigma}$ degrees of freedom:

$$\tilde{H} = -\frac{1}{2}W \sum_{\mathbf{q},\sigma} r_{\mathbf{q},\sigma} + W \sum_{\mathbf{q}_1,\mathbf{q}_2} r_{\mathbf{q}_1,\uparrow} r_{\mathbf{q}_2,\downarrow} + \sum_{\mathbf{q}_1,\mathbf{q}_2,\alpha,\beta} J^* \mathbf{S}_d \cdot \boldsymbol{\sigma}_{\alpha\beta} \phi_{\mathbf{q}_1\alpha}^\dagger \phi_{\mathbf{q}_2\beta} \quad (7.9)$$

For bath interaction strength close to the critical value ($W \lesssim W_c$), the fixed point coupling value J^* is much smaller than W . In order to obtain the gapless excitations of the system arising from the presence of the impurity site, we integrate out the impurity dynamics via a Schrieffer-Wolff transformation. The perturbation term $\mathcal{V}$ then consists of Hamiltonian terms that modify the impurity configuration,

$$\mathcal{V} = \sum_{\mathbf{q}_1,\mathbf{q}_2} J^* S_d^+ \phi_{\mathbf{q}_1\downarrow}^\dagger \phi_{\mathbf{q}_2\uparrow} + \text{h.c.}, \quad (7.10)$$

while the “non-interacting” Hamiltonian is

$$H_D = -\frac{1}{2}W \sum_{\mathbf{q},\sigma} r_{\mathbf{q},\sigma} + W \sum_{\mathbf{q}_1,\mathbf{q}_2} r_{\mathbf{q}_1,\uparrow} r_{\mathbf{q}_2,\downarrow}. \quad (7.11)$$

The renormalisation in the dynamics of $b_{k,\sigma}^\dagger$ are given by

$$\Delta \tilde{H} = \mathcal{P}_L \mathcal{V} \mathcal{P}_H \frac{1}{E_L - H_D} \mathcal{V} \mathcal{P}_L, \quad (7.12)$$

where E_L is the energy of the ground state of H_D , $-(E_L - H_D)$ is the excitation energy of the state generated upon applying the perturbation $\mathcal{V}$, and $\mathcal{P}_{L(H)}$ projects onto the low(high)-energy subspace. For the present Hamiltonian, the low-energy state (associated with E_L) is the one that

minimises the bath interaction term $\tilde{H}_{\text{cbath-int}}$; since the coupling W is negative, the minimal configuration is one in which the local operators $\sum_{\mathbf{q}} r_{\mathbf{q},\sigma}$ (for $\sigma = \pm 1$) are unity. This leads to the following low-energy state:

$$|L\rangle = \frac{1}{N} \sum_{\mathbf{q}_1, \mathbf{q}_2} \phi_{\mathbf{q}_1, \uparrow}^\dagger \phi_{\mathbf{q}_2, \downarrow}^\dagger |0\rangle \quad (7.13)$$

where $|0\rangle$ is the $r_{\mathbf{q}\uparrow} = r_{\mathbf{q}\downarrow} = 0$ state. The state $|L\rangle$ is at zero energy. High-energy states are obtained by applying, on the state $|L\rangle$, the excitation operator $\phi_{\mathbf{q}_1, \sigma}^\dagger \phi_{\mathbf{q}_2, \bar{\sigma}}$ or its hermitian conjugate. The operator converts a double and a hole to spin states, incurring a total excitation cost of $-W = |W|$.

We now calculate the various terms in $\Delta\tilde{H}$:

$$\begin{aligned} \mathcal{P}_L \mathcal{V} \mathcal{P}_H \frac{1}{E_L - H_D} \mathcal{V} \mathcal{P}_L &= \left(\frac{1}{2} + S_d^z \right) \sum_{\mathbf{q}_1, \mathbf{q}_2} \frac{J^{*2}}{-4W} \phi_{\mathbf{q}_1, \downarrow}^\dagger \phi_{\mathbf{q}_2, \uparrow} \left(\phi_{\mathbf{q}_2, \uparrow}^\dagger \phi_{\mathbf{q}_1, \downarrow} + (1 - \delta_{\mathbf{q}_1, \mathbf{q}_2}) \phi_{\mathbf{q}_1, \uparrow}^\dagger \phi_{\mathbf{q}_2, \downarrow} \right) \\ &= - \sum_{\mathbf{q}_1, \mathbf{q}_2} \frac{J^{*2}}{4W} \left[r_{\mathbf{q}_1 \downarrow} (1 - r_{\mathbf{q}_2 \uparrow}) - (1 - \delta_{\mathbf{q}_1, \mathbf{q}_2}) \phi_{\mathbf{q}_1, \uparrow}^\dagger \phi_{\mathbf{q}_1, \downarrow}^\dagger \phi_{\mathbf{q}_2, \downarrow} \phi_{\mathbf{q}_2, \uparrow} \right] \\ \mathcal{P}_L \mathcal{V} \mathcal{P}_H \frac{1}{E_L - H_D} \mathcal{V} \mathcal{P}_L &= \sum_{\mathbf{q}_1, \mathbf{q}_2} \frac{J^{*2}}{-4W} \left[r_{\mathbf{q}_1 \uparrow} (1 - r_{\mathbf{q}_2 \downarrow}) - (1 - \delta_{\mathbf{q}_1, \mathbf{q}_2}) \phi_{\mathbf{q}_1, \downarrow}^\dagger \phi_{\mathbf{q}_1, \uparrow}^\dagger \phi_{\mathbf{q}_2, \uparrow} \phi_{\mathbf{q}_2, \downarrow} \right] \end{aligned} \quad (7.14)$$

In both terms, we set $S_d^z = 0$ owing to local SU(2) symmetry on the impurity site. In the same way, we treat the effects of the dispersion term (eq. 7.7) to second order:

$$\begin{aligned} \mathcal{P}_L \mathcal{V} \mathcal{P}_H \frac{1}{E_L - H_D} \mathcal{V} \mathcal{P}_L &= \sum_{\mathbf{q}, \sigma} \frac{\varepsilon_{\mathbf{N}_1 + \mathbf{q}}^2}{-W} \left[\psi_{\mathbf{q}, \sigma}^\dagger \phi_{\mathbf{q}, \sigma} \phi_{\mathbf{q}, \sigma}^\dagger \psi_{\mathbf{q}, \sigma} + \phi_{\mathbf{q}, \sigma}^\dagger \psi_{\mathbf{q}, \sigma} \psi_{\mathbf{q}, \sigma}^\dagger \phi_{\mathbf{q}, \sigma} \right] \\ &= \sum_{\mathbf{q}, \sigma} \frac{\varepsilon_{\mathbf{N}_1 + \mathbf{q}}^2}{-W} \left[s_{\mathbf{q}, \sigma} (1 - r_{\mathbf{q}, \sigma}) + r_{\mathbf{q}, \sigma} (1 - s_{\mathbf{q}, \sigma}) \right]. \end{aligned} \quad (7.15)$$

At the values of $\mathbf{q}$ for which $\varepsilon_{\mathbf{N}_1 + \mathbf{q}} > 0$, we can set the number operator $s_{\mathbf{q}, \sigma} = 0$ (these positive energy modes are unoccupied in the ground state), while for $\varepsilon_{\mathbf{N}_1 + \mathbf{q}} < 0$, we have $s_{\mathbf{q}, \sigma} = 1$:

$$\mathcal{P}_L \mathcal{V} \mathcal{P}_H \frac{1}{E_L - H_D} \mathcal{V} \mathcal{P}_L = \sum_{\mathbf{q}(\varepsilon_{\mathbf{N}_1 + \mathbf{q}} > 0), \sigma} \frac{\varepsilon_{\mathbf{N}_1 + \mathbf{q}}^2}{-W} r_{\mathbf{q}, \sigma} + \sum_{\mathbf{q}(\varepsilon_{\mathbf{N}_1 + \mathbf{q}} < 0), \sigma} \frac{\varepsilon_{\mathbf{N}_1 + \mathbf{q}}^2}{-W} (1 - r_{\mathbf{q}, \sigma}). \quad (7.16)$$

The complete second-order renormalised Hamiltonian is

$$\Delta\tilde{H} = \sum_{\mathbf{q}, \sigma} \epsilon_{\mathbf{q}} r_{\mathbf{q}, \sigma} + \mathcal{U} \sum_{\mathbf{q}, \sigma} r_{\mathbf{q}, \sigma} r_{\mathbf{q}, \bar{\sigma}} + \mathcal{U} \sum_{\mathbf{q}_1 \neq \mathbf{q}_2, \sigma} \left[r_{\mathbf{q}_1, \sigma} r_{\mathbf{q}_2, \bar{\sigma}} + \phi_{\mathbf{q}_1, \bar{\sigma}}^\dagger \phi_{\mathbf{q}_1, \sigma}^\dagger \phi_{\mathbf{q}_2, \sigma} \phi_{\mathbf{q}_2, \bar{\sigma}} \right]. \quad (7.17)$$

where $\epsilon_{\mathbf{q}} = \text{sign}(\varepsilon_{\mathbf{N}_1 + \mathbf{q}}) \frac{\varepsilon_{\mathbf{N}_1 + \mathbf{q}}^2}{-W}$ and $\mathcal{U} = \frac{J^{*2}}{4W}$.

The $\mathbf{q}_1 = \mathbf{q}_2$ component of the Hamiltonian shows the emergence of the exactly solvable Hatsugai-Kohmoto model [53, 54] at the critical point. The correlation term $\mathcal{U} \sum_{\mathbf{q}, \sigma} r_{\mathbf{q}, \sigma} r_{\mathbf{q}, \bar{\sigma}}$ leads

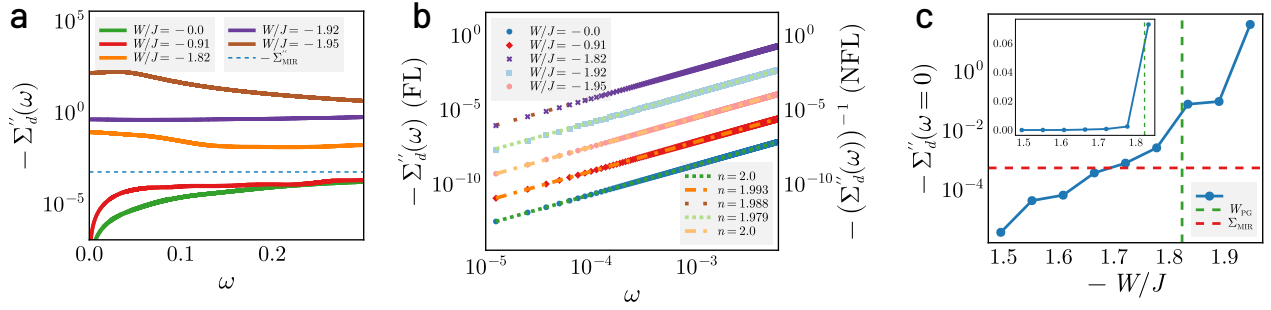


Figure 2.11: (a) Imaginary part of impurity self-energy $\Sigma''(\omega)$ as a function of frequency ω for Fermi liquid (FL, $|W/J| < 1.79$) and pseudogapped phases (NFL, $|W/J| \geq 1.79$). $\Sigma''(\omega)$ falls to zero as $\omega \rightarrow 0$ for the FL, while it attains a peak in the pseudogap. All $\Sigma''(\omega)$ for the NFL are observed to lie above the Mott-Ioffe-Regel (MIR) bound (dashed blue line), while those for the FL lie below. (b) Scaling of $\Sigma''(\omega)$ with frequency for Fermi liquid and pseudogapped phases. The FL self-energy fits to $\Sigma'' \sim \omega^\alpha$ with $\alpha \approx 2$, vanishing as $\omega \rightarrow 0$, while the NFL self-energy grows as $\Sigma'' \approx a - \omega^\beta$ for small ω , with $\beta \approx 2$. Remarkably, the NFL exponent remains mostly unchanged through the entirety of the PG phase. (c) Variation of the zero-frequency imaginary self-energy $-\Sigma''(\omega = 0)$ with ω in the FL and pseudogap phases (entry into the PG is marked by the vertical dashed line). The inset shows the same but in linear scale, in order to display the dramatic rise (by almost 30 times) on entering the PG.

to a transfer of spectral weight across the Fermi surface and separates the available states into three classes:

$$\langle n_{\mathbf{q}} \rangle = \begin{cases} 2, & \epsilon_{\mathbf{q}} < -|\mathcal{U}/2|, \\ 1, & |\mathcal{U}/2| > \epsilon_{\mathbf{q}} > -|\mathcal{U}/2|, \\ 0, & |\mathcal{U}/2| < \epsilon_{\mathbf{q}}. \end{cases} \quad (7.18)$$

In contrast to a Fermi liquid, there is a highly degenerate single-occupied region in the middle, and this gives rise to non-Fermi liquid excitations.

We now discuss the effects of the $\mathbf{q}_1 \neq \mathbf{q}_2$ component. The first term partially lifts the degeneracy of the central singly-occupied region and allows only zero magnetisation configurations. The second term creates gapped excitations involving the regions of zero and double occupancy; these represent subdominant pairing fluctuations of the nodal non-Fermi liquid. We recall that similar fluctuations across the Fermi surface were observed in our calculations of charge correlations in Fig. 2.9 (a), and are consistent with pairing fluctuations observed from quantum Monte Carlo simulations and found to give rise to quantum-critical behavior and PG physics [36].

Consequently, the resulting NFL metal of the lattice model involves long-lived excitations of multiple $\mathbf{k}$ -states, and manifest in the form of a divergent one-particle self-energy at the non-interacting FS [77]: $\Sigma_{\mathbf{q}}(\omega) = -\mathcal{U}^2/4(\omega - \epsilon_{\mathbf{q}})$, such that $\Sigma_{\epsilon_{\mathbf{q}}=0}(\omega \rightarrow 0) \rightarrow \infty$. This zero-frequency self-energy pole presages the transition into a Mott insulating phase, where it marks a hard gap in the spectral function for charge excitations. This is consistent with our findings for the lattice (Fig. 2.9(c)) and impurity self-energies (Fig. 2.11(c)). The nodal Mott metal thus comprises of a Greens function zero at the Fermi energy together with an anisotropic massless Dirac dispersion, leading to a non-

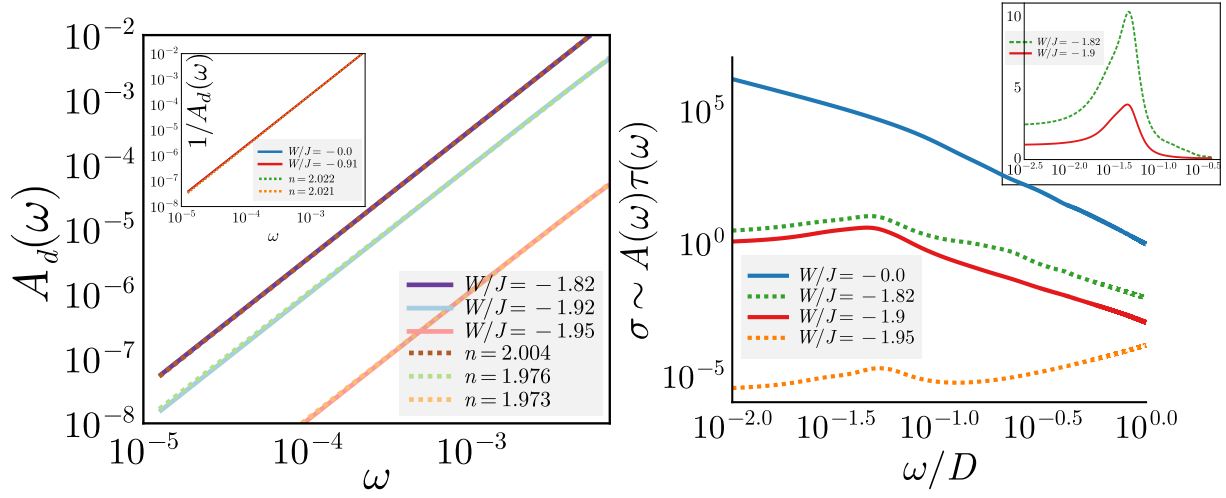


Figure 2.12: Left: Impurity spectral function $A_d(\omega)$ in the local Fermi liquid (inset) and pseudogap (main). The PG spectral function is fit to a power law (with the $\omega = 0$ contribution subtracted, see Fig. 2.11(c) for the full spectral function). The exponent of the power law comes out to be $n \approx 2$ throughout the PG. The local FL spectral function (inset) is fit to a lorentzian (by fitting the inverse to a power law). Right: Optical conductivity $\sigma(\omega)$ for excitations of the impurity model, approximated through the product of the carrier density (taken from the density of states $A(\omega)$) and the lifetime $\tau(\omega)$ (obtained from the inverse scattering rate $-1/\Sigma''(\omega)$, where Σ'' is the imaginary part of the self-energy). The Fermi liquid (blue) shows a sharp Drude peak at $\omega = 0$ due to the presence of long-lived quasiparticles. The PG (green, red, orange) shows a shifted Drude peak at a non-zero frequency which is reminiscent of the mid-infrared (MIR) peak seen experimentally in the cuprates [66–68]. Inset shows the robustness of the position of the MIR peak through the PG, indicating it as a universal feature of the Mott metal.

zero Chern number [78–80]. This topological index survives into the Mott insulator.

For small but non-zero values of $\omega - \epsilon_{\mathbf{k}}$, we obtain a quasiparticle residue that vanishes with ω , $Z_{\text{imp}} \sim \omega^2/\mathcal{U}^2$. The scattering rate of this singular NFL possesses a sharp peak at the FS ($\omega = \epsilon_{\mathbf{k}_F}$):

$$\Gamma \sim \mathcal{U}^2 \delta(\omega - \epsilon_{\mathbf{k}}), \quad (7.19)$$

consistent with the sharply peaked Lorentzian $\Sigma''_{\mathbf{k}}(\omega) \sim \omega^{-2}$ captured in Fig. 2.11(a). This quantum critical NFL metal is an example of a strongly coupled scale-invariant form of quantum matter. The exact solution for eq.(?) reveals the presence of low-energy excitations comprised of holons and doublons [53]. These features point to the nodal NFL as a long-ranged and multipartite entangled, strongly interacting scale-invariant state of quantum matter [11, 81–83] that are completely disconnected from the quasiparticles of the FL. Following the arguments laid out in [14], at finite temperatures, such a scale-invariant highly entangled non-Fermi liquid is likely associated with Planckian dissipation (i.e., $\Sigma''(T) \sim k_B T$) and a resistivity that varies linearly with temperature ($\rho \sim T$) [?, 84, 85]. This sits well with the possibility that this state is also a highly efficient scrambler [74–76]. Additionally, as mentioned above, we observe that the nodal metal possesses pairing fluctuations that can become dominant upon doping [77].

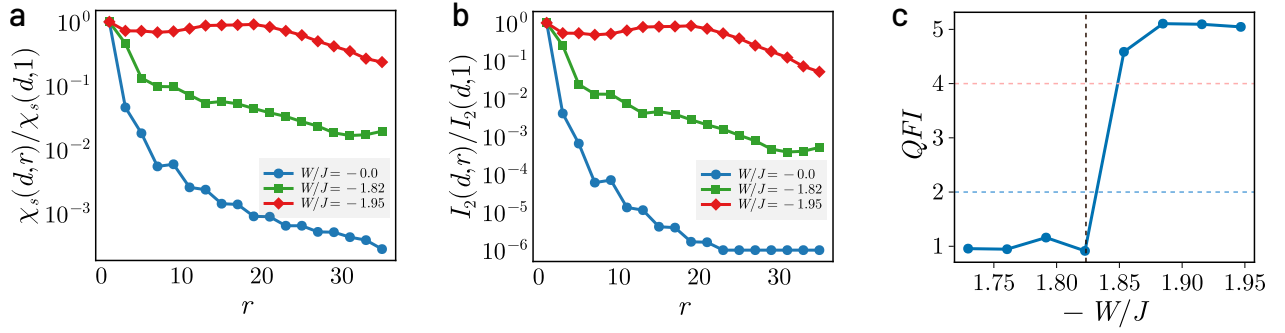


Figure 2.13: (a) Spin correlation $\langle \mathbf{S}_d \cdot \mathbf{S}_r \rangle$ and (b) mutual information $I_2(d, r)$ between the impurity spin and conduction bath local spin density as a function of the distance $\mathbf{r}$ between them, normalised against the value at $r = 1$. Both decay very quickly in the FL phase (blue), but show long-ranged behaviour in the NFL phase (green and red), extending to the edges of the system at the critical point (red). (c) Evolution of the Quantum Fisher Information F_Q for a nearest-neighbour spin-flip operator $O = \sum_{i \in \text{odd}} (S_i^+ S_{i+1}^- + \text{h.c.})$ through the first Lifshitz transition and the pseudogap. The vertical dashed line marks the onset of the PG. To the left of it, the QFI in the Fermi liquid phase shows at most bipartite entanglement ($F_Q < 2$ (below blue dashed line)), while the PG shows the presence of multipartite entanglement upto 5 parties ($F_Q > 4$ (red dashed line)).

2.8 Heisenberg Model as a low-energy description of the tiled Mott insulator

For simplicity, we consider two impurity sites labelled 1 and 2 associated with two nearest-neighbour auxiliary models. The ground state subspace is four-fold degenerate:

$$|\Psi_L\rangle = \{|\sigma_1, \sigma_2\rangle\}, \quad \sigma_i = \pm 1, \quad (8.1)$$

where σ_i is the spin state of site i . This ground state is derived from the following "zeroth order" Hamiltonian that emerges in the local moment phase of the auxiliary models when all scattering processes between the impurity and conduction bath are RG-irrelevant:

$$H_0 = -\frac{U}{2} \sum_{i=1,2} (n_{i\uparrow} - n_{i\downarrow})^2; \quad (8.2)$$

the local correlation on the impurity site becomes the largest scale in the problem in this phase and pushes the $|n_i = 2\rangle$ and $|n_i = 0\rangle$ states to high energies. This then defines the high-energy subspace for our calculation:

$$|\Psi_H\rangle = |C_1, C_2\rangle, \quad (8.3)$$

where C_i can take values 0 or 2, indicating that the state i is either empty or full, respectively. Both the double and hole states exist at a charge gap of the order of $U/2$ above the low-energy singly-occupied subspace defined by the states $|\Psi_L\rangle$.

In order to allow virtual fluctuations that can lift the large ground state degeneracy and lower the energy, we consider (perturbatively) the effects of an irrelevant single-particle hybridisation that

connects the nearest-neighbour sites. This perturbation Hamiltonian is therefore of the form

$$H_t = \sum_{\omega} V(\omega) \mathcal{P}(\omega) \sum_{\sigma} \left(c_{1\sigma}^{\dagger} c_{2\sigma} + \text{h.c.} \right), \quad (8.4)$$

where $V(\omega)$ only acts on states at the energy scale ω ; the renormalisation of V is encoded in the fact that $V(\omega)$ is largest for the excited states and vanishes at low-energies: $V(\omega \rightarrow 0) = 0$.

In order to obtain a low-energy effective Hamiltonian for the impurity sites arising from this hybridisation, we integrate out H_t via a Schrieffer-Wolff transformation. This leads to the following second-order Hamiltonian:

$$H_{\text{eff}} = \mathcal{P}_L H_t G \mathcal{H}_t \mathcal{P}_L. \quad (8.5)$$

The operator $\mathcal{P}_L$ projects onto the low-energy subspace $|\Psi_L\rangle$ - this ensures that we remain in the low-energy subspace at the beginning and at the end of the total process. The Greens function $G = (E_L - H_0)^{-1}$ incorporates the excitation energy to go from the low-energy subspace $|\Psi\rangle_L$ (of energy E_L) to the excited subspace $|\Psi\rangle_H$ of energy $E_L + U/2$. Substituting the form of the perturbation Hamiltonian and the excitation energy into the above expression gives

$$H_{\text{eff}} = \frac{V_H^2}{-U/2} \sum_{\sigma, \sigma'} \left[c_{1\sigma}^{\dagger} c_{2\sigma} c_{2\sigma'}^{\dagger} c_{1\sigma'} + c_{2\sigma}^{\dagger} c_{1\sigma} c_{1\sigma'}^{\dagger} c_{2\sigma'} \right]. \quad (8.6)$$

where $V_H \equiv V(\omega \rightarrow U/2)$ is the impurity-bath hybridisation at energy scales of the order of the Mott gap, in the sense of an RG flow. Terms with consecutive creation or annihilation operators on the same site are prohibited because each site is singly-occupied in the ground state. It is now easy to cast this Hamiltonian into a more recognizable form. For $\sigma' = \sigma$, we get

$$\sum_{\sigma} \delta_{\sigma, \sigma'} c_{1\sigma}^{\dagger} c_{2\sigma} c_{2\sigma'}^{\dagger} c_{1\sigma'} = \sum_{\sigma} (n_{1\sigma} - n_{1\sigma} n_{2\sigma}), \quad (8.7)$$

while $\sigma = -\sigma' = \pm 1$ gives

$$\sum_{\sigma} \delta_{\sigma, -\sigma'} c_{1\sigma}^{\dagger} c_{2\sigma} c_{2\sigma'}^{\dagger} c_{1\sigma'} = - (S_1^+ S_2^- + \text{h.c.}). \quad (8.8)$$

For the latter expression, we introduced the local spin-flip operators $S_i^{\pm}$. The expression above it can also be cast into spin variables, using the equations

$$\begin{aligned} \frac{1}{2} \sum_{\sigma} n_{i\sigma} &= \frac{1}{2}, \\ \frac{1}{2} \sum_{\sigma} \sigma n_{i\sigma} &= S_i^z, \end{aligned} \quad (8.9)$$

where the first equation is simply the condition of half-filling at each site, and the second equation is the definition of the local spin operator in z -direction. Adding and subtracting the equations gives $n_{i\sigma} = \frac{1}{2} + \sigma S_i^z$.

Substituting everything back into eq. 8.6 and dropping constant terms gives

$$H_{\text{eff}} = 2 \frac{V_H^2}{U/2} (2S_1^z S_2^z + S_1^+ S_2^- + S_1^- S_2^+) = J_{\text{eff}} \mathbf{S}_1 \cdot \mathbf{S}_2, \quad (8.10)$$

where the effective antiferromagnetic Heisenberg coupling is $J_{\text{eff}} = \frac{8V_H^2}{U}$.

2.9 Effective Hamiltonian for excitons in the Mott insulator

The Mott insulating phase is characterised by a large onsite correlation U that forbids long-lived coherent charge propagation. This suggests a natural separation of the total Hamiltonian into a large zeroth order part arising from the atomic limit of the Hubbard model, followed by a “small” perturbative part due to irrelevant hopping processes:

$$H_0 = -\frac{U}{2} \sum_i (n_{i\uparrow} - n_{i\downarrow})^2, \quad H_1 = -t \sum_{\langle i,j \rangle, \sigma} (c_{i,\sigma}^\dagger c_{j,\sigma} + \text{h.c.}) . \quad (9.1)$$

The zeroth order ground state is of course half-filled, each site being singly-occupied:

$$|\Psi_0\rangle = \otimes_i |\sigma_i\rangle , \quad (9.2)$$

where σ_i can be ± 1 and indicates that each site hosts a local moment that can be either up or down, but doubles or holes are prohibited. This also reflects the large degeneracy ($\sim 2^N$) associated with such a state.

We first note that a non-zero hopping probability t indicates that the ground state cannot be just $|\Psi_0\rangle$, simply because $|\Psi_0\rangle$ is not an eigenstate of H_1 . In order to consider hopping processes that start from the half-filled ground state, we define a projector $P_H = \otimes_i P_H(i) = \otimes_i (n_{i\uparrow} + n_{i\downarrow} - 2n_{i\uparrow}n_{i\downarrow})$ that annihilates the hole or double state at any site. The only allowed processes are therefore

$$H_{\text{eff}} = H_1 P_H = -t \sum_{\langle i,j \rangle, \sigma} (c_{i,\sigma}^\dagger P_H(i) c_{j,\sigma} P_H(j) + \text{h.c.}) . \quad (9.3)$$

The action of the projector on any local operator is easy to calculate:

$$\begin{aligned} c_{j,\sigma} P_H(j) &= c_{j\sigma} (n_{j\sigma} + n_{j\bar{\sigma}} - 2n_{j\sigma}n_{j\bar{\sigma}}) = c_{j\sigma} (1 - n_{j\bar{\sigma}}) , \\ c_{i,\sigma}^\dagger P_H(i) &= c_{i,\sigma}^\dagger (n_{j\sigma} + n_{j\bar{\sigma}} - 2n_{j\sigma}n_{j\bar{\sigma}}) = c_{i,\sigma}^\dagger n_{i\bar{\sigma}} , \end{aligned} \quad (9.4)$$

The effective Hamiltonian for such hopping processes from the correlated ground state is therefore

$$H_{\text{eff}} = -t \sum_{\langle i,j \rangle, \sigma} (c_{i,\sigma}^\dagger c_{j\sigma} n_{i\bar{\sigma}} (1 - n_{j\bar{\sigma}}) + \text{h.c.}) . \quad (9.5)$$

In order to identify the propagating degrees of freedom, we define an exciton creation operator $b^\dagger$:

$$b_{i,\delta}^\dagger = B \sum_\sigma c_{i\sigma}^\dagger c_{i+\delta,\sigma} n_{i\bar{\sigma}} (1 - n_{i+\delta,\bar{\sigma}}) , \quad (9.6)$$

where δ is a primitive lattice vector. [CHECK THE ALGEBRA.] This operator creates an exciton: a bound holon-doublon pair between the sites i and $i + \delta$. In terms of these operators, the Hamiltonian becomes

$$H_{\text{eff}} = -tB \sum_{i,\delta} (b_{i,\delta}^\dagger + \text{h.c.}) . \quad (9.7)$$

2.10 Pseudogap as a strongly coupled phase of quantum matter

We now unveil an organising principle that leads to the remarkable properties observed above for the strongly interacting NFL of the PG phase. The existence of a sharp connected FS at $T = 0$ can be understood as the existence of a topologically protected manifold of gapless chiral excitations in $\mathbf{k}$ -space at the FS [56]. The FS is characterised by a topological index corresponding to an anomaly in the quantum many-body theory for electrons, and can be understood as a generalised symmetry of such a system [57, 86]. A theorem by Luttinger and Ward [87] shows that a count of the physical charge (known as Luttinger's volume) is identical to the topological index (a so-called homotopy charge known as Luttinger's count) even in the presence of electronic interactions that do not disturb the FS. We will now argue that the emergence of antinodal Luttinger surfaces involve a disconnection of the FS (into Fermi arcs) and that, by following La Nave et al. [57], the accompanying change in its topological properties leads to the existence of gapless NFL excitations that are non-local in nature.

The antinodal Luttinger surfaces arise from the splitting of double poles of the single-particle Greens function on the FS into poles lying on opposite complex half-planes, together with zeros that are pinned at the FS. These changes in the analytic structure of the single-particle Greens function have important consequences. First, the emergent zeros break a $\mathbb{Z}_2$ symmetry of the FS [49, 50, 57]. This symmetry is guaranteed within FL theory because the presence of quasiparticles vanishingly close to the FS allows the interchange of spin and charge degrees of freedom, promoting the separate $SU(2)$ symmetries of spin and charge to the larger symmetry of $O(4) = SU(2) \times SU(2) \times \mathbb{Z}_2$ [49]. The insertion of zeros of the Greens function at the Fermi surface in the PG, and the associated splitting of the Greens function, breaks this $\mathbb{Z}_2$ symmetry by placing the pole for the spin excitation in one half of the complex plane while placing that for the charge excitation in the other half [50, 88].

Second, they signal a divergent electronic self-energy as a function of the wavevector $\mathbf{k}$, render ill-behaved the Luttinger-Ward functional of the interacting electronic problem, and violate the generalised symmetry encoded within it. The changes in the pole structure change the Luttinger count topological invariant, while the zeros give rise to an additional topological contribution (linked to the Adler-Bell-Jackiw-type chiral anomaly) [55, 80, 89, 90]. As a consequence of the half-filled particle-hole symmetric nature of the system at hand, Luttinger's volume is preserved upon taking into account topological contributions from both the Luttinger count *and* the zeros [48]. Importantly, La Nave et al. [57] argue, following recent developments in understanding generalised symmetries [91], that the additional anomaly arising from the Luttinger surfaces guarantees the existence of gapless NFL excitations that are non-local in nature.

Thus, the drastic change in nature of the real-space excitations - from locally well-defined Landau quasiparticles of the FL to the increasingly nonlocal excitations of the NFL Fermi arcs in the PG phase - appears to be dictated by a topological principle and, therefore, robust under renormalisation. This signals the NFL Fermi arcs of the PG as an emergent phase of strongly interacting quantum matter - which we dub the *Mott metal* - that is the parent metal of the MI. Such an identification of the pseudogap as a distinct phase is backed up by angle-resolved photoemission spectroscopy (ARPES) experiments [17]; the anomalous momentum, temperature and doping dependence of the gap function observed in the cuprates suggests that the pseudogap maybe described by an order parameter distinct from that of the superconductor [92, 93]. The same topological principle connects the nonlocal unparticle-like gapless excitations [11, 81–83] of the scale-invariant nodal NFL of the

HK model observed precisely at the Mott critical point to those of the rest of the PG phase, e.g., a universal scaling of $\Sigma''_{\mathbf{k}}(\omega) \sim (a + \omega^2)^{-1}$ (Fig. 2.11(b)) and local density of states $A_d(\omega) \sim \omega^2$ (Fig. 2.12) of the NFL throughout the PG phase, such that the value of $\Sigma''_{\mathbf{k}}(\omega = 0)$ value continues to grow upon violating the MIR bound (Fig. 2.11(c)).

2.11 Conclusions

Our analysis reveals that the Mott transition proceeds via a continuous evolution through a PG regime characterized by a singular NFL metal - the Mott metal - with deconfined holon–doublon excitations confined to nodal Fermi arcs, and a scaling of its spectral features with universal power-law exponents. As the system approaches criticality, this metallic phase exhibits increasingly non-local correlations and a divergent self-energy, signalling the breakdown of Landau quasiparticles, formation of local moments and the onset of long-range quantum entanglement. Anchored in two-channel Kondo dynamics at intermediate scales and governed by Hatsugai–Kohmoto physics near the critical endpoint, the Mott metal provides a unified framework for understanding anomalous metallicity in strongly correlated systems. Its fate under finite doping presents a compelling direction for future investigation. More complicated models can be studied by suitably extending the tiling approach and choosing an appropriate auxiliary model. Electron-phonon coupling can be taken into account by studying an Anderson-Holstein model that couples the impurity site to local lattice modes [94]. The effects of disorder can be studied by averaging quantities over various disorder configurations of the conduction bath. The versatility of the unitary RG solver and the impurity model-based construction makes it convenient to explore a large variety of lattice models.

2.12 Appendix: Effective action for the Hubbard-Heisenberg model and the eSIAM

In order to better understand the similarities and differences between the tiling approach outlined here and the approach adopted in dynamical mean-field approximations, we compute the effective action for the local impurity as well as the impurity-zeroth site system, in both the auxiliary and bulk models. This will be done in the limit of infinite coordination number, in order to create a tractable effective theory.

2.12.1 Local theory for the Hubbard-Heisenberg model

We will first work with the Hubbard-Heisenberg model. Let us recall that within the eSIAM, most of the dynamics is governed by the physics of the impurity site and the bath site coupled to the impurity. Accordingly, we will first obtain an effective action within the Hubbard-Heisenberg model for the small system consisting of a local site and all its nearest-neighbours.

We choose a certain local site that we call 0, whose nearest-neighbours will be denoted as $\{\bar{0}\}$. The action for the full Hubbard-Heisenberg model can be formally separated into three parts:

$$S_{\text{H-H}} = S_{0,\bar{0}} + S^{(0,\bar{0})} + S_{\text{int}} , \quad (12.1)$$

where $S_{0,\bar{0}}$ represents the part of the action that involves only the sites 0 and $\{\bar{0}\}$, $S^{(0,\bar{0})}$ represents the part that has all the sites apart from 0 and $\{\bar{0}\}$, while S_{int} contains all terms that connect these two parts. These three parts have the forms

$$S_{0,\bar{0}} = \sum_{i=0,\bar{0}} \sum_{\sigma} \int_0^{\beta} d\tau c_{i\sigma}^{\dagger}(\tau) \partial_{\tau} c_{i\sigma}(\tau) - t \sum_{\bar{0},\sigma} \int_0^{\beta} d\tau \left[c_{0\sigma}^{\dagger}(\tau) c_{\bar{0}\sigma}(\tau) + \text{h.c.} \right] \\ - \frac{U}{2} \sum_{i=0,\bar{0}} \int_0^{\beta} d\tau (n_{i\uparrow}(\tau) - n_{i\downarrow}(\tau))^2 + J \sum_{\bar{0}} \int_0^{\beta} d\tau \vec{S}_0(\tau) \cdot \vec{S}_{\bar{0}}(\tau), \quad (12.2)$$

$$S^{(0,\bar{0})} = \sum_{j \neq (0,\bar{0})} \sum_{\sigma} \int_0^{\beta} d\tau c_{j\sigma}^{\dagger}(\tau) \partial_{\tau} c_{j\sigma}(\tau) - t \sum_{\langle j,l \rangle \neq (0,\bar{0}),\sigma} \int_0^{\beta} d\tau \left[c_{j\sigma}^{\dagger}(\tau) c_{l\sigma}(\tau) + \text{h.c.} \right] \\ - \frac{U}{2} \sum_{j \neq (0,\bar{0})} \int_0^{\beta} d\tau (n_{j\uparrow}(\tau) - n_{j\downarrow}(\tau))^2 + J \sum_{\langle j,l \rangle \neq (0,\bar{0})} \int_0^{\beta} d\tau \vec{S}_l(\tau) \cdot \vec{S}_j(\tau), \quad (12.3)$$

$$S_{\text{int}} = -t \sum_{i \in \bar{0}} \sum_{j \in \text{NN of } i} \sum_{\sigma} \int_0^{\beta} d\tau \left(c_{i\sigma}^{\dagger}(\tau) c_{j\sigma}(\tau) + \text{h.c.} \right) + J \sum_{i \in \bar{0}} \sum_{j \in \text{NN of } i} \int_0^{\beta} d\tau \vec{S}_i(\tau) \cdot \vec{S}_j(\tau) \\ = S_{\text{int}}^{\text{hop}} + S_{\text{int}}^{\text{spin}} \quad (12.4)$$

In order to obtain an effective theory $S_{\text{eff}}^{0,\bar{0}}$ purely for the local system $(0, \bar{0})$, we need to trace over all the other degrees of freedom. This partial trace will be carried out over the states of the so-called ‘‘cavity’’ system $S^{(0,\bar{0})}$. The effective action can be constructed by tracing over scattering processes of all orders that leave the final action diagonal in the system $(0, \bar{0})$. The formal expression can be written as

$$S_{\text{eff}}^{0,\bar{0}} = S_{0,\bar{0}} + \sum_{n=1}^{\infty} \langle (S_{\text{int}}^{\text{hop}})^{2n} \rangle_{(0,\bar{0})} + \sum_{n=1}^{\infty} \langle (S_{\text{int}}^{\text{spin}})^{2n} \rangle_{(0,\bar{0})} \quad (12.5)$$

where, as mentioned before, the average is carried out in the cavity system $S^{(0,\bar{0})}$. Each hopping term in the expansion is of the form

$$\left(S_{\text{int}}^{\text{hop}} \right)^{2n} \\ = t^{2n} \sum_{\sigma} \sum_{(i_1, i'_1), \dots, (i_n, i'_n)} \sum_{(j_1, j'_1), \dots, (j_n, j'_n)} \int_0^{\beta} d\tau_1 \dots d\tau_n d\tau'_1 \dots d\tau'_{j_n} \\ c_{i_1\sigma}^{\dagger}(\tau_1) c_{i_2\sigma}^{\dagger}(\tau_2) \dots c_{i_n\sigma}^{\dagger}(\tau_n) c_{i'_1\sigma}(\tau'_1) c_{i'_2\sigma}(\tau'_2) \dots c_{i'_n\sigma}(\tau'_n) \left\langle c_{j_1\sigma}(\tau_1) c_{j_2\sigma}(\tau_2) \dots c_{j_n\sigma} c_{j'_1\sigma}^{\dagger}(\tau'_1) \dots c_{j'_n\sigma}^{\dagger}(\tau'_n) \right\rangle_{(0,\bar{0})} \\ = t^{2n} \sum_{\sigma} \sum_{(i_1, i'_1), \dots, (i_n, i'_n)} \sum_{(j_1, j'_1), \dots, (j_n, j'_n)} \int_0^{\beta} d\tau_1 \dots d\tau'_{j_n} c_{i_1\sigma}^{\dagger}(\tau_1) \dots c_{i_n\sigma}^{\dagger}(\tau_n) c_{i'_1\sigma}(\tau'_1) \dots c_{i'_n\sigma}(\tau'_n) \\ G_n^{(0,\bar{0})}(\tau_1 \dots \tau_n; \tau'_1 \dots \tau'_n), \quad (12.6)$$

where the indices i_1 through i_n and their primed counterparts run through $\bar{0}$, while $j_l (\in \{j_1, \dots, j'_n\})$ runs through the nearest-neighbours of the corresponding i_l index. The n -particle Greens function $G_n^{(0,\bar{0})}(\tau_1 \dots \tau_n; \tau'_1 \dots \tau'_n)$ for the cavity system is defined in the usual fashion

$$G_n^{(0,\bar{0})}(\tau_1 \dots \tau_n; \tau'_1 \dots \tau'_n) = \left\langle c_{j_1\sigma}(\tau_1) c_{j_2\sigma}(\tau_2) \dots c_{j_n\sigma} c_{j'_1\sigma}^\dagger(\tau'_1) \dots c_{j'_n\sigma}^\dagger(\tau'_n) \right\rangle_{(0,\bar{0})} \quad (12.7)$$

In the limit of infinite coordination number, only the $n = 1$ term survives:

$$\begin{aligned} \lim_{Z \rightarrow \infty} \sum_{n=1}^{\infty} \langle (S_{\text{int}}^{\text{hop}})^{2n} \rangle_{(0,\bar{0})} &= t^2 \sum_{\sigma} \sum_{i,i'} \int_0^{\beta} d\tau d\tau' c_{i\sigma}^\dagger(\tau) c_{i'\sigma}(\tau') \sum_{j,j'} G_1^{(0,\bar{0})}(j\tau; j'\tau') \\ &= \sum_{\sigma} \sum_{i,i'} \int_0^{\beta} d\tau d\tau' \Delta_{ii',\sigma}(\tau - \tau') c_{i\sigma}^\dagger(\tau) c_{i'\sigma}(\tau'), \end{aligned} \quad (12.8)$$

where we have defined the bath hybridisation function $\Delta_{ii',\sigma}(\tau - \tau')$ as

$$\Delta_{ii',\sigma}(\tau - \tau') = t^2 \sum_{j \in \text{NN of } i} \sum_{j' \in \text{NN of } i'} G_1^{(0,\bar{0})}(j\sigma\tau; j'\sigma\tau'). \quad (12.9)$$

Under similar approximations, the spin term in the action reduces to

$$\lim_{Z \rightarrow \infty} \sum_{n=1}^{\infty} \langle (S_{\text{int}}^{\text{spin}})^{2n} \rangle_{(0,\bar{0})} = \sum_{i,i'} \int_0^{\beta} d\tau d\tau' \chi(\tau - \tau') \vec{S}_i(\tau) \cdot \vec{S}_{i'}(\tau'), \quad (12.10)$$

where the susceptibility $\chi_{ii'}(\tau - \tau')$ is defined as

$$\chi_{ii'}(\tau - \tau') = J^2 \sum_{j \in \text{NN of } i} \sum_{j' \in \text{NN of } i'} \vec{S}_j(\tau) \cdot \vec{S}_{j'}(\tau'). \quad (12.11)$$

The effective action for the $0, \bar{0}$ system, in the limit of large coordination number, simplifies to

$$\begin{aligned} S_{\text{eff}}^{0,\bar{0}} &= \sum_{i=0,\bar{0}} \sum_{\sigma} \int_0^{\beta} d\tau c_{i\sigma}^\dagger(\tau) \partial_{\tau} c_{i\sigma}(\tau) - t \sum_{\bar{0},\sigma} \int_0^{\beta} d\tau \left[c_{0\sigma}^\dagger(\tau) c_{\bar{0}\sigma}(\tau) + \text{h.c.} \right] \\ &\quad - \frac{U}{2} \sum_{i=0,\bar{0}} \int_0^{\beta} d\tau (n_{i\uparrow}(\tau) - n_{i\downarrow}(\tau))^2 + J \sum_{\bar{0}} \int_0^{\beta} d\tau \vec{S}_0(\tau) \cdot \vec{S}_{\bar{0}}(\tau) \\ &\quad + \sum_{i,i'} \int_0^{\beta} d\tau d\tau' \left[\sum_{\sigma} \Delta_{ii',\sigma}(\tau - \tau') c_{i\sigma}^\dagger(\tau) c_{i'\sigma}(\tau') + \chi_{ii'}(\tau - \tau') \vec{S}_i(\tau) \cdot \vec{S}_{i'}(\tau') \right] \end{aligned} \quad (12.12)$$

One can also write down an effective action purely for the local site 0. The sites $\bar{0}$ will now be a part of the environment action $S^{(0)}$ instead of the system action, but since the system is thermodynamically large, the cavity action can be assumed to remain the same, such the expectation values are calculated in the same system. As a result, the hybridisation and susceptibility functions remain

unchanged. S_{int} is now made of terms that connect 0 and $\bar{0}$, so the transformation from $S^{(0,\bar{0})}$ can be made by replacing the set $\{j\}$ with the set $\bar{0}$, and the set $\bar{0}$ itself will be replaced by the local site 0. We quote the final form of the local effective action:

$$S_{\text{eff}}^0 = \sum_{\sigma} \int_0^{\beta} d\tau c_{0\sigma}^{\dagger}(\tau) \partial_{\tau} c_{0\sigma}(\tau) - \frac{U}{2} \int_0^{\beta} d\tau (n_{0\uparrow}(\tau) - n_{0\downarrow}(\tau))^2 + \int_0^{\beta} d\tau d\tau' \left[\sum_{\sigma} \Delta_{00,\sigma}(\tau - \tau') c_{i\sigma}^{\dagger}(\tau) c_{i'\sigma}(\tau') + \chi_{00}(\tau - \tau') \vec{S}_i(\tau) \cdot \vec{S}_{i'}(\tau') \right] \quad (12.13)$$

2.12.2 Local theory for the extended SIAM

The Hamiltonian for the extended SIAM model is shown in eq. ?? . We will denote the impurity with the label d and the bath sites with the label c . Among the bath sites, we will represent the site coupled to the impurity with z . The full action has the form

$$S_{\text{ES}} = \sum_{d,c} \sum_{\sigma} \int_0^{\beta} d\tau c_{i\sigma}^{\dagger}(\tau) \partial_{\tau} c_{i\sigma}(\tau) - t \sum_{\langle i,j \rangle \in c,\sigma} \int_0^{\beta} d\tau \left[c_{i\sigma}^{\dagger}(\tau) c_{j\sigma}(\tau) + \text{h.c.} \right] - V \sum_{\sigma} \int_0^{\beta} d\tau \left[c_{d\sigma}^{\dagger}(\tau) c_{z\sigma}(\tau) + \text{h.c.} \right] + J \int_0^{\beta} d\tau \vec{S}_d(\tau) \cdot \vec{S}_z(\tau) - \frac{U}{2} \int_0^{\beta} d\tau (n_{d\uparrow}(\tau) - n_{d\downarrow}(\tau))^2 - \frac{W}{2} \int_0^{\beta} d\tau (n_{z\uparrow}(\tau) - n_{z\downarrow}(\tau))^2, \quad (12.14)$$

As in the Hubbard-Heisenberg model, we first obtain an effective action for the pair of sites (d, z) . As in the previous calculation, this will again generate a hybridisation function $\Delta_{zz,\sigma}$ for the bath site nearest to the zeroth site. No susceptibility will be generated however, because there is no spin-exchange coupling within the bath. The net result is

$$S_{\text{eff}}^{d,z} = \sum_{d,c} \sum_{\sigma} \int_0^{\beta} d\tau c_{i\sigma}^{\dagger}(\tau) \partial_{\tau} c_{i\sigma}(\tau) - V \sum_{\sigma} \int_0^{\beta} d\tau \left[c_{d\sigma}^{\dagger}(\tau) c_{z\sigma}(\tau) + \text{h.c.} \right] + J \int_0^{\beta} d\tau \vec{S}_d(\tau) \cdot \vec{S}_z(\tau) - \frac{U}{2} \int_0^{\beta} d\tau (n_{d\uparrow}(\tau) - n_{d\downarrow}(\tau))^2 - \frac{W}{2} \int_0^{\beta} d\tau (n_{z\uparrow}(\tau) - n_{z\downarrow}(\tau))^2 + \int_0^{\beta} d\tau d\tau' \sum_{\sigma} \Delta_{zz,\sigma}(\tau - \tau') c_{z\sigma}^{\dagger}(\tau) c_{z\sigma}(\tau') \quad (12.15)$$

This hybridisation $\Delta_{zz,\sigma}$ is calculated in the cavity model of the eSIAM, obtained by removing the impurity and the zeroth site from the full model; it is a completely non-interacting system.

One can go one step further and also remove the zeroth site in order to obtain a theory for the impurity site. With this choice, the cavity model now also includes the zeroth site, and hence contains the correlation term associated with W . The one-particle connection V will lead to a modified hybridisation $\mathcal{F}$ for this effective theory; the Kondo terms will generate a susceptibility. The resultant

action is

$$S_{\text{eff}}^d = \sum_{\sigma} \int_0^{\beta} d\tau \, c_{d\sigma}^{\dagger}(\tau) \partial_{\tau} c_{d\sigma}(\tau) - \frac{U}{2} \int_0^{\beta} d\tau \, (n_{d\uparrow}(\tau) - n_{d\downarrow}(\tau))^2 \\ + \int_0^{\beta} d\tau d\tau' \left[\sum_{\sigma} \mathcal{F}_{\sigma}(\tau - \tau') c_{i\sigma}^{\dagger}(\tau) c_{i'\sigma}(\tau') + \chi_d(\tau - \tau') \vec{S}_i(\tau) \cdot \vec{S}_{i'}(\tau') \right] \quad (12.16)$$

where $\mathcal{F}$ is defined in terms of the local correlator of the zeroth site

$$\mathcal{F}_{\sigma}(\tau - \tau') = V^2 G_1^{(d)}(z_{\sigma}\tau; z_{\sigma}\tau') = V^2 \langle c_{z\sigma}(\tau) c_{z\sigma}^{\dagger}(\tau') \rangle_{(d)} \quad (12.17)$$

with the average computed in the interacting cavity model that has the impurity site removed. The interaction arises from the W -term. The susceptibility is also calculated in this interacting cavity model, and follows the same definition as in the bulk model:

$$\chi_d(\tau - \tau') = J^2 \vec{S}_z(\tau) \cdot \vec{S}_z(\tau') . \quad (12.18)$$

Chapter 3

Competing tendencies In Heavy Fermion Systems: Kondo screening vs. Mott localisation

3.1 Bilayer extended Hubbard model: Impurity model

[60] We approach the heavy-fermion problem by starting from a bilayer extended Hubbard model, consisting of two layers (f and c). Towards studying this lattice model, we adopt a two-layer impurity problem that hosts a correlated impurity site in each layer (S_f and S_d):

$$H_{\text{aux}} = H_{\text{lp}} + H_f + H_d + H_{fd} , \quad (1.1)$$

where H_{lp} is the Hamiltonian for the non-interacting itinerant electrons of either layer,

$$H_{\text{lp}} = -t \sum_{\sigma, \alpha} \sum_{\langle i, j \rangle} \left(c_{i, \sigma, \alpha}^\dagger c_{j, \sigma, \alpha} + \text{h.c.} \right) - \sum_{i, \sigma, \alpha} \mu_\alpha n_{i, \sigma, \alpha} , \quad (1.2)$$

such that α sums over the two layers f and d . H_f and H_d describe the dynamics of the correlated impurity sites (and their local neighbourhood) in each layer:

$$\begin{aligned} H_f = & -\frac{U_f}{2} (n_{f, \uparrow} - n_{f, \downarrow})^2 + \sum_{Z \in \text{NN}} \left[-V \sum_{\sigma} \left(c_{f, \sigma}^\dagger c_{Z, \sigma, f} + \text{h.c.} \right) + \frac{J}{2} \sum_{\alpha, \beta} \mathbf{S}_f \cdot \boldsymbol{\sigma}_{\alpha\beta} c_{Z, \alpha, f}^\dagger c_{Z, \beta, f} \right. \\ & \left. - \frac{W_f}{2} (n_{Z, \uparrow, f} - n_{Z, \downarrow, f})^2 \right] , \\ H_d = & -\eta_d n_d - \frac{U_d}{2} (n_{d, \uparrow} - n_{d, \downarrow})^2 + \sum_{Z \in \text{NN}} \left[-V \sum_{\sigma} \left(c_{d, \sigma}^\dagger c_{Z, \sigma, d} + \text{h.c.} \right) + \frac{J}{2} \sum_{\alpha, \beta} \mathbf{S}_d \cdot \boldsymbol{\sigma}_{\alpha\beta} c_{Z, \alpha, d}^\dagger c_{Z, \beta, d} \right] , \end{aligned} \quad (1.3)$$

where $c_{v, \sigma, \alpha}^\dagger$ can refer to creation operator for either the f -layer. Finally, H_{fd} represents the inter-layer hybridisation:

$$H_{fd} = -V_\perp \sum_{\sigma} \left(c_{d, \sigma}^\dagger c_{f, \sigma} + \text{h.c.} \right) . \quad (1.4)$$

3.2 Tiling Into Bilayer Extended Hubbard Model

Tiling the impurity model leads to a bilayer extended Hubbard model on the 2D square lattice. In order to tile our auxiliary model to restore translation invariance, we identify the cluster which acts as the “unit” cell of translation. As discussed in the text surrounding eq. 3.9, a good choice for the unit cell is the impurity site and its surrounding correlated environment. Placing the impurity sites at $\mathbf{r}_d$ in each layer, we get

$$H_{\text{clust}}(\mathbf{r}_d) = H_f(\mathbf{r}_d) + H_d(\mathbf{r}_d) + H_{fc}(\mathbf{r}_d) . \quad (2.1)$$

The details of tiling a cluster Hamiltonian were already shown in Sec. 1.3.2; the general idea is to use manybody translation operators (details in Sec. 1.3.1) to center the cluster Hamiltonian at various sites of the lattice, such that combining all such terms leads to a periodised Hamiltonian. we write down the final result here:

$$\begin{aligned} H_{\text{latt}} &= \sum_{\mathbf{r}} T^\dagger(\mathbf{r}) H_{\text{clust}}(\mathbf{r}_d) T(\mathbf{r}) \\ &= -\frac{2V}{Z} \sum_{\langle i,j \rangle, \sigma} \left(c_{i,\sigma}^\dagger c_{j,\sigma} + \text{h.c.} \right) - \sum_i \left[\eta_d n_{i,d} + \frac{U_d}{2} (n_{i,\uparrow,d} - n_{i,\downarrow,d})^2 \right] + \frac{2J}{Z} \sum_{\langle i,j \rangle} \mathbf{S}_{i,d} \cdot \mathbf{S}_{j,d} \\ &\quad - \frac{2V}{Z} \sum_{\langle i,j \rangle, \sigma} \left(f_{i,\sigma}^\dagger f_{j,\sigma} + \text{h.c.} \right) - \frac{U_f - W_f}{2} \sum_i (n_{i,\uparrow,f} - n_{i,\downarrow,f})^2 + \frac{2J}{Z} \sum_{\langle i,j \rangle} \mathbf{S}_{i,f} \cdot \mathbf{S}_{j,f} \\ &\quad - V_\perp \sum_{i,\sigma} (f_{i,\sigma}^\dagger c_{i,\sigma} + \text{h.c.}) \end{aligned} \quad (2.2)$$

Tiling the impurity model therefore leads to a bilayer extended Hubbard model, with effective couplings that depend on the auxiliary model couplings. The value of η_d defines the effective chemical potential of the d -layer; a non-zero value moves the layer away from half-filling. The f -layer is pinned at half-filling.

Following Sec. 1.5.1, the retarded k -space Greens function in either layer can be expressed in terms of auxiliary model Greens functions:

$$G(\mathbf{k}, \sigma, \alpha; \omega) = \sum_{\nu} C_{\nu, k\alpha} \mathcal{G}(\mathcal{T}_{\nu, \sigma, \alpha}; \omega - \mathcal{E}_\nu(\mathbf{k})) \quad (2.3)$$

where $\alpha = f, d$ denotes the layer in which the Greens function is being calculated, the modes ν indicate the operators which diagonalise the Hamiltonian H_{1p} , with energies $\mathcal{E}_\nu(\mathbf{k})$:

$$H_{1p} = -\frac{2V}{Z} \sum_{\langle i,j \rangle, \sigma} \left(c_{i,\sigma}^\dagger c_{j,\sigma} + \text{h.c.} + f_{i,\sigma}^\dagger f_{j,\sigma} + \text{h.c.} \right) - V_\perp \sum_{i,\sigma} (f_{i,\sigma}^\dagger c_{i,\sigma} + \text{h.c.}) - \sum_i \eta_d n_{i,d} , \quad (2.4)$$

and the $\mathcal{T}$ -matrices are defined in eq. 5.11:

$$\mathcal{T}_{\nu, \sigma, \alpha} = \left[\left(\sum_{\sigma'} c_{\alpha\sigma'}^\dagger + \text{h.c.} \right) + (S_\alpha^+ + \text{h.c.}) + (C_\alpha^+ + \text{h.c.}) \right] c_{\nu, \sigma} , \quad (2.5)$$

In order to calculate $\mathcal{E}_v(\mathbf{k})$, we introduce the k -space modes via a Fourier transform:

$$c_{i,\sigma}^\dagger = \frac{1}{\sqrt{N}} \sum_{\mathbf{k}} e^{-i\mathbf{k} \cdot \mathbf{r}_i} c_{\mathbf{k},\sigma}^\dagger, \quad f_{i,\sigma}^\dagger = \frac{1}{\sqrt{N}} \sum_{\mathbf{k}} e^{-i\mathbf{k} \cdot \mathbf{r}_i} f_{\mathbf{k},\sigma}^\dagger. \quad (2.6)$$

In terms of these modes, the 1-particle Hamiltonian decouples in k -space:

$$H_{1p} = \sum_{\mathbf{k},\sigma} \left[(\epsilon_{\mathbf{k}} - \eta_d)(n_{\mathbf{k},\sigma,d} + n_{\mathbf{k},\sigma,f}) - V_\perp (c_{\mathbf{k},\sigma}^\dagger f_{\mathbf{k},\sigma} + \text{h.c.}) \right]. \quad (2.7)$$

The diagonal modes are given by

$$v_{\mathbf{k},\sigma,\pm}^\dagger = C_{\pm,\mathbf{k}} c_{\mathbf{k},\sigma}^\dagger + \sqrt{1 - C_{\pm,\mathbf{k}}^2} f_{\mathbf{k},\sigma}^\dagger, \quad (2.8)$$

the coefficients obtained from diagonalising the matrix

$$H_{1p}(\mathbf{k}) = \begin{pmatrix} \epsilon_{\mathbf{k}} & -V_\perp \\ -V_\perp & \epsilon_{\mathbf{k}} - \eta_d \end{pmatrix}. \quad (2.9)$$

The energies are given by $\mathcal{E}_\pm(\mathbf{k}) = \epsilon_{\mathbf{k}} - \eta_d/2 \pm \Delta$, with $\epsilon_{\mathbf{k}}$ being $-\frac{4V}{Z}(\cos k_x + \cos k_y)$ for a 2D square lattice and $\Delta = \sqrt{V_\perp^2 + \eta_d^2}/4$. Substituting into the expression for the lattice Greens function, we get

$$\begin{aligned} G(\mathbf{k}, d; \omega) = & C_{+,\mathbf{k}} \mathcal{G}(\mathcal{T}_{\mathbf{k}+,d}, \mathcal{T}_{\mathbf{k},d}^\dagger; \omega - \mathcal{E}_+(\mathbf{k})) + C_{-,\mathbf{k}} \mathcal{G}(\mathcal{T}_{\mathbf{k}-,d}, \mathcal{T}_{\mathbf{k},d}^\dagger; \omega - \mathcal{E}_-(\mathbf{k})) \\ & + C_{+,\mathbf{k}}^* \mathcal{G}(\mathcal{T}_{\mathbf{k},d}, \mathcal{T}_{\mathbf{k}+,d}^\dagger; \omega - \mathcal{E}_+(\mathbf{k})) + C_{-,\mathbf{k}}^* \mathcal{G}(\mathcal{T}_{\mathbf{k},d}, \mathcal{T}_{\mathbf{k}-,d}^\dagger; \omega - \mathcal{E}_-(\mathbf{k})). \end{aligned} \quad (2.10)$$

Following eq. 5.33, we can similarly write down an expression the local lattice Greens function for the f - and d -electrons. For that, we define the local diagonal modes $v_\pm$ obtained from the local Hamiltonian

$$H_{\text{loc}} = \begin{pmatrix} 0 & -V_\perp \\ -V_\perp & 0 - \eta_d \end{pmatrix}, \quad (2.11)$$

and their associated energies $\varepsilon_\pm$. Within the impurity model, these local bonding/antibonding modes can be constructed on the impurity sites ($v_\pm$) or on the neighbouring bath sites, ($v_\pm^Z$). The local Greens function can be expressed in terms of these modes:

$$\begin{aligned} G_{dd}(\omega) = & \frac{1}{2} \sum_{\nu} C_{\nu,d} [\mathcal{G}(v\sigma, d\sigma; \omega - \varepsilon_\nu) + \mathcal{G}(d\sigma, v\sigma; \omega - \varepsilon_\nu)] \\ & + \frac{1}{2} \sum_{\nu^Z} C_{\nu^Z,Z_d} [\mathcal{G}(\mathcal{T}_{Z_d,\sigma}, \mathcal{T}_{\nu^Z,\sigma}; \omega - \varepsilon_{\nu^Z}) + \mathcal{G}(\mathcal{T}_{\nu^Z,\sigma}, \mathcal{T}_{Z_d,\sigma}; \omega - \varepsilon_{\nu^Z})]. \end{aligned} \quad (2.12)$$

The presence of local hybridisation between the d - and f - orbitals makes the inter-orbital Greens function important as well. Following eq. 5.35, we have

$$\begin{aligned} G_{fd}(\omega) = & \frac{1}{2} \sum_{\nu} [C_{\nu,f} \mathcal{G}(v\sigma, d\sigma; \omega - \varepsilon_\nu) + C_{\nu,d} \mathcal{G}(f\sigma, v\sigma; \omega - \varepsilon_\nu)] \\ & + \frac{1}{2} \sum_{\nu^Z} [C_{\nu^Z,Z_f} \mathcal{G}(\mathcal{T}_{\nu^Z,\sigma}, \mathcal{T}_{Z_d,\sigma}; \omega - \varepsilon_{\nu^Z}) + C_{\nu^Z,Z_d} \mathcal{G}(\mathcal{T}_{Z_f,\sigma}, \mathcal{T}_{\nu^Z,\sigma}; \omega - \varepsilon_{\nu^Z})]. \end{aligned} \quad (2.13)$$

Using these, we can construct the hybridised Greens function associated with the states v :

$$G_{v+}(\omega) = C_{+,d}^2 G_{dd}(\omega) + C_{+,f}^2 G_{ff}(\omega) + C_{+,d} C_{+,f} [G_{df}(\omega) + G_{fd}(\omega)] , \quad (2.14)$$

3.3 Unitary RG Analysis of The Bilayer Lattice-Embedded ESIAM

In the limit of large U_α , we carry out a Schrieffer-Wolff transformation and work with the following low-energy Hamiltonian:

$$H_{\text{aux}} = \sum_{\mathbf{k}, \sigma, \alpha} \epsilon_{\mathbf{k}, \alpha} n_{\mathbf{k}, \sigma, \alpha} + \sum_{\alpha} \sum_{Z \in \text{NN}} \left[\frac{1}{2} J_{\alpha} \sum_{\sigma, \sigma'} \mathbf{S}_{\alpha} \cdot \boldsymbol{\sigma}_{\alpha \beta} c_{Z, \sigma, \alpha}^{\dagger} c_{Z, \sigma', \alpha} - \frac{W_{\alpha}}{2} (n_{Z, \uparrow, \alpha} - n_{Z, \downarrow, \alpha})^2 \right] + J \mathbf{S}_f \cdot \mathbf{S}_d . \quad (3.1)$$

The Kondo coupling and bath interaction acquire k -dependence:

$$J(\mathbf{k}, \mathbf{q}) = \frac{J}{2} [\cos(\mathbf{k}_x - \mathbf{q}_x) + \cos(\mathbf{k}_y - \mathbf{q}_y)] , \quad (3.2)$$

$$W(\mathbf{k}, \mathbf{q}, \mathbf{k}', \mathbf{q}') = \frac{W}{2} [\cos(\mathbf{k}_x - \mathbf{q}_x + \mathbf{k}'_x - \mathbf{q}'_x) + \cos(\mathbf{k}_y - \mathbf{q}_y + \mathbf{k}'_y - \mathbf{q}'_y)] .$$

The particle-hole transformed state $\bar{\mathbf{q}}$ associated with $\mathbf{q}$ is defined as $\bar{\mathbf{q}} = \mathbf{q} + (\pi, \pi)$ if $\mathbf{q}$ is in the first or third quadrant and $\mathbf{q} + (\pi, -\pi)$ is in the second or fourth quadrant.

In order to study the low-energy physics of the impurity model, we iteratively integrate out high-energy degrees of freedom using the unitary RG method. Let the momentum states being decoupled from the two layers be $|\mathbf{q}, \sigma, f\rangle$ and $|\mathbf{q}, \sigma, d\rangle$ if they are above the Fermi energy (unoccupied in the low-energy configuration) and $|\bar{\mathbf{q}}, \sigma, f\rangle$ and $|\bar{\mathbf{q}}, \sigma, d\rangle$ if they are below the Fermi energy (occupied in the low-energy configuration). For every shell of conduction bath states of width ΔD integrated out, the Hamiltonian couplings renormalise by folding in the effects of virtual excitations to these states. We already have the renormalisation group equations for the couplings J_{α} in the case of $J = 0$:

$$\frac{\Delta J_f(\mathbf{k}_1, \mathbf{k}_2)}{\Delta D} = - \int_{UV} d\mathbf{q} \rho(\mathbf{q}) [J_f(\mathbf{k}_1, \mathbf{q}) J_f(\mathbf{q}, \mathbf{k}_2) + 4 J_f(\mathbf{q}, \bar{\mathbf{q}}) W_f(\mathbf{k}_1, \mathbf{q}, \bar{\mathbf{q}}, \mathbf{k}_2)] G_f^{(0)}(\mathbf{q}) , \quad (3.3)$$

where $\mathbf{q}$ sums over the UV states being integrated out, $\rho(\mathbf{q})$ is the density of states for the momentum states being integrated out and $G_f^{(0)}$ is the propagator for the excited intermediate state (for $J = 0$):

$$G_f^{(0)}(\mathbf{q}) = \frac{1}{\omega - |\epsilon_f(\mathbf{q})|/2 + J_f(\mathbf{q})/4 + W_f(\mathbf{q})/2} . \quad (3.4)$$

An identical equation holds for ΔJ_d , with all f -quantities replaced with d -quantities. Switching on J modifies the excitation energy in the propagator:

$$G_f(\mathbf{q}) = \frac{1}{1/G_f^{(0)} - J \vec{S}_f \cdot \vec{S}_d} . \quad (3.5)$$

This expression can be simplified by noting that the operator in the denominator has only two eigenvalues, $-3/4$ in the singlet sector and $1/4$ in the triplet sector. Defining the projectors $\mathcal{P}_0$ and $\mathcal{P}_1$ for the two sectors, we have

$$\vec{S}_f \cdot \vec{S}_d = \frac{-3}{4} \mathcal{P}_0 + \frac{1}{4} \mathcal{P}_1, \quad \mathcal{P}_0 + \mathcal{P}_1 = \mathbb{I}. \quad (3.6)$$

Using these relations, we can write

$$\begin{aligned} G_f(\mathbf{q}) &= \frac{1}{1/G_f^{(0)} + 3J/4} \mathcal{P}_0 + \frac{1}{1/G_f^{(0)} - J/4} \mathcal{P}_1 \\ &= \left(\frac{1/4}{1/G_f^{(0)} + 3J/4} + \frac{3/4}{1/G_f^{(0)} - J/4} \right) \mathbb{I} + \left(\frac{1}{1/G_f^{(0)} + 3J/4} - \frac{1}{1/G_f^{(0)} - J/4} \right) \vec{S}_f \cdot \vec{S}_d. \end{aligned} \quad (3.7)$$

Noting that only the first operator renormalises the Kondo interaction, we get the following modified RG equation for the Kondo coupling J_α :

$$\begin{aligned} \frac{\Delta J_\alpha}{\Delta D} &= \int_{UV} d\mathbf{q} \rho(\mathbf{q}) [J_\alpha(\mathbf{k}_1, \mathbf{q}) J_\alpha(\mathbf{q}, \mathbf{k}_2) + J_\alpha(\mathbf{q}, \bar{\mathbf{q}}) W_\alpha(\mathbf{k}_1, \mathbf{q}, \bar{\mathbf{q}}, \mathbf{k}_2)] \left(\frac{1}{4} \frac{1}{1/G_\alpha^{(0)}(\mathbf{q}) + 3J/4} \right. \\ &\quad \left. + \frac{3}{4} \frac{1}{1/G_\alpha^{(0)}(\mathbf{q}) - J/4} \right), \end{aligned} \quad (3.8)$$

We now turn to the renormalisation of J , arising from hybridisation of f - and d -layers. In order to capture coherent scattering processes between the two layers, we project onto the following rotated basis of excited states:

$$\begin{aligned} |H\rangle_\pm &= \frac{1}{\sqrt{2}} \left(|H(\mathbf{q})\rangle_f \pm |H(\mathbf{q})\rangle_d \right), \\ \mathcal{P}(\mathbf{q})_H &= |H\rangle_+ \langle H|_+ + |H\rangle_- \langle H|_- , \end{aligned} \quad (3.9)$$

where $|H(\mathbf{q})\rangle_\alpha$ is the state obtained upon exciting both states in the layer α : $|H(\mathbf{q})\rangle_\alpha = c_{\mathbf{q},\alpha}^\dagger c_{\bar{\mathbf{q}},\alpha} |L\rangle_f |L\rangle_d$, and $\mathcal{P}(\mathbf{q})_H$ projects onto this rotated excited basis. We focus on the spin-flip component $J S_f^+ S_d^-$ of the Hamiltonian. There are two kinds of processes that renormalise this coupling. We first consider one that involves a spin-flip of the d -layer followed by a spin-flip of the f -layer:

$$\Delta H = \frac{1}{N^2} \sum_{\mathbf{q} \in UV} \langle L | \frac{1}{2} J_f(\bar{\mathbf{q}}, \mathbf{q}) S_f^+ c_{\mathbf{q}\downarrow,f}^\dagger c_{\mathbf{q}\uparrow,f} \mathcal{P}(\mathbf{q})_H G \mathcal{P}(\mathbf{q})_H \frac{1}{2} J_d(\mathbf{q}, \bar{\mathbf{q}}) S_d^- c_{\mathbf{q}\uparrow,d}^\dagger c_{\bar{\mathbf{q}}\downarrow,d} | L \rangle , \quad (3.10)$$

where $|L\rangle = |L\rangle_f |L\rangle_d$ is the initial configuration for both layers. G is the propagator for the excited state:

$$G = \frac{1}{\omega - H_D}, \quad (3.11)$$

where H_D is the diagonal part of the Hamiltonian corresponding to the excited(intermediate) state. The diagonal part consists of the kinetic energy and the Ising part of the Kondo interaction:

$$H_D = \sum_{\mathbf{q}, \sigma \in \mathbf{q}_{\pm}, \alpha} \epsilon_{\alpha}(\mathbf{q}) \tau_{\mathbf{q}, \sigma} + \frac{1}{2} \sum_{\alpha} J_{\alpha} S_{\alpha}^z \sum_{\mathbf{q}, \sigma} \sigma n_{\mathbf{q}, \sigma, \alpha} + J \vec{S}_f \cdot \vec{S}_d . \quad (3.12)$$

Calculating the matrix elements of the propagator in the rotated basis gives

$$\mathcal{P}(\mathbf{q})_H G \mathcal{P}(\mathbf{q})_H = \frac{1}{2} (G_f + G_d) (|H\rangle_+ \langle H|_+ + |H\rangle_- \langle H|_-) + \frac{1}{2} (G_f - G_d) (|H\rangle_+ \langle H|_- + |H\rangle_- \langle H|_+) , \quad (3.13)$$

where

$$G_{\alpha}(\mathbf{q}) = \frac{1}{\omega - \frac{1}{2} [\epsilon_{\alpha}(\mathbf{q}) - \epsilon_{\alpha}(\bar{\mathbf{q}})] + \frac{1}{2} J_{\alpha}(\mathbf{q}) + W_{\alpha}(\mathbf{q}) - \frac{1}{4} J} . \quad (3.14)$$

In eq. 3.10, since the first process is an excitation into a d -state and the second process is a de-excitation from an f -state, the expression can be simplified into

$$\begin{aligned} \langle L | S_f^+ c_{\mathbf{q}-\downarrow, f}^{\dagger} c_{\mathbf{q}+\uparrow, f} \mathcal{P}(\mathbf{q})_H G \mathcal{P}(\mathbf{q})_H S_d^- c_{\mathbf{q}+\uparrow, d}^{\dagger} c_{\mathbf{q}-\downarrow, d} | L \rangle &= S_f^+ S_d^- \langle H_f | \mathcal{P}(\mathbf{q})_H G \mathcal{P}(\mathbf{q})_H | H_d \rangle \\ &= \frac{1}{2} (G_f + G_d) S_f^+ S_d^- . \end{aligned} \quad (3.15)$$

The net Hamiltonian renormalisation is finally a sum over the contribution from each mode:

$$\Delta H = \frac{1}{8N^2} \sum_{\mathbf{q} \in UV} J_f(\bar{\mathbf{q}}, \mathbf{q}) J_d(\mathbf{q}, \bar{\mathbf{q}}) (G_f(\mathbf{q}) + G_d(\mathbf{q})) S_f^+ S_d^- . \quad (3.16)$$

Converting the sums to integrals (assuming a density of states $\rho(\mathbf{q})$) gives the final expression

$$\frac{\Delta H}{\Delta D} = \frac{1}{8N} S_f^+ S_d^- \int_{UV} d\mathbf{q} \rho(\mathbf{q}) J_f(\bar{\mathbf{q}}, \mathbf{q}) J_d(\mathbf{q}, \bar{\mathbf{q}}) (G_f(\mathbf{q}) + G_d(\mathbf{q})) . \quad (3.17)$$

The time-reversed of this process can be obtained simply by exchanging the d -interaction with the f -interaction. Since the renormalisation itself is symmetric, the contribution is equal to what we obtained here. The total renormalisation in the coupling for the term $S_f^+ S_d^-$ is therefore

$$\begin{aligned} \frac{\Delta J}{\Delta D} &= \frac{1}{2} \int_{UV} d\mathbf{q} \rho(\mathbf{q}) J_f(\bar{\mathbf{q}}, \mathbf{q}) J_d(\mathbf{q}, \bar{\mathbf{q}}) (G_f(\mathbf{q}) + G_d(\mathbf{q})) \\ &= \frac{1}{2} \int_{UV} d\mathbf{q} \rho(\mathbf{q}) \left[\frac{J_f(\bar{\mathbf{q}}, \mathbf{q}) J_d(\mathbf{q}, \bar{\mathbf{q}})}{\omega - \frac{1}{2} \bar{\epsilon}_f(\mathbf{q}) + \frac{1}{2} J_f(\mathbf{q}) + W_f(\mathbf{q}) - \frac{1}{4} J} + \frac{J_f(\bar{\mathbf{q}}, \mathbf{q}) J_d(\mathbf{q}, \bar{\mathbf{q}})}{\omega - \frac{1}{2} \bar{\epsilon}_d(\mathbf{q}) + \frac{1}{2} J_d(\mathbf{q}) + W_d(\mathbf{q}) - \frac{1}{4} J} \right] , \end{aligned} \quad (3.18)$$

where $\bar{\epsilon}_{\alpha}(\mathbf{q}) = \epsilon_f(\mathbf{q}) - \epsilon_f(\bar{\mathbf{q}})$ is the total excitation energy for the pair of states $\mathbf{q}, \bar{\mathbf{q}}$.

Since the bath interaction doesn't renormalise, it's functional form remains unchanged and can be substituted into the RG equations. In fact, we can write the expressions in a compact matrix form for efficient computation:

$$\frac{\Delta J_{\alpha}}{\Delta D} = -J_{\alpha} \mathcal{G}_{\alpha} J_{\alpha}^{\dagger} - 4 \mathcal{W}_{\alpha} \text{Tr} [\mathcal{G}'_{\alpha}] , \quad (3.19)$$

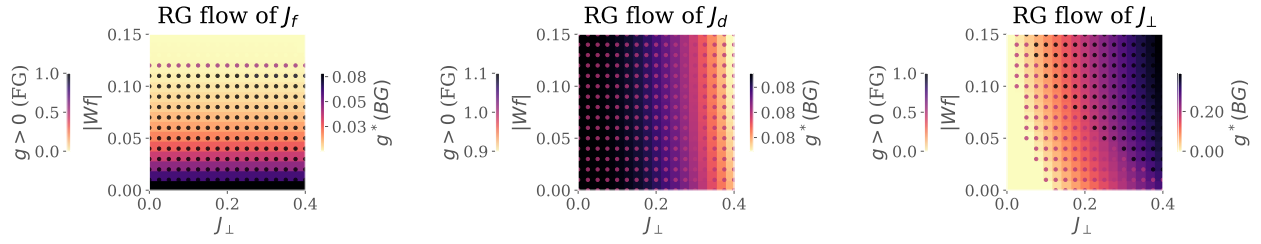


Figure 3.1: RG phase diagrams for various Kondo couplings: f -layer (left), d -layer (middle) and inter-layer (right). For each plot, the background color (quantified by the right colorbar) tracks the fixed point value of the coupling as a function of the inter-layer couplings $J_\perp$ and the f -layer bath interaction $|W_f|$. The marker color complements this by tracking whether the coupling is relevant, intermediate-coupling or irrelevant in the RG sense, by taking the values 1, 1/2 or 0 respectively.

where all objects are matrices. J_α is the Kondo coupling matrix. $\mathcal{G}_\alpha$ is a diagonal matrix with matrix elements

$$\mathcal{G}_\alpha[q, q] = \begin{cases} \rho(q) \left(\frac{1}{8} \frac{1}{1/G_\alpha^{(0)}(\mathbf{q}) + \mu/2 + 3J/4} + \frac{3}{8} \frac{1}{1/G_\alpha^{(0)}(\mathbf{q}) + \mu/2 - J/4} \right), & \mathbf{q} \in \text{PS}, \\ \rho(q) \left(\frac{1}{8} \frac{1}{1/G_\alpha^{(0)}(\mathbf{q}) - \mu/2 + 3J/4} + \frac{3}{8} \frac{1}{1/G_\alpha^{(0)}(\mathbf{q}) - \mu/2 - J/4} \right), & \mathbf{q} \in \text{HS}, \end{cases} \quad (3.20)$$

if q is in the UV subspace and zero otherwise. $\mathcal{W}_\alpha$ is the bath interaction matrix whose matrix elements are given by $\mathcal{W}_\alpha[k, k'] = \frac{W}{2} [\cos(k_x - k'_x) + \cos(k_y - k'_y)]$. Finally, $\mathcal{G}'_\alpha$ is another diagonal matrix whose matrix elements are defined as $\mathcal{G}'_\alpha[q, q] = \mathcal{G}[q, q]J_\alpha[q, \bar{q}]$.

The RG equation for the inter-layer coupling J can be written as

$$\frac{\Delta J}{\Delta D} = \frac{1}{2} \mathbf{J} \cdot \mathbf{G}, \quad (3.21)$$

where $\mathbf{J}$ and $\mathbf{G}$ are vectors defined by the elements $\mathbf{J}[q] = J_f(\bar{\mathbf{q}}, \mathbf{q})J_d(\mathbf{q}, \bar{\mathbf{q}})$ and

$$\mathbf{G}[q] = \rho(\mathbf{q}) \left[\frac{1}{\omega - \frac{1}{2}\bar{\epsilon}_f(\mathbf{q}) + \frac{1}{2}J_f(\mathbf{q}) + W_f(\mathbf{q}) - \frac{1}{4}J} + \frac{1}{\omega - \frac{1}{2}\bar{\epsilon}_d(\mathbf{q}) + \frac{1}{2}J_d(\mathbf{q}) + W_d(\mathbf{q}) - \frac{1}{4}J} \right]. \quad (3.22)$$

3.4 Renormalisation Group Flows of Kondo Couplings: RG Phase Diagram

We first analyse the model by obtaining unitary RG flows of three kinds of Kondo couplings. Fig. 3.1 shows the fixed point values of the couplings under renormalisation to a low-energy theory. We view these values as a function of the f -layer bath interaction W_f and inter-layer Kondo coupling $J_\perp$. In each panel, the right (left) colorbar quantifies the background (marker) color of the plot. The background displays the fixed point value of each coupling whereas the marker color tracks whether the couplings is relevant (1), intermediate coupling (0.5) or irrelevant (0).

The Kondo coupling J_d (middle panel) of the d -layer remains relevant for the entirety of the chosen parameter regime. The Kondo coupling J_f (left panel) of the f -layer becomes RG-irrelevant for larger values of $|W_f|$; this leads to an in-plane Mott transition of the kind observed in our earlier work, where the local Fermi liquid excitations of the f -impurity give way to a pseudogapped non-Fermi liquid Mott metal phase which ultimately destabilises into a local moment phase. The RG flow of J_f does not appear to be significantly affected by the variation of $J_\perp$. Concomitant with the irrelevance of J_f under increasing $|W_f|$, the inter-plane coupling $J_\perp$ (right panel) becomes RG-relevant, as seen from the darkening markers as we go to higher $|W_f|$.

These simulations motivate an RG-based phase diagram with the following regimes: (i) weak $J_\perp$, weak $|W_f|$: decoupled metallic layers, each layer participating in transport independently, (ii) strong $J_\perp$, strong $|W_f|$: Kondo insulator regime, where the local moments of the f -layer hybridise with the itinerant electrons of the d -layer to form a band insulator with the chemical potential situated in the hybridisation gap, and (iii) weak $J_\perp$, strong $|W_f|$: the so-called RKKY regime, where the f -layer forms local moments which decouple from the d -layer and are then free to order magnetically, and (iv) strong $J_\perp$, weak $|W_f|$: this is similar to the Kondo insulator regime, except here the f -electrons remain itinerant and it is they who hybridise with the d -electrons to again form two bands separated by a hybridisation gap.

3.5 Results at half-filling

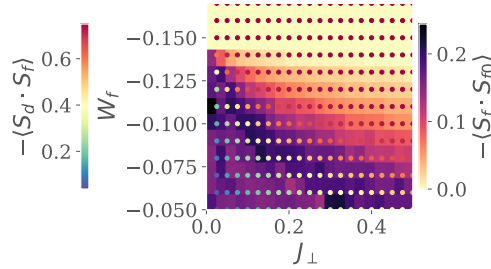


Figure 3.2: Left: Real-space spin correlations for the auxiliary model. Background colour (right colourbar) represents that between the f -impurity and its nearest-neighbour sites, while the marker colour (left colourbar) represents that between the f -impurity and the d -impurity. While the former shrinks as $J_\perp$ or $|W_f|$ is increased, the latter grows. This marks a transfer of entanglement from intra-plane channel into inter-plane channel. Right:

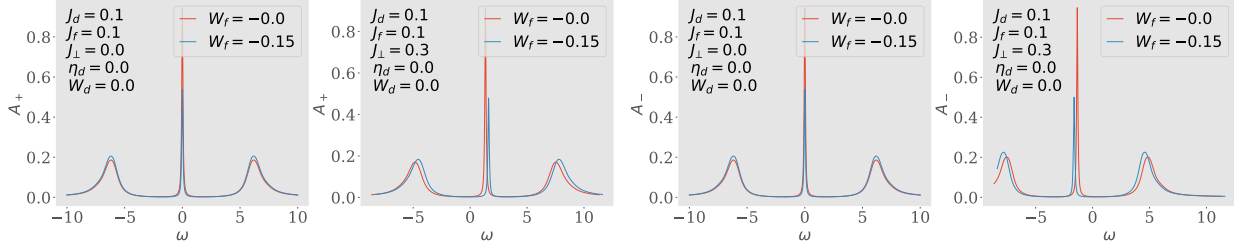


Figure 3.3: Spectral functions of the tiled model, for excitations in the bonding (two on the left) and antibonding (two on the right) basis, defined by the Greens function $G_{k,\pm} = G(k_d \pm k_f; \omega)$. There is no hybridisation for $J_\perp = 0$ (first and third plots), and the only effect of W_f is to shrink the height of the central resonance. Introducing a non-zero $J_\perp$ splits the central resonance into an upper band (second from left) and a lower band (first from right). This leads to a Kondo insulator, where $\omega = 0$ resides inside the hybridisation gap.

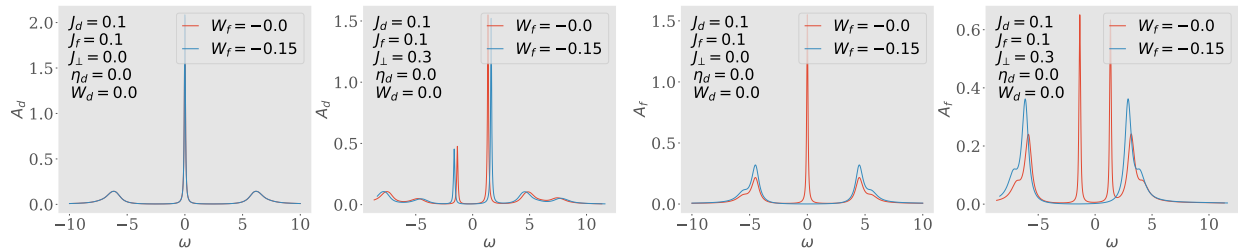


Figure 3.4: Spectral functions A_d and A_f of the auxiliary model, for excitations in the d - and f -layers respectively. Both show a central Kondo resonance for $J_\perp = 0$ which splits into two bands upon adding a non-zero $J_\perp$. Increasing $|W_f|$ to large values gaps out the local Fermi liquid excitations of A_f .

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